

Executive Summary

This audit report was prepared by Quantstamp, the leader in blockchain security.

Type	Prediction Market	Documentation quality	High	
Timeline	2026-07-27 through 2026-08-07	Test quality	High	
Language	Solidity	Total Findings	11	Fixed: 5 Acknowledged: 6
Methods	Architecture Review, Unit Testing, Functional Testing, Computer-Aided Verification, Manual Review	High severity findings ⓘ	0	
Source Code	Polymarket/polymarket-v2 ↗ #11627f7 ↗	Medium severity findings ⓘ	0	
Auditors	- Jennifer Wu Auditing Engineer - Tim Sigl Auditing Engineer - Paul Clemson Auditing Engineer	Low severity findings ⓘ	2	Fixed: 1 Acknowledged: 1
		Undetermined severity findings ⓘ	0	
		Informational findings ⓘ	9	Fixed: 4 Acknowledged: 5

Summary of Findings

Quantstamp conducted a Polymarket V2 diff review comparing `v1.0.0` against commit `11627f7` . The review focused on changes that affect oracle resolution, migration, NegRisk accounting, and combinatorial payouts.

The main changes are:

- OracleAggregator redesign:** the oracle layer was redesigned around `OracleAggregator` , configurable reporters, disputers, arbitrators, finalizers, on-chain market data, and the new `OOReporterModule` adapter for UMA's managed `OOReporter` . The previous `OptimisticOracleModule` flow was removed. Reporter modules such as `ChainlinkReporterModule` rely on `OracleAggregator.reportResult()` for per-request vote deduplication rather than storing local report state.
- Resolution-chain bits in IDs:** condition and event IDs encode the intended resolution chain. This makes chain-specific resolution part of the ID model and changes the operational meaning of resolver permissions. The change remains compatible with existing condition IDs in use on the Polygon chain because the Polygon resolution-chain value is zero.
- Migration changes:** migration supports users who migrate after legacy markets have resolved. The migration path can redeem resolved legacy collateral into the V2 vault during migration instead of blocking late users.
- NegRisk one-winner model:** NegRisk follows the one-winner assumption used by legacy NegRisk markets.
- Automatic NegRisk result derivation:** NegRisk results are derived automatically from aggregate event state. Once one real condition resolves YES, sibling real conditions derive NO. If every real condition resolves NO, synthetic Other derives YES.
- Combinatorial payout changes:** combinatorial flows support collateral-return operations across multi-leg positions, including basket conversions and event-level NegRisk composition.

These changes affect core assumptions around ID uniqueness, resolution authority, migration settlement, NegRisk payout derivation, and combinatorial accounting.

The audit identified two low severity issues and several informational severity issues. The low findings are `POL-1` and `POL-2`. `POL-1` is the primary solvency issue because the upgraded `NegRiskModule` derives synthetic Other as full YES once every real condition is resolved. If a pre-upgrade event already had a partial real-condition result, synthetic Other would no longer be computed as the remaining payout, allowing redemptions to mint more pUSD than the event's burned collateral. `POL-2` is a low-severity oracle configuration issue where a result can be reported to a mismatched target module, blocking the intended aggregator resolution path and emitting misleading resolution events, with admin recovery still available. The remaining informational findings are mostly defense-in-depth recommendations. We recommend addressing these issues and validating each fix with additional tests.

Fix-Review Update 2026-08-20:

During the fix review, the client addressed the reported issues and suggestions through fixes or acknowledgements. The changes include hardened result-bridge routing, real-condition-only NegRisk resolution counting with `v1.0.0` compatibility, migration alias protections that prevent

legacy markets from being bound to existing or duplicate V2 result state, module/route validation for oracle reports, and additional guards around liveness and mutable request configuration.

Fix-Review Update 2026-08-25:

Repository: `https://github.com/Polymarket/polymarket-v2` Commit: `7656bb2a6c4d64de4084ea50b1c17854e88bc056`

We reviewed an additional fix commit for PR-341, focused on maintaining backwards compatibility with `CTFExchangeV2` by ensuring that taker BUY orders are filled based on the collateral spent. No security issues were identified in relation to this change, but a handful of operational considerations related to the change were added under the section "Exchange Order Matching".

ID	DESCRIPTION	SEVERITY	STATUS
POL-1	Synthetic Other Derivation Does Not Check <code>resultsSum</code> Against Pre-Upgrade Partial Results	• Low ⓘ	Acknowledged
POL-2	Results Can Be Reported To A Mismatched Target Module Without Any Module Id Check	• Low ⓘ	Fixed
POL-3	Bridged Synthetic Other Can Push <code>conditionsResolved</code> Past Event Arity	• Informational ⓘ	Fixed
POL-4	Migration Registration Can Alias Existing Native V2 Result State	• Informational ⓘ	Fixed
POL-5	Divergent Reporter Votes Can Exhaust the Reporter Set without Triggering Arbitration	• Informational ⓘ	Acknowledged
POL-6	Excessive Liveness Windows Can Effectively Disable Normal Finalization	• Informational ⓘ	Fixed
POL-7	Empty <code>PositionManager</code> Batch Calls Can Emit Unauthorized No-Op Transfer Events	• Informational ⓘ	Acknowledged
POL-8	<code>ChainlinkReporterModule</code> Accepts Non-Positive Asset Prices	• Informational ⓘ	Acknowledged
POL-9	Rule Updates Can Diverge After <code>00Reporter</code> Has Resolved	• Informational ⓘ	Acknowledged
POL-10	Resolved Oracle Requests Can Still Have Their Configuration Mutated	• Informational ⓘ	Fixed
POL-11	Pause Mapping Rekeyed From <code>ConditionId</code> To <code>EventId</code> Strands Existing Subcondition Pauses	• Informational ⓘ	Acknowledged

Assessment Breakdown

Quantstamp's objective was to evaluate the repository for security-related issues, code quality, and adherence to specification and best practices.

i

Disclaimer

Only features that are contained within the repositories at the commit hashes specified on the front page of the report are within the scope of the audit and fix review. All features added in future revisions of the code are excluded from consideration in this report.

Possible issues we looked for included (but are not limited to):

- Transaction-ordering dependence
- Timestamp dependence
- Mishandled exceptions and call stack limits
- Unsafe external calls
- Integer overflow / underflow
- Number rounding errors
- Reentrancy and cross-function vulnerabilities
- Denial of service / logical oversights
- Access control
- Centralization of power

- Business logic contradicting the specification
- Code clones, functionality duplication
- Gas usage
- Arbitrary token minting

Methodology

1. Code review that includes the following
 1. Review of the specifications, sources, and instructions provided to Quantstamp to make sure we understand the size, scope, and functionality of the smart contract.
 2. Manual review of code, which is the process of reading source code line-by-line in an attempt to identify potential vulnerabilities.
 3. Comparison to specification, which is the process of checking whether the code does what the specifications, sources, and instructions provided to Quantstamp describe.
2. Testing and automated analysis that includes the following:
 1. Test coverage analysis, which is the process of determining whether the test cases are actually covering the code and how much code is exercised when we run those test cases.
 2. Symbolic execution, which is analyzing a program to determine what inputs cause each part of a program to execute.
3. Best practices review, which is a review of the smart contracts to improve efficiency, effectiveness, clarity, maintainability, security, and control based on the established industry and academic practices, recommendations, and research.
4. Specific, itemized, and actionable recommendations to help you take steps to secure your smart contracts.

Scope

Files Included

Repo: `https://github.com/Polymarket/polymarket-v2`

Included Paths: `src/`

Extensions: `sol`

Operational Considerations

Deployment And Chain Configuration

- **Each module deployment must use the correct `RESOLUTION_CHAIN` for its chain:** `OracleModule.onlyResolver()` checks the `resolutionChain` encoded in the condition ID against the module's immutable `RESOLUTION_CHAIN`; it does not check `block.chainid`. If a spoke module is deployed with the hub chain's `RESOLUTION_CHAIN`, resolver-role callers on the spoke can locally resolve hub-resolution condition IDs, including migrated condition IDs that the spoke cannot identify as migrated. Operations should ensure every Binary and NegRisk module deployment uses the intended chain-specific `RESOLUTION_CHAIN`.

Oracle Request Lifecycle

- **OracleAggregator request setup and config changes are high-trust operations:** `OracleAggregator.initializeRequest()` relies on the operator to supply the correct market type, target contract, reporter and disputer modules, thresholds, arbitrator, finalizer, and liveness window. The contract checks that the request is structurally valid. It does not check that the selected modules can actually report, dispute, arbitrate, or finalize the market. Operators should check the request state before changing config. Before swapping an arbitrator, operators should confirm no request under the event is in `ResolutionStatus.ArbitrationRequested`; otherwise the outgoing arbitrator loses authority while the incoming arbitrator has no in-flight arbitration context.
- **Rule changes after votes or UMA settlement need explicit reconciliation:** `OracleAggregator.updateRequestRules()` can update rules before a request is marked `Resolved`, but existing reporter votes are not invalidated. For UMA-backed requests, the aggregator-side rule update can also succeed after UMA has already resolved because `OOReporterModule` treats `RequestAlreadyResolved` and `RequestNotRegistered` as non-reverting and emits `OOReporterRulesForwardFailed`. Operations should avoid substantive rule changes after votes exist or UMA has settled. If a rule change is still needed, operators should reconcile the old votes and UMA state, monitor `OOReporterRulesForwardFailed`, and use manual/admin resolution when needed.
- `OOReporterModule` **can finalize only the result already proposed in** `OracleAggregator`: `OOReporterModule.finalize()` passes the settled UMA result into `OracleAggregator.finalize()`. The aggregator then checks that the result hashes to the stored `proposedResultHash`. If another reporter result reached the threshold, or the aggregator proposal no longer matches UMA's settled result, finalization through `OOReporterModule` reverts. Operations should route that case through the configured arbitrator/admin path or change the finalizer before resolution.

Reporter Modules

- **UMA `OOReporter` request rules are part of the request namespace:** `OOReporterModule` passes UMA a Polymarket `requestId`, but UMA also computes a reporter key from `priceIdentifier` and `requestRules`. Only one request can use the same identifier/rules pair in the same `OOReporter` contract. If another enabled requester registers that pair first, Polymarket's later `OOReporterModule` registration reverts even when the Polymarket `requestId` is new. Operations should keep UMA requester access narrow and make request rules or ancillary data unique per market before registration.
- **Chainlink reports depend on final `DataStore` candles:** `ChainlinkReporterModule.report()` reads the configured start and end timestamp prices from `DataStore` and does not check `block.timestamp >= endTimestamp`. Operations should ensure the authorized `priceSource` writes only final candle values for those timestamp keys.
- **EOA reporter compromise risk depends on reporter and finalizer configuration:** `EOAReporterModule` stores `authorizedReporters` only during `initializeReporterModule()`, so key rotation requires registering a replacement module and removing the old one. If the EOA module is configured with `reporterThreshold = 1`, a compromised reporter can cast the module's

vote alone. A zero `finalizer` and zero `livenessWindow` can make that vote finalizable in the same block. Operations should keep the finalizer controlled, use a nonzero liveness window, and pause the affected markets during reporter rotation.

Migration And NegRisk

- **Legacy NegRisk events should be frozen before migration preparation:** `prepareMigrationEvent()` reads the legacy question count once and encodes that count into the V2 `eventId`. If the legacy event later gains another question, the original V2 event keeps the old arity and synthetic Other represents a broader bucket than the updated legacy event. Operations should treat the legacy NegRisk question set as frozen before calling `prepareMigrationEvent()` and should not prepare the same legacy event again under a different arity.
- **Atomic NegRisk needs a documented recovery path if synthetic Other is ever correct:** `ATOMIC_NEGRISK` should only be used for exhaustive events where one real condition must win. Markets where all real conditions can be NO should use `INCREMENTAL_NEGRISK`, or the fallback outcome should be modeled as a real condition instead of relying on synthetic Other. If an atomic market is misconfigured and synthetic Other is the correct outcome, `OracleAggregator` cannot resolve it directly because normal finalization and admin/arbitrator override reject `winnerIndex >= eventId.arity()`. Operations should document the manual recovery path before deployment.

Combinatorial Operations

- **Combinatorial collateral return depends on balanced off-chain batches:** `Router.combinatorialCollateralReturn()` transfers collateral and positions into `CombinatorialModule` before executing the supplied operations. The router does not verify that the operation list consumes every transferred asset. Operations should generate `_transfers` and `_operations` from the same leg computation and treat any residual module balance as unsafe, because later permissionless module calls can consume stranded collateral or positions.

Exchange Order Matching

- **Operators are trusted to execute market buy orders at fair prices:** A market buy is represented by an order with `takerAmount == 0`. This removes the signed minimum-receive or limit-price constraint. The signed `makerAmount` still caps the buyer's total collateral expenditure, and settlement accounting ensures that the operator-supplied amounts are internally consistent, but the contract does not prevent the operator from purchasing a small number of shares at an unfavorable price. Operators must therefore enforce fair execution, slippage controls, and routing policies off-chain.
- **Existing partially filled V3 BUY orders may not carry their full unspent collateral budget after taker BUY order fill behaviour change:** Previously, BUY order `remaining` was reduced based on the shares received, converted at the order's limit price. The new behaviour changes BUY order `remaining` to track collateral spent. The upgrade reuses the same stored `remaining` value, so old partially filled BUY orders can end up with less fillable budget on-chain than the buyer actually left unspent. Currently this potential edge case only applies to orders involving combinatorial positions, as other market types should not have outstanding orders on the exchange. The operator should account for this mixed accounting state when continuing those orders.
- **Dust order signatures require trusted operator handling:** Price-improved BUY fills can leave economically unfillable `remaining` amounts, while order signatures do not expire and cannot be invalidated on-chain. Operators must therefore reliably identify and discard these residual orders to prevent stale signatures from remaining indefinitely eligible for execution.

Key Actors And Their Capabilities

The following authorized roles exist across the various contracts in the codebase:

PositionManager

Owner

- The owner is trusted to handle upgrading the `PositionManager` contract to a new implementation.
- The owner is trusted to grant the `admin` as necessary.

Admin

- The admin is trusted to add and remove valid modules within the system.
- The admin is trusted to add and remove the "cross module authority" of a module within the system.
- The admin is trusted to revoke the `admin` role as necessary.
- The admin has power to grant and revoke the `operator` and `creator` roles as necessary, however they are currently unused within the `PositionManager` contract.

BinaryModule

Owner

- The owner is trusted to handle upgrading the `BinaryModule` contract to a new implementation.
- The owner is trusted to grant the `admin` as necessary.

Creator

- The creator is trusted to handle migrating legacy CTF conditions to V2.

Operator

- The operator can migrate positions on behalf of users if they have approved the `BinaryModule` as spender of their legacy condition tokens.

Bridge

- The bridge is responsible for calling the `mintFromBridge()` and `burnFromBridge()` functions.

Resolver

- Report results for conditions via `reportResult()`.

NegRiskModule

Owner

- The owner is trusted to handle upgrading the `NegRiskModule` contract to a new implementation.
- The owner is trusted to grant the `admin` role as necessary.

Creator

- The creator is trusted to handle migrating legacy event IDs to V2.

Operator

- The operator can migrate positions on behalf of users if they have approved the `NegRiskModule` as spender of their legacy condition tokens.

Bridge

- The bridge is responsible for calling the `mintFromBridge()` and `burnFromBridge()` functions.

Resolver

- Report results for conditions via `reportResult()`.

Exchange

Owner

- The owner is trusted to handle upgrading the `Exchange` contract to a new implementation.
- The owner is trusted to grant and revoke the `admin` role as necessary.

Admin

- The admin is trusted to grant and revoke the `operator` role as necessary.
- The admin is trusted to pause and unpause trading whenever necessary.
- The admin is trusted to set the number of blocks a user must wait before pausing their own trade eligibility via `setUserPauseBlockInterval()`.

Operator

- The operator is responsible for matching user trade orders via `matchOrders()`.
- The operator can pre-approve a user's order by validating the provided signature via `preapproveOrder()`.
- The operator can also invalidate pre-approved orders.
- The operator can renounce the `operator` role via `renounceOperatorRole()`.

OracleAggregator

Owner

- The owner is trusted to handle upgrading the `OracleAggregator` contract to a new implementation.
- The owner is trusted to grant the `admin` as necessary.

Admin

- The admin is responsible for pausing and unpausing the `OracleAggregator` contract as necessary.
- The admin is trusted to revoke the `admin` role as necessary.
- The admin has power to grant and revoke the `operator` and `rule manager` roles as necessary.

Operator

- The operator is responsible for creating price requests for a given event via the `initializeRequest()` function.

Rule Manager

- The rule manager is responsible for writing product specifications through `MarketDataRegistry.setProductSpecification()`.

OOReporterModule

Owner

- The owner is trusted to handle upgrading the `OOReporterModule` contract to a new implementation.
- The owner is trusted to grant the `admin` as necessary.

Admin

- The admin is trusted to set the `OracleAggregator` instance the module is linked to.
- The admin is trusted to grant and revoke the `operator` role as necessary.

Operator

- The operator can register UMA `OOReporter` requests through `createRequest()`.

ChainlinkReporterModule

Admin

- The admin is trusted to pause and unpause the `ChainlinkReporterModule` contract as necessary.
- The admin is responsible for setting the correct `priceSource`.
- The admin is responsible for setting the correct `feedId` for a given asset pair.

Bridge

The `owner`, `admin` and `bridge` roles within the bridge logic are expected to be implemented by the concrete transporter contract (such as LZ or CCIP), rather than being implemented within the `BridgeBase` contract.

Owner

- The owner is responsible for setting which modules are supported on a given destination chain via the `setModuleSupported()` function.

Admin

- The admin is responsible for pausing and unpausing the ability to send and receive tokens across the bridge.

AutoRedeemer

Owner

- Add an admin who can manage operators.

Admin

- Add and remove operators as needed.

Operator

- Redeem resolved positions on behalf of users.

Findings

POL-1

Synthetic Other Derivation Does Not Check `resultsSum` Against Pre-Upgrade Partial Results

• Low ⓘ

Acknowledged

Update

The client acknowledged the issue and explained that no market has ever resolved to a partial result.

No neg-risk condition has ever resolved to a partial result, so no pre-upgrade event can hit this. Ack, no code change.

File(s) affected: `src/modules/NegRiskModule.sol`

Description: The way the synthetic Other condition is resolved changes with this upgrade:

1. Before, `_finalizeNegriskResolution()` stored Other as the exact remainder of the event, `result[0] = RESULT_DENOMINATOR - resultsSum_` and `result[1] = resultsSum_`, once the last real condition resolved. Any value that the real conditions had already paid out was subtracted, so the event paid out 1000000 in total no matter what the individual results were.
2. Now, Other is no longer stored. `NegRiskModule.getResult()` derives it when it is read, and returns full YES whenever `conditionIndex_ == conditionCount_ && conditionsResolved[eventId_] == conditionCount_`, without reading `resultsSum[eventId_]` at all.

The new form is only correct if every real condition resolved fully YES or fully NO, and the matching check on the write side, in `_finalizeNegriskResolution()`, does test `resultsSum[eventId_] == 0` before treating Other as derivable. The new `_storeResult()` override limits every condition to fully YES or fully NO, so events created after the upgrade cannot reach this case. `NegRiskModule` is upgraded in place, though, and the deployed version accepts any split that sums to 1000000, so the storage left behind does not have to follow the new rule. If an event carried a `resultsSum` that is neither 0 nor 1000000 at the time of the upgrade, Other would resolve to YES on top of a payout that was already made, which breaks the solvency invariant across the event's conditions and lets the event's positions be redeemed for more collateral than was deposited into it. Since `redeem()` mints the payout rather than paying it out of a balance, the surplus is newly created collateral that no USDC backs.

The team confirmed that no neg-risk condition has ever resolved to a partial result, so no such event exists today and the upgrade does not trigger this case as it stands. The deployed version still allows such a result to be stored until the upgrade lands.

Exploit Scenario:

Assuming an event with a partial result existed at upgrade time:

1. A three-condition incremental neg-risk event resolves condition A with `result = [300000, 700000]`, leaving `resultsSum[eventId_] == 300000` and `conditionsResolved[eventId_] == 1`.
2. The `NegRiskModule` proxy is upgraded, and the slots holding `result`, `resultsSum`, and `conditionsResolved` are unchanged.
3. Conditions B and C resolve NO, raising `conditionsResolved[eventId_]` to three.
4. `getResult()` on the synthetic Other returns `[1000000, 0]`, because the branch tests only `conditionsResolved[eventId_] == conditionCount_`.
5. An attacker pre-transfers 1 collateral to the module and calls `horizontalSplit()`, which mints a YES position for A, B, C and Other.
6. The attacker redeems all four YES positions: 0.3 from A, 0 from B and C, and 1.0 from Other. `redeem()` mints the payout, so 1.3 collateral is created against the 1 that was burned.
7. Neither `horizontalSplit()` nor `redeem()` writes `result`, `resultsSum` or `conditionsResolved`, so the attacker repeats the round with the full balance and grows it by a factor of 1.3 each time.

Recommendation: Derive the synthetic Other as the remainder rather than as unconditional YES, so the payout stays solvent for any pre-upgrade state:

```
if (conditionIndex_ == conditionCount_ && conditionsResolved[eventId_] == conditionCount_) {
    result_ = new uint256[](2);
    result_[0] = RESULT_DENOMINATOR - resultsSum[eventId_];
    result_[1] = resultsSum[eventId_];
    return result_;
}
```

This matches the current behavior when `resultsSum[eventId_]` is 0, and `getResult()` is a view function, so it is not bound by the fully YES or fully NO rule enforced in `_storeResult()`. As an alternative, run an off-chain script shortly before the upgrade that reads `resultsSum(eventId)` for every neg-risk event on the deployed module and confirms that no event holds a value other than 0 or 1000000, and only upgrade once that check passes.

POL-2

Results Can Be Reported To A Mismatched Target Module Without Any Module Id Check

Low Fixed

Update

The client fixed the issue in 8d23f5a9eb608440c5bb7e497e74c27c60dd72f2 by adding `_validateRequestRoute()`, which checks that `market.moduleId == eventId.moduleId()` and that `targetContract` matches the expected resolution contract for that module. The fix also adds validation in `BinaryModule.reportResult()` and `NegRiskModule.reportResult()` that the reported condition belongs to the module. This is a defensive measure against operator error; the extra SLOAD is acceptable because this path is not part of the trading or matching hot path.

File(s) affected: `src/oracle/OracleAggregator.sol`, `src/modules/BinaryModule.sol`, `src/modules/NegRiskModule.sol`, `src/modules/abstract/BaseModule.sol`

Description: `OracleAggregator.initializeRequest()` never compares the module ID encoded in the supplied event ID against the `targetContract` the request will report to. `_validateEventId()` checks only the arity field. The two modules then guard the mismatch unevenly: `NegRiskModule.reportResult()` rejects a foreign event ID because `_validateEventId()` calls `conditionCount()`, which returns 0 when the encoded module ID is not `ModuleIds.NEGRISK`, whereas `BinaryModule.reportResult()` performs no equivalent check and `BaseModule._storeResult()` performs no module-ownership check on the condition. A request created for a neg-risk event ID but pointed at `BinaryModule` therefore finalizes successfully: the binary module stores the payout under a condition ID it does not own, and the aggregator marks the request `Resolved`. The module that actually holds the positions never receives a result, and since `requestConfigs` has no setter for `targetContract` and `initializeRequest()` rejects repeated registrations, the event ID can never be resolved through the aggregator again. The intended oracle resolution path is therefore blocked, but the positions are not stranded: an admin holding the resolver role can report the correct result to the owning module directly. `_storeResult()` also emits `ConditionResolved` from the wrong module, so an indexer keyed on the event rather than on the emitting address records the condition as resolved when it is not.

Reaching this state requires an operator error, as `initializeRequest()` is gated by `onlyOperator`.

Recommendation: Reject condition IDs whose encoded module ID is not the module's own at the start of `BinaryModule.reportResult()`, so a misdirected result cannot be silently absorbed regardless of caller, and add the same explicit check to `NegRiskModule.reportResult()` so the guarantee does not rest on a side effect of `conditionCount()`. Additionally, cross-check the event ID against the target in `initializeRequest()` e.g. `( _params.eventId.moduleId() == IPositionManagerModule(_params.targetContract).moduleId() )` so the misconfiguration surfaces at registration rather than at finalization.

POL-3

Bridged Synthetic Other Can Push conditionsResolved Past Event Arity

Informational Fixed

Update

The client fixed in c0df06571c76c94b7a5d998849209200898a6e43 and 973a9668aef5ebe21af445b5deb3118bb0d403a1. The bridge now restricts result bridging to the resolution chain and rejects inbound result messages unless they come from the configured resolution-chain source. The NegRisk accounting fix also excludes directly reported synthetic Other from `conditionsResolved`, so synthetic Other no longer pushes the real-condition counter past event arity.

The fix also preserves compatibility with v1.0.0 state where synthetic Other may already have been stored and counted.

File(s) affected: `src/modules/NegRiskModule.sol`

Description: `NegRiskModule` lets bridge callers report the synthetic Other condition directly. In the affected implementation, that path stores the synthetic result and increments `conditionsResolved`, the same counter used for real-condition reports.

If synthetic Other is bridged before every real condition has been stored, later bridge reports for the real conditions can push `conditionsResolved[eventId]` above `conditionCount(eventId)`. This can happen on a spoke, and it can also happen on the hub if a spoke pushes a synthetic Other result back to the hub.

On bridge chains, `conditionsResolved` no longer means number of real conditions resolved. It also weakens the module's own invariants because `conditionsResolved <= conditionCount` is not always true after bridge-sourced synthetic Other reports.

Recommendation: Consider disallowing direct bridging of synthetic Other. For real-winner outcomes, bridging the winning real condition is enough because each destination chain can derive the remaining real conditions and synthetic Other as NO. Disallowing direct synthetic Other bridging only changes the all-NO outcome path, where synthetic Other otherwise derives after the destination has received every real NO result.

POL-4

Migration Registration Can Alias Existing Native V2 Result State

Informational Fixed

✓ Update

The client fixed the reported stale-result path in `4ee20a3a258fe1ff15926a83b898dd5d2866a3b0`. Migration registration now requires the derived V2 result slot to be empty before binding a legacy market to that V2 ID. Binary checks the derived condition. NegRisk checks synthetic Other and each real condition. This prevents a reported native V2 result from becoming the payout source for a migrated market.

The client also fixed a related legacy-alias path in `c69833a7332c43dac0a0bd129aca696798e75f7e`. `migratePositions()` now checks the registered legacy condition identity, so a second legacy condition ID that aliases to an already registered migration condition cannot reuse that registration.

File(s) affected: `src/modules/migration/BinaryMigrationMixin.sol`, `src/modules/migration/NegRiskMigrationMixin.sol`, `src/modules/migration/BaseMigrationMixin.sol`, `src/modules/BinaryModule.sol`, `src/modules/NegRiskModule.sol`, `src/modules/abstract/BaseModule.sol`

Description: Migration registration maps legacy markets into V2 IDs. Binary migration registers one legacy CTF condition under one derived V2 `conditionId`. NegRisk migration registers one legacy event under one derived V2 `eventId`, then maps each legacy CTF subcondition to `eventId.computeConditionId(i)`.

The current migration flow intentionally allows users to migrate after a legacy market has resolved, so late users are not stranded. In `v1.0.0`, `migratePositions()` reverted when the mapped V2 condition already had a stored result. The current flow removed that block. Binary registration checks the legacy condition shape and duplicate migration registration. NegRisk registration checks the legacy event shape and duplicate event registration. Neither registration path rejects a derived V2 condition or event that already has native result state.

The alias case is extremely unlikely, but possible because the structured V2 ID stores only the 128-bit `baseHash`. Native and migrated IDs share the same module namespace within each module: binary with binary, and NegRisk with NegRisk for the same event arity. If a colliding legacy condition or event is prepared after the native V2 result already exists, registration succeeds normally.

After registration, `migratePositions()` mints positions against the shared ID. If the legacy condition has resolved, `_redeemIfResolved()` sees the existing `result[conditionId]` and skips storing the legacy payout. The migrated position then settles through `BaseModule.redeem()` using the unrelated native result.

The conditions for unbacked `pUSD` are very specific and require operational mismanagement. The native and legacy IDs must collide, the native V2 result must exist before legacy preparation, the native and legacy outcomes must be opposite for the affected condition, and the creator must register the colliding legacy condition or event. A public attacker cannot freely choose this collision because migration preparation is permissioned by `onlyCreator`. The likelihood is extremely low, but the impact can be critical if migrated positions settle against an unrelated native result.

Recommendation: As a defense-in-depth measure, do not allow migration registration for a V2 condition or event that already has result state. At registration time, `prepareMigrationCondition()` should reject `result[conditionId].length != 0`, and `prepareMigrationEvent()` should reject if any derived real condition already has stored result state or if the derived event already has aggregate result state. This check covers the narrow ordering where a colliding native V2 market already exists and has been resolved by the oracle before migration preparation starts. It avoids registering a migration route whose resolution path would later skip result storage because `_redeemIfResolved()` skips `_finalizeMigrationResolution()` when the shared result slot is already populated. `migratePositions()` may still be intended after resolution for properly registered migration conditions, but registration itself should not bind a legacy market to already-resolved native V2 state.

POL-5

Divergent Reporter Votes Can Exhaust the Reporter Set without Triggering Arbitration

• Informational ⓘ

Acknowledged

i Update

The client acknowledged the issue and explained that admin/arbitrator can resolve deadlocks.

Rejected the auto-arbitration recommendation. Admin/arbitrator can break deadlocks via `resolveResult`; we don't expect decentralized reporter sets in the current operational model.

File(s) affected: `src/oracle/OracleAggregator.sol`

Description: `OracleAggregator` counts votes independently for each result hash, and each registered reporter module may vote only once per request. A proposal is created only when one result receives `reporterThreshold` matching votes.

Arbitration is triggered only when:

- A different result reaches `reporterThreshold` after a proposal has already been created; or
- The configured dispute threshold is reached after a proposal has been created.

Consequently, divergent votes can exhaust all eligible reporters before any result reaches the reporter threshold.

For example, consider a request configured with two reporter modules and a reporter threshold of two:

1. Reporter A submits result X.
2. Reporter B submits result Y.
3. Each result receives one vote, so neither reaches the threshold.
4. Both reporters have consumed their only vote for the request.

The request remains in the `Active` state without a `proposedResultHash`. No `ArbitrationTriggered` event is emitted, and the arbitrator's `onArbitrationTriggered` hook is not called.

The ordinary resolution paths are then unavailable:

- Additional reports from either reporter revert because the reporter has already voted.
- `disputeResult` reverts with `NoProposalToChallenge` because no proposal exists.
- `finalize` reverts with `ThresholdNotMet`.

The request will require an admin or configured arbitrator to break the deadlock via the `resolveResult()` function. However, off-chain systems must reconstruct the stalled state from individual `ResultReported` and reporter-registration events to know that a request is deadlocked.

Recommendation: Consider emitting the `ArbitrationRequested` event if all registered reporter modules have voted without the threshold being met. Alternatively, ensure the off-chain systems are robust enough to notify the `OracleAggregator`'s admin or arbitrators when a request is deadlocked, so they can intervene on time.

POL-6

Excessive Liveness Windows Can Effectively Disable Normal Finalization

• Informational ⓘ

Fixed



Update

The client fixed the issue in `5b62de8a161325b194de2b75e2584cdaa5870d63` by adding a maximum liveness window.

File(s) affected: `src/oracle/OracleAggregator.sol`

Description: `OracleAggregator` accepts any `uint32` value as its `livenessWindow`. When a result reaches the reporter threshold, the contract calculates the end of the dispute period using the configured value:

```
state.disputeWindowEnd = uint40(block.timestamp + cfg.livenessWindow);
```

The maximum `uint32` value is `4,294,967,295` seconds, or approximately 136 years. A request configured with such a value would remain inside its dispute window for longer than the expected lifetime of the protocol, effectively preventing the normal `finalize()` path from being used. This value can be supplied during request initialization or subsequently configured through `setLivenessWindow()`.

Furthermore, changing the configured `livenessWindow` does not update an existing proposal's `disputeWindowEnd`. If an excessive window has already been applied to a proposal, correcting the request configuration will affect only future proposals. The existing request must instead be resolved by the configured arbitrator or an admin through `resolveResult`.

Recommendation: Consider defining and enforcing a reasonable protocol-level maximum liveness window in both `initializeRequest()` and `setLivenessWindow()`.

For example:

```
uint32 internal constant MAX_LIVENESS_WINDOW = 30 days;

require(_params.livenessWindow <= MAX_LIVENESS_WINDOW, InvalidConfig());
```

POL-7

Empty `PositionManager` Batch Calls Can Emit Unauthorized No-Op Transfer Events

• Informational ⓘ

Acknowledged



Update

The client acknowledged the issue and noted that the empty batch calls do not change balances.

No balance changes, event-only. Ack.

File(s) affected: `src/positionManager/PositionManager.sol`

Description: `PositionManager.batchMint()` and `PositionManager.batchBurn()` authorize callers through `onlyModuleByPositionIds`, which validates that each supplied position ID belongs to the calling module. When `_positionIds` is empty, the authorization loop has no element to check.

As a result, any address can call `batchMint()` or `batchBurn()` with empty position and amount arrays. `batchMint` still requires a nonzero recipient, but the module-authorization loop has no position ID to validate. No balances change, but an empty ERC1155 `TransferBatch` event is still emitted.

Recommendation: As a defense-in-depth check, consider rejecting empty arrays in `batchMint()` and `batchBurn()`.

POL-8

ChainlinkReporterModule Accepts Non-Positive Asset Prices

• Informational ⓘ Acknowledged

✓ Update

The client explained that non-positive prices are intended and required for supported market types such as perps, where values can be negative. Based on this clarification, this issue is considered a false positive.

Intended and valid behavior. Ack. Examples include Perps where it can go negative

File(s) affected: `src/oracle/modules/reporters/ChainlinkReporterModule.sol`

Description: `ChainlinkReporterModule.report()` reads the start and end candle prices as signed `int192` values and settles YES when `endPrice >= startPrice`. The signed comparison is correct, including for negative numbers. The issue is that the module also accepts zero and negative prices as valid inputs for asset-price markets.

For directional asset markets such as BTC/USD, a zero or negative candle price could be bad source data, a wrong feed, or a DataStore writer mistake. In that scenario, the safer behavior is likely to revert and resolve the market manually through the fallback path. If such a value is written today, the module still settles from it. For example, `startPrice = -10` and `endPrice = -5` settles YES because `-5 >= -10`, even though both prices are not expected for a normal asset-price candle.

Recommendation: Reject `startPrice <= 0` or `endPrice <= 0` in `report()` when the module is used for asset-price candles. If the DataStore contains a zero or negative asset price, the Chainlink reporter should revert and the market should be resolved manually through the configured fallback path.

POL-9

Rule Updates Can Diverge After OOReporter Has Resolved

• Informational ⓘ Acknowledged

i Update

The client acknowledged the issue and explained that clarification updates should remain pushable after UMA resolves because other reporters may still be active.

Intended → clarifications should remain pushable after UMA resolves since other reporters may still be active. Not blocking UMA.

File(s) affected: `src/oracle/OracleAggregator.sol`, `src/oracle/modules/OOReporterModule.sol`

Description: `OracleAggregator.updateRequestRules()` is the Polymarket-side rule update path. It records the new rule in `MarketDataRegistry` before forwarding the update to reporter modules. For UMA-backed requests, `OOReporterModule.updateRules()` forwards the update to UMA `OOReporter.updateRequestRules()`.

UMA `OOReporter` rejects rule updates after the UMA request has resolved. The adapter treats `RequestAlreadyResolved` as non-reverting, emits `OOReporterRulesForwardFailed`, and does not revert the aggregator call. Because the aggregator writes its local rule before forwarding, a rule update can remain in `OracleAggregator` even though UMA/MOOV2 did not accept the same update.

This can make the resolution record misleading. The Polymarket-side registry can present the new rule as the current canonical rule while UMA's settled result was produced under the previous rule set, so readers and downstream systems may assume the reported UMA result applied rules that UMA never accepted.

This can only happen during the intermediate period where UMA has already resolved the request, but the corresponding `OracleAggregator` request has not yet been marked `Resolved`.

Recommendation: If this behavior is intended, document `OOReporterRulesForwardFailed` as the signal that the rule update was recorded only in `OracleAggregator` and was not mirrored to UMA/MOOV2. If exact rule-history mirroring is required, do not treat `RequestAlreadyResolved` as non-reverting.

POL-10

Resolved Oracle Requests Can Still Have Their Configuration Mutated

• Informational ⓘ Fixed

✓ Update

The client fixed the issue in `e02a6c2a53101862e7b988f3d9d76ad7ad5a6ddd` by adding `_requireMutableRequestConfig()`, which gates request-configuration setters so they can only update unresolved requests.

File(s) affected: `src/oracle/OracleAggregator.sol`

Description: `OracleAggregator` blocks rule updates after a request is resolved, but other request configuration fields remain mutable. After a request reaches `Resolved`, an operator or admin can still add/remove reporter modules, add/remove disputer modules, swap the arbitrator module, change the finalizer, and change the liveness window. These paths only check that the event has a request config and that the new values pass local validation. They do not check `resolutionStates[requestId].status`.

Recommendation: Reject request configuration mutations once the request is resolved, including reporter/disputer module changes, arbitrator changes, finalizer changes, and liveness-window changes. If post-resolution edits are intentionally allowed, document that the current request config is not a historical snapshot of the config that governed resolution.

POL-11

Pause Mapping Rekeyed From `ConditionId` To `EventId` Strands Existing Subcondition Pauses

• Informational ⓘ Acknowledged

i Update

The client acknowledged the issue and explained that `OracleAggregator` has not been deployed and `pauseResolution` has never been used on a subcondition.

OracleAggregator not deployed and pauseResolution never used on a subcondition.

File(s) affected: `src/modules/abstract/OracleModule.sol`

Description: From the last deployed state to the audited state, `OracleModule.resolutionPausedAt` changes from `mapping(ConditionId => uint256)` to `mapping(EventId => uint256)`, and `pauseResolution()` / `unpauseResolution()` now take an `EventId`, while `onlyResolver` reads `resolutionPausedAt[_id.eventId()]`.

The variable stays at slot 1, so the storage layout does not shift, but the key width does. `ConditionId` is a `bytes31` — an `EventId` followed by the 2-byte `conditionIndex` — and `EventId` is a `bytes29`, and both are right-padded to 32 bytes before being hashed with the slot. An entry written before the upgrade is therefore still reachable afterwards only when its `conditionIndex` is 0.

Two cases follow for pauses that are active at upgrade time. A pause on subcondition 0 is re-read as the event-level flag and widens: every sibling under the event becomes paused, including conditions the admin never froze. A pause on a subcondition with `conditionIndex >= 1` lands on a slot nothing reads or writes any more — `onlyResolver` no longer consults it and `unpauseResolution` cannot clear it — so a condition an admin deliberately froze silently becomes resolvable again. The second case only bites if subcondition 0 was left unpaused, because if it was paused too, the event-level read covers the whole event and the leftover entries are dead storage. The team confirmed that `pauseResolution` has never been used on an individual subcondition, so no such entry exists today and neither case is triggered by the upgrade as it stands.

Recommendation: Re-confirm immediately before upgrading that no `resolutionPausedAt` entry is set on a subcondition with a non-zero `conditionIndex`. If the event-level flag is wanted long term, add it in a new storage slot and leave the `ConditionId`-keyed mapping in place rather than repurposing slot 1, so existing pauses stay readable and clearable.

Auditor Suggestions

S1 Empty Combinatorial Return Leaves Pre-Transferred Assets Claimable

Fixed

✓ Update

The client fixed the issue in `f76e4997d16bbf0e222fac7457a1c9ca41a6741e` by checking that the operation list is non-empty before proceeding.

File(s) affected: `src/routers/Router.sol`

Description: In `Router.combinatorialCollateralReturn()`, the `_operations` array is not validated to be non-empty. When it is empty, both pre-transfer blocks still execute, the operation loop does not run, and `CombinatorialCollateralReturned` is emitted as though the call succeeded, so the caller donates the entire `_transfers.collateralAmount` and every transferred position to the `CombinatorialModule`, where any subsequent caller can claim them through `split()` or `merge()`. Consider reverting when `_operations.length` is zero.

Recommendation: Revert in `combinatorialCollateralReturn()` when `_operations.length == 0`, before transferring any collateral or positions to the `CombinatorialModule`.

S2 Miscellaneous Improvements

Acknowledged

Update

The client acknowledged the suggestion and provided the following explanation:

(1) redundant `i != 0` guard in `BaseMigrationMixin._migratePositions`; (2) README lists `arbitratorModule: address(0)` as valid but `initializeRequest` rejects it. All Ack

File(s) affected: `src/modules/migration/BaseMigrationMixin.sol`, `src/oracle/OracleAggregator.sol`, `src/oracle/modules/README.md`

Description: The following changes would improve the overall quality of the codebase:

1. In `BaseMigrationMixin._migratePositions()`, the sortedness check `if (i != 0 && uint256(legacyConditionId_) < uint256(previousLegacyConditionId))` carries a redundant `i != 0` guard. On the first iteration `previousLegacyConditionId` is still `bytes32(0)`, so the comparison is `uint256(legacyConditionId_) < 0`, which is false for every value including zero, and the guard can be removed without changing behavior.
2. `src/oracle/modules/README.md` lists `arbitratorModule: address(0)` as a valid configuration, but `initializeRequest()` and `setArbitratorModule()` both reject a zero arbitrator with `InvalidConfig()`. An operator following the documentation therefore hits a revert, and the zero-address branch in `_triggerArbitration()` is unreachable. Consider stating in the documentation that a non-zero arbitrator is required, and removing the unreachable branch.

Recommendation: Consider applying the code improvements as described.

Definitions

- **High severity** – High-severity issues usually put a large number of users' sensitive information at risk, or are reasonably likely to lead to catastrophic impact for client's reputation or serious financial implications for client and users.
- **Medium severity** – Medium-severity issues tend to put a subset of users' sensitive information at risk, would be detrimental for the client's reputation if exploited, or are reasonably likely to lead to moderate financial impact.
- **Low severity** – The risk is relatively small and could not be exploited on a recurring basis, or is a risk that the client has indicated is low impact in view of the client's business circumstances.
- **Informational** – The issue does not pose an immediate risk, but is relevant to security best practices or Defence in Depth.
- **Undetermined** – The impact of the issue is uncertain.
- **Fixed** – Adjusted program implementation, requirements or constraints to eliminate the risk.
- **Mitigated** – Implemented actions to minimize the impact or likelihood of the risk.
- **Acknowledged** – The issue remains in the code but is a result of an intentional business or design decision. As such, it is supposed to be addressed outside the programmatic means, such as: 1) comments, documentation, README, FAQ; 2) business processes; 3) analyses showing that the issue shall have no negative consequences in practice (e.g., gas analysis, deployment settings).

Appendix

File Signatures

The following are the SHA-256 hashes of the reviewed files. A file with a different SHA-256 hash has been modified, intentionally or otherwise, after the security review. You are cautioned that a different SHA-256 hash could be (but is not necessarily) an indication of a changed condition or potential vulnerability that was not within the scope of the review.

Files

Repo: `https://github.com/Polymarket/polymarket-v2`

- `ffc...724 ./src/abstract/ERC1155TokenReceiver.sol`

- 145...120 ./src/abstract/test/ERC1155TokenReceiver.t.sol
- 8f7...c96 ./src/auth/InitializableRoles.sol
- 7c7...9af ./src/auth/Roles.sol
- f6b...ae9 ./src/bridge/abstract/BridgeBase.sol
- 833...36c ./src/bridge/interfaces/IBridge.sol
- eb9...571 ./src/collateral/CollateralOfframp.sol
- a37...a47 ./src/collateral/CollateralOnramp.sol
- 5a6...cc1 ./src/collateral/CollateralToken.sol
- bd2...afb ./src/collateral/PermissionedRamp.sol
- 140...8d6 ./src/collateral/abstract/CollateralErrors.sol
- 9f9...076 ./src/collateral/abstract/Pausable.sol
- 0c7...672 ./src/collateral/interfaces/ICollateralToken.sol
- f52...3a2 ./src/exchange/Exchange.sol
- cf1...20d ./src/exchange/OrderStructs.sol
- 666...934 ./src/libraries/BridgePayloads.sol
- cf8...206 ./src/libraries/CrossChainTypes.sol
- 7d6...efb ./src/libraries/Ids.sol
- 848...91c ./src/libraries/ModuleIds.sol
- c36...012 ./src/modules/BinaryModule.sol
- 75c...5d8 ./src/modules/CombinatorialModule.sol
- 6dc...042 ./src/modules/NegRiskModule.sol
- f2c...a9c ./src/modules/abstract/BaseModule.sol
- 827...30d ./src/modules/abstract/ModuleErrors.sol
- 56e...aba ./src/modules/abstract/OracleModule.sol
- 822...813 ./src/modules/migration/BaseMigrationMixin.sol
- 8d4...b4c ./src/modules/migration/BinaryMigrationMixin.sol
- 512...6d5 ./src/modules/migration/NegRiskMigrationMixin.sol
- 05f...ff7 ./src/oracle/OracleAggregator.sol
- eb0...d63 ./src/oracle/abstract/OracleAggregatorErrors.sol
- 937...7f0 ./src/oracle/abstract/OracleAggregatorEvents.sol
- 5b0...845 ./src/oracle/abstract/OracleModuleBase.sol
- 4a7...367 ./src/oracle/interfaces/IArbitratorModule.sol
- f68...d8d ./src/oracle/interfaces/IBinaryReporter.sol
- fa8...071 ./src/oracle/interfaces/IDisputerModule.sol
- 417...96a ./src/oracle/interfaces/IOracleAggregator.sol
- 068...792 ./src/oracle/interfaces/IReporterModule.sol
- 468...4a0 ./src/oracle/libraries/OptimisticOraclePayoutLib.sol
- d97...89d ./src/oracle/mixins/Auth.sol
- db7...818 ./src/oracle/mixins/MarketDataRegistry.sol
- de9...c31 ./src/oracle/mixins/Pausable.sol
- 7b6...b40 ./src/oracle/modules/OOReporterModule.sol
- 296...066 ./src/oracle/modules/reporters/ChainlinkReporterModule.sol
- 7cb...a31 ./src/oracle/modules/reporters/EOAReporterModule.sol
- db9...b9c ./src/positionManager/IPositionManagerModule.sol
- 365...beb ./src/positionManager/PositionManager.sol
- 125...580 ./src/routers/BridgeRouter.sol
- 0ab...4c4 ./src/routers/CtfRouter.sol
- 058...e76 ./src/routers/Router.sol
- 4fe...a6b ./src/utils/AutoRedeemer.sol

Test Suite Results

Ran 5 tests for src/collateral/test/CollateralOnramp.t.sol:CollateralOnrampTest_admin
[PASS] test_addAdmin() (gas: 48314)
[PASS] test_removeAdmin() (gas: 37749)
[PASS] test_revert_addAdmin_unauthorized() (gas: 20109)
[PASS] test_revert_pause_unauthorized() (gas: 15939)
[PASS] test_revert_removeAdmin_unauthorized() (gas: 20208)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 2.83ms (790.04µs CPU time)

Ran 4 tests for src/collateral/test/CollateralOnramp.t.sol:CollateralOnrampTest_pause
[PASS] test_pause_emitsEvent() (gas: 43276)
[PASS] test_revert_wrapUSDC_paused() (gas: 121001)
[PASS] test_revert_wrapUSDCe_paused() (gas: 120979)
[PASS] test_unpause() (gas: 198724)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 1.79ms (580.29µs CPU time)

Ran 2 tests for src/collateral/test/CollateralOnramp.t.sol:CollateralOnrampTest_wrap
[PASS] test_wrapUSDC() (gas: 167156)
[PASS] test_wrapUSDCe() (gas: 167139)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 1.57ms (395.88µs CPU time)

Ran 4 tests for src/modules/test/BinaryMigration.t.sol:BinaryMigrationTest_softViewAliasReverts
[PASS] test_getResult_unknownCanonicalReturnsEmpty() (gas: 14315)
[PASS] test_hasResult_unknownCanonicalReturnsFalse() (gas: 13387)
[PASS] test_revert_getResult_nonCanonical() (gas: 7024)
[PASS] test_revert_hasResult_nonCanonical() (gas: 7047)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 7.93ms (714.29µs CPU time)

Ran 1 test for src/collateral/dev/CollateralSetup.t.sol:CollateralSetupTest
[PASS] test_setup() (gas: 46404)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 1.45ms (260.13µs CPU time)

Ran 1 test for src/modules/test/BinaryMigration.t.sol:BinaryMigrationTest_cantina10
[PASS] test_migratePositions_doesNotInvokeReceiverHook() (gas: 427299)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 8.53ms (1.01ms CPU time)

Ran 2 tests for src/collateral/test/CollateralToken.t.sol:CollateralTokenTest_burn
[PASS] test_burn() (gas: 56604)
[PASS] test_revert_unauthorized() (gas: 20551)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 1.50ms (216.67µs CPU time)

Ran 2 tests for src/collateral/test/CollateralToken.t.sol:CollateralTokenTest_initialize
[PASS] test_initialize() (gas: 14998)
[PASS] test_revert_alreadyInitialized() (gas: 21032)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 1.45ms (159.88µs CPU time)

Ran 2 tests for src/collateral/test/CollateralToken.t.sol:CollateralTokenTest_mint
[PASS] test_mint() (gas: 71067)
[PASS] test_revert_unauthorized() (gas: 21014)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 1.48ms (204.75µs CPU time)

Ran 1 test for src/collateral/test/CollateralToken.t.sol:CollateralTokenTest_permit2
[PASS] test_permit2NoInfiniteAllowance() (gas: 15863)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 1.42ms (154.13µs CPU time)

Ran 7 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_resolver
[PASS] test_addResolver_thenReport() (gas: 114536)
[PASS] test_pauseResolution() (gas: 47124)
[PASS] test_pauseResolver() (gas: 46184)
[PASS] test_revert_reportResult_resolutionPaused() (gas: 56307)
[PASS] test_revert_reportResult_resolverPaused() (gas: 54112)
[PASS] test_unpauseResolution() (gas: 38119)
[PASS] test_unpauseResolver() (gas: 37563)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 15.50ms (1.97ms CPU time)

Ran 8 tests for src/collateral/test/CollateralToken.t.sol:CollateralTokenTest_roles
[PASS] test_addMinter() (gas: 46564)
[PASS] test_addWrapper() (gas: 46610)
[PASS] test_removeMinter() (gas: 38666)
[PASS] test_removeWrapper() (gas: 38632)
[PASS] test_revert_addMinter_unauthorized() (gas: 20598)

```
[PASS] test_revert_addWrapper_unauthorized() (gas: 20599)
[PASS] test_revert_removeMinter_unauthorized() (gas: 20622)
[PASS] test_revert_removeWrapper_unauthorized() (gas: 20598)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 2.17ms (877.00µs CPU time)
```

```
Ran 3 tests for src/utils/test/AutoRedeemer.t.sol:AutoRedeemerTest_deployment
[PASS] test_deployment() (gas: 67742)
[PASS] test_revert_cannotReinitialize() (gas: 23540)
[PASS] test_revert_implementationInitializerDisabled() (gas: 3549574)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 19.82ms (1.00ms CPU time)
```

```
Ran 8 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_bridge
[PASS] test_burnFromBridge() (gas: 107032)
[PASS] test_burnFromBridge_batch() (gas: 162876)
[PASS] test_mintFromBridge() (gas: 92192)
[PASS] test_mintFromBridge_thenRedeem() (gas: 259631)
[PASS] test_mintFromBridge_zeroAmount() (gas: 63252)
[PASS] test_onERC1155BatchReceived_normalTransfer() (gas: 152324)
[PASS] test_onERC1155Received_normalTransfer() (gas: 105946)
[PASS] test_revert_mintFromBridge_unauthorized() (gas: 45921)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 14.16ms (2.87ms CPU time)
```

```
Ran 3 tests for src/utils/test/AutoRedeemer.t.sol:AutoRedeemerTest_events
[PASS] test_redeemEmitsPayout() (gas: 474257)
[PASS] test_redeemLegacyBinaryEmitsPayout() (gas: 482253)
[PASS] test_redeemLegacyNegRiskEmitsPayout() (gas: 830212)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 10.62ms (5.55ms CPU time)
```

```
Ran 4 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_resolverModifier
[PASS] test_revert_reportResult_resolutionPaused() (gas: 56284)
[PASS] test_revert_reportResult_resolverPaused() (gas: 54156)
[PASS] test_revert_reportResult_wrongResolutionChain() (gas: 23886)
[PASS] test_revert_reportResult_wrongSender() (gas: 23283)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 21.16ms (932.13µs CPU time)
```

```
Ran 7 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_eventOperations
[PASS] test_convertOnEvent() (gas: 589171)
[PASS] test_convertOnEvent_recipientsFollowConditionIndexExcludingEverySource() (gas: 2269836)
[PASS] test_revert_convertOnEvent_invalidRecipientCount() (gas: 217986)
[PASS] test_revert_splitOnEvent_eventAlreadyPresent() (gas: 217962)
[PASS] test_revert_splitOnEvent_invalidRecipientCount() (gas: 197497)
[PASS] test_splitMergeOnEvent() (gas: 777636)
[PASS] test_splitOnEvent_recipientsFollowConditionIndexIncludingOther() (gas: 721314)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 38.12ms (17.74ms CPU time)
```

```
Ran 6 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_direct
[PASS] test_merge_direct() (gas: 380621)
[PASS] test_redeem_direct() (gas: 413979)
[PASS] test_revert_merge_noPreTransfer() (gas: 90347)
[PASS] test_revert_redeem_noPreTransfer() (gas: 160928)
[PASS] test_revert_split_noPreTransfer() (gas: 97794)
[PASS] test_split_direct() (gas: 236432)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 17.05ms (4.80ms CPU time)
```

```
Ran 1 test for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_split
[PASS] test_split() (gas: 293536)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 6.04ms (468.21µs CPU time)
```

```
Ran 4 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_upgrade
[PASS] test_revert_upgrade_incompatibleModuleId() (gas: 6832533)
[PASS] test_revert_upgrade_incompatibleUsdce() (gas: 5696892)
[PASS] test_revert_upgrade_unauthorized() (gas: 5692437)
[PASS] test_upgradeToAndCall_preservesState() (gas: 5781104)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 6.75ms (2.32ms CPU time)
```

```
Ran 12 tests for src/modules/test/BinaryMigration.t.sol:BinaryMigrationTest_migrate
[PASS] test_migrate() (gas: 521038)
[PASS] test_migrate_alreadyResolved_redeemsToVault() (gas: 495392)
[PASS] test_migrate_asymmetricComplementaryBalances_sameCall() (gas: 494356)
[PASS] test_migrate_fromOperator() (gas: 531949)
[PASS] test_migrate_repeatedConditionEntries_clampsToCurrentBalance() (gas: 532415)
[PASS] test_prepareMigrationCondition_emitsMigrationConditionRegistered() (gas: 90409)
```

```
[PASS] test_revert_InvalidFromAddress() (gas: 376991)
[PASS] test_revert_amountsLengthMismatch() (gas: 95592)
[PASS] test_revert_invalidArrayLength() (gas: 95615)
[PASS] test_revert_migrationConditionIdAlias() (gas: 99029)
[PASS] test_revert_migrationNotRegistered() (gas: 205013)
[PASS] test_revert_unsortedMigrationConditions() (gas: 384205)
Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 41.52ms (33.70ms CPU time)
```

```
Ran 6 tests for src/collateral/test/CollateralToken.t.sol:CollateralTokenTest_unwrap
[PASS] test_revert_invalidAsset(address) (runs: 256,  $\mu$ : 28718,  $\sim$ : 28718)
[PASS] test_revert_unauthorized() (gas: 25606)
[PASS] test_revert_unwrap_noPreTransfer() (gas: 108268)
[PASS] test_unwrapUSDC() (gas: 177524)
[PASS] test_unwrapUSDC_pretransfer() (gas: 172908)
[PASS] test_unwrapUSDCe() (gas: 179171)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 35.25ms (33.93ms CPU time)
```

```
Ran 7 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_extract
[PASS] test_extract() (gas: 517134)
[PASS] test_extract_fullDecomposition() (gas: 717329)
[PASS] test_revert_extract_requiresNoSide() (gas: 277524)
[PASS] test_revert_extract_unprepared() (gas: 197326)
[PASS] test_revert_extract_wrongToLength() (gas: 277318)
[PASS] test_revert_indexOutOfRange() (gas: 255884)
[PASS] test_revert_singleCondition() (gas: 233884)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 13.84ms (4.49ms CPU time)
```

```
Ran 4 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_getPayout
[PASS] test_fullPayout() (gas: 114983)
[PASS] test_partialPayout() (gas: 135197)
[PASS] test_revert_conditionNotResolved() (gas: 43306)
[PASS] test_revert_invalidOutcomeIndex() (gas: 110173)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 14.22ms (3.62ms CPU time)
```

```
Ran 2 tests for src/collateral/test/CollateralToken.t.sol:CollateralTokenTest_upgrade
[PASS] test_revert_unauthorized() (gas: 1768699)
[PASS] test_upgradeToAndCall() (gas: 1773033)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 1.94ms (592.13 $\mu$ s CPU time)
```

```
Ran 3 tests for src/collateral/test/CollateralToken.t.sol:CollateralTokenTest_validAsset
[PASS] test_revert_invalidAsset() (gas: 21574)
[PASS] test_validAsset_usdc() (gas: 129912)
[PASS] test_validAsset_usdce() (gas: 129869)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 1.97ms (691.29 $\mu$ s CPU time)
```

```
Ran 5 tests for src/modules/test/BinaryMigration.t.sol:BinaryMigrationTest_operatorBatchMigrate
[PASS] test_batchMigratePositions() (gas: 582109)
[PASS] test_revert_amountsLengthMismatch() (gas: 98069)
[PASS] test_revert_invalidArrayLength() (gas: 98153)
[PASS] test_revert_positionIdsLengthMismatch() (gas: 98009)
[PASS] test_revert_unauthorized() (gas: 95549)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 7.92ms (3.42ms CPU time)
```

```
Ran 1 test for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_getLegs
[PASS] test_getLegs() (gas: 250428)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 5.17ms (549.00 $\mu$ s CPU time)
```

```
Ran 4 tests for src/collateral/test/CollateralToken.t.sol:CollateralTokenTest_view
[PASS] test_decimals() (gas: 10645)
[PASS] test_immutables() (gas: 19992)
[PASS] test_name() (gas: 15069)
[PASS] test_symbol() (gas: 15001)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 2.05ms (754.96 $\mu$ s CPU time)
```

```
Ran 3 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_viaRouter
[PASS] test_merge_viaRouter() (gas: 405687)
[PASS] test_redeem_viaRouter() (gas: 437513)
[PASS] test_split_viaRouter() (gas: 291501)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 11.78ms (2.07ms CPU time)
```

```
Ran 4 tests for src/modules/test/BinaryMigration.t.sol:BinaryMigrationTest_prepareCondition
[PASS] test_revert_alreadyPrepared() (gas: 90024)
```



```
[PASS] test_revert_getLegacyPositionId_invalidOutcomeIndex() (gas: 90103)
[PASS] test_revert_questionNotPrepared() (gas: 26255)
[PASS] test_revert_wrongOutcomeCount() (gas: 65879)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 4.93ms (1.92ms CPU time)

Ran 1 test for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_getPayoutCoverage
[PASS] test_revert_getPayout_invalidOutcomeIndex() (gas: 87052)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 11.84ms (515.50µs CPU time)

Ran 4 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_bridge
[PASS] test_addBridge_removed() (gas: 24694)
[PASS] test_burnFromBridge_removed() (gas: 20438)
[PASS] test_mintFromBridge_removed() (gas: 21260)
[PASS] test_noUnbindingMintPathRemains() (gas: 64280)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 6.35ms (1.79ms CPU time)

Ran 5 tests for src/modules/test/BinaryMigration.t.sol:BinaryMigrationTest_resolveMigrationCondition
[PASS] test_reportResult_migrationConditionSettlesCollateralToVault() (gas: 745260)
[PASS] test_resolveMigrationCondition() (gas: 475801)
[PASS] test_revert_notResolvedOnLegacy() (gas: 93365)
[PASS] test_revert_resolutionPaused() (gas: 319476)
[PASS] test_unpauseAllowsResolution() (gas: 504055)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 7.99ms (5.09ms CPU time)

Ran 2 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_hasResult
[PASS] test_hasResult_false() (gas: 39703)
[PASS] test_hasResult_true() (gas: 109525)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 5.76ms (1.07ms CPU time)

Ran 3 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_canonicalization
[PASS] test_permutationYieldsSameId() (gas: 32552)
[PASS] test_revert_invalidModule() (gas: 15361)
[PASS] test_revert_invalidOutcomeIndex() (gas: 17564)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 7.81ms (1.33ms CPU time)

Ran 1 test for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_initialize
[PASS] test_revert_cannotReinitialize() (gas: 23546)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 6.03ms (408.13µs CPU time)

Ran 1 test for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_legacyPositionId
[PASS] test_getLegacyPositionId_nonMigrated() (gas: 17031)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 5.17ms (662.83µs CPU time)

Ran 12 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_compress
[PASS] test_compress_fractionalDustCannotExtractAfterRemainingLegResolves() (gas: 658328)
[PASS] test_compress_fractionalDustDoesNotLeakCollateral() (gas: 593265)
[PASS] test_compress_noPosition_allTrueWorthless() (gas: 395724)
[PASS] test_compress_noPosition_falseConditionMakesCollateral() (gas: 375302)
[PASS] test_compress_noPosition_fractionalConditionMintsCollateralAndReducedNo() (gas: 483349)
[PASS] test_compress_noPosition_trueConditionReduces() (gas: 414523)
[PASS] test_compress_yesPosition_allTrue_collateral() (gas: 439537)
[PASS] test_compress_yesPosition_falseConditionMakesWorthless() (gas: 341855)
[PASS] test_compress_yesPosition_fractionalConditionAfterUnresolvedScalesDown() (gas: 434908)
[PASS] test_compress_yesPosition_fractionalConditionScalesDown() (gas: 439125)
[PASS] test_compress_yesPosition_trueConditionRemoved() (gas: 416685)
[PASS] test_revert_nothingToCompress() (gas: 273481)
Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 15.25ms (10.48ms CPU time)

Ran 12 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_getPayout
[PASS] test_getPayout_noAllTrue() (gas: 365237)
[PASS] test_getPayout_noAnyFalse() (gas: 364885)
[PASS] test_getPayout_noFractionalComplement() (gas: 405620)
[PASS] test_getPayout_noLaterFalseBeforeAllResolved() (gas: 292786)
[PASS] test_getPayout_supportsManyFullPayoutLegs() (gas: 2268599)
[PASS] test_getPayout_supportsThirteenFractionalLegs() (gas: 1743171)
[PASS] test_getPayout_yesAllTrue() (gas: 365065)
[PASS] test_getPayout_yesAnyFalse() (gas: 358267)
[PASS] test_getPayout_yesFractionalProduct() (gas: 405384)
[PASS] test_getPayout_yesFractionalRoundsAtEnd() (gas: 401563)
[PASS] test_getPayout_yesLaterFalseBeforeAllResolved() (gas: 292829)
[PASS] test_revert_unresolved() (gas: 300149)
Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 36.20ms (22.14ms CPU time)
```

Ran 1 test for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_merge
[PASS] test_merge() (gas: 410837)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 14.04ms (927.67µs CPU time)

Ran 19 tests for src/utils/test/AutoRedeemer.t.sol:AutoRedeemerTest_redeem
[PASS] test_redeemAllUnapproved() (gas: 578141)
[PASS] test_redeemBatch() (gas: 840084)
[PASS] test_redeemBatchManyUsers() (gas: 3813068)
[PASS] test_redeemBinary() (gas: 477739)
[PASS] test_redeemCombinatorial() (gas: 607737)
[PASS] test_redeemDoesNotForwardStrandedBalance() (gas: 505129)
[PASS] test_redeemLegacyBinary() (gas: 510026)
[PASS] test_redeemLegacyBinaryDoesNotForwardStrandedBalance() (gas: 489456)
[PASS] test_redeemLegacyBinarySkipsUnapproved() (gas: 839971)
[PASS] test_redeemLegacyNegRisk() (gas: 871268)
[PASS] test_redeemLegacyNegRiskSkipsUnapproved() (gas: 1184609)
[PASS] test_redeemNegRisk() (gas: 518420)
[PASS] test_redeemSkipsUnapproved() (gas: 745729)
[PASS] test_redeemSkipsUnapprovedFirst() (gas: 686146)
[PASS] test_redeemSkipsUnapprovedInMiddle() (gas: 1110905)
[PASS] test_revertLegacyBinaryLengthMismatch() (gas: 20605)
[PASS] test_revertLegacyNegRiskLengthMismatch() (gas: 20539)
[PASS] test_revertLengthMismatch() (gas: 20791)
[PASS] test_revertUnauthorized() (gas: 19444)
Suite result: ok. 19 passed; 0 failed; 0 skipped; finished in 67.78ms (59.92ms CPU time)

Ran 5 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_convertToYesBasket
[PASS] test_convertToYesBasket() (gas: 602592)
[PASS] test_convertToYesBasket_singleCondition() (gas: 310772)
[PASS] test_revert_convertToYesBasket_requiresNoSide() (gas: 277585)
[PASS] test_revert_convertToYesBasket_unprepared() (gas: 197378)
[PASS] test_revert_convertToYesBasket_wrongToLength() (gas: 278010)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 13.72ms (7.63ms CPU time)

Ran 5 tests for src/collateral/test/CollateralToken.t.sol:CollateralTokenTest_wrap
[PASS] test_revert_invalidAsset(address) (runs: 256, µ: 28675, ~: 28675)
[PASS] test_revert_unauthorized() (gas: 25563)
[PASS] test_revert_wrap_noPreTransfer() (gas: 77636)
[PASS] test_wrapUSDC() (gas: 139830)
[PASS] test_wrapUSDCe() (gas: 139875)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 49.18ms (47.88ms CPU time)

Ran 4 tests for
src/test/integration/CombinatorialCollateralReturnUpgrade.t.sol:CombinatorialCollateralReturnUpgradeTest
[PASS] test_buildCallsAndExecute_preservesOwnershipAndAdmin() (gas: 254212)
[PASS] test_buildCalls_revertsForCombinatorialImmutableMismatch() (gas: 9358600)
[PASS] test_buildCalls_revertsWhenCombinatorialConfigTargetIsStale() (gas: 139634)
[PASS] test_buildCalls_revertsWhenRouterConfigTargetIsStale() (gas: 139579)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 6.14ms (2.17ms CPU time)

Ran 2 tests for src/utils/test/AutoRedeemer.t.sol:AutoRedeemerTest_upgrade
[PASS] test_revert_upgradeUnauthorized() (gas: 3559245)
[PASS] test_upgradeToAndCall_preservesState() (gas: 3589482)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 24.14ms (13.78ms CPU time)

Ran 1 test for src/modules/test/BinaryMigration.t.sol:BinaryMigrationTest_aliasKeyRegression
[PASS] test_revert_resolveMigrationCondition_aliasOutcomeByte() (gas: 90195)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 40.80ms (308.79µs CPU time)

Ran 1 test for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_initialize
[PASS] test_revert_cannotReinitialize() (gas: 23585)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 72.72ms (3.45ms CPU time)

Ran 2 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_erc1155Receiver
[PASS] test_onERC1155BatchReceived_normalTransfer() (gas: 237948)
[PASS] test_onERC1155Received_normalTransfer() (gas: 212464)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 62.62ms (54.96ms CPU time)

Ran 15 tests for
src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_conditionDefinitionBinding
[PASS] test_convertToYesBasketBindsEveryPrefix() (gas: 621489)

```
[PASS] test_inverseOpBindsDerivedDefinition() (gas: 612195)
[PASS] test_inverseOpPricesUnconsumedSideImmediately() (gas: 782995)
[PASS] test_prepareCondition_acceptsPreexistingIdenticalDefinition() (gas: 40740)
[PASS] test_prepareCondition_idempotentForSameDefinition() (gas: 106508)
[PASS] test_revert_convertToYesBasketAdoptsCollidingBasketDefinition() (gas: 293680)
[PASS] test_revert_convertToYesBasketAdoptsCollidingLaterBasketDefinition() (gas: 420546)
[PASS] test_revert_injectAdoptsCollidingReducedDefinition() (gas: 255871)
[PASS] test_revert_mergeFromYesBasketAdoptsCollidingBasketDefinition() (gas: 252253)
[PASS] test_revert_mergeOnConditionAdoptsCollidingChildDefinition() (gas: 234884)
[PASS] test_revert_mergeOnEventAdoptsCollidingChildDefinition() (gas: 212976)
[PASS] test_revert_prepareConditionCollidingDefinition() (gas: 44978)
[PASS] test_revert_prepareConditionCollidingDefinitionOfDifferentLength() (gas: 40417)
[PASS] test_revert_splitOnConditionAdoptsCollidingChildDefinition() (gas: 279439)
[PASS] test_revert_wrapAdoptsCollidingDefinition() (gas: 212570)
Suite result: ok. 15 passed; 0 failed; 0 skipped; finished in 97.31ms (69.24ms CPU time)

Ran 2 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_view
[PASS] test_getConditionId(bytes) (runs: 256, μ: 13544, ~: 13518)
[PASS] test_getPositionId(bytes32,uint256) (runs: 256, μ: 939, ~: 939)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 175.36ms (168.54ms CPU time)

Ran 1 test for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_initialize
[PASS] test_revert_cannotReinitialize() (gas: 23524)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 10.66ms (2.00ms CPU time)

Ran 9 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_configValidation
[PASS] test_initializeRequest_sentinelDisputerConfig() (gas: 276150)
[PASS] test_initializeRequest_zeroLivenessAllowsSameBlockFinalization() (gas: 520038)
[PASS] test_revert_initializeRequest_disputerModulesBelowThreshold() (gas: 263959)
[PASS] test_revert_initializeRequest_duplicateReportersBelowThreshold() (gas: 263351)
[PASS] test_revert_initializeRequest_reporterModulesBelowThreshold() (gas: 261145)
[PASS] test_revert_initializeRequest_zeroArbitrator() (gas: 77229)
[PASS] test_revert_initializeRequest_zeroDisputerThreshold() (gas: 78984)
[PASS] test_revert_initializeRequest_zeroReporterThreshold() (gas: 70217)
[PASS] test_revert_initializeRequest_zeroTargetContract() (gas: 69740)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 72.31ms (50.38ms CPU time)

Ran 1 test for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_migrateAmountsMismatch
[PASS] test_revert_migratePositions_operator_amountsMismatch() (gas: 31882)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 10.20ms (851.21μs CPU time)

Ran 2 tests for src/routers/test/BridgeRouter.t.sol:BridgeRouterTest_bridgeCollateral
[PASS] test_bridgeCollateralTo_emits_explicit_recipient() (gas: 497640)
[PASS] test_emits_initiator_event() (gas: 495836)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 24.99ms (15.45ms CPU time)

Ran 2 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_prepareEventCoverage
[PASS] test_constructor_zeroNegRiskAdapter() (gas: 6810929)
[PASS] test_getEventId_isDeterministic() (gas: 16920)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 14.70ms (5.83ms CPU time)

Ran 5 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:PreapprovedTest_matching_Migratable
[PASS] test_partialFill() (gas: 916043)
[PASS] test_preapprovedMakerComplementary() (gas: 596714)
[PASS] test_preapprovedTakerComplementary() (gas: 596708)
[PASS] test_respectsFilledStatus() (gas: 623940)
[PASS] test_respectsUserPause() (gas: 513452)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 45.34ms (28.51ms CPU time)

Ran 5 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_dispute
[PASS] test_disputeResult_differentModulesReachThreshold() (gas: 1206894)
[PASS] test_disputeResult_sameModuleDifferentRequests() (gas: 1353392)
[PASS] test_dispute_triggersArbitration() (gas: 696813)
[PASS] test_revert_disputeResult_alreadyVoted() (gas: 1130514)
[PASS] test_revert_dispute_zeroLivenessClosesImmediately() (gas: 626867)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 29.74ms (19.60ms CPU time)

Ran 3 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_inject
[PASS] test_inject() (gas: 579212)
[PASS] test_revert_inject_requiresNoSide() (gas: 231788)
[PASS] test_revert_inject_unprepared() (gas: 150368)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 57.56ms (47.51ms CPU time)
```


Ran 7 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_reportResult
[PASS] test_allNoResolvesSyntheticOther() (gas: 334807)
[PASS] test_finalYesOutcome() (gas: 317509)
[PASS] test_reportResult() (gas: 139622)
[PASS] test_revert_multipleYesOutcomes() (gas: 189819)
[PASS] test_revert_reportResult_eventPaused() (gas: 57567)
[PASS] test_revert_resolverSyntheticOtherYes() (gas: 34698)
[PASS] test_revert_unauthorized() (gas: 26794)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 35.99ms (25.33ms CPU time)

Ran 3 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:PreapprovedTest_preapproval_Migratable
[PASS] test_invalidatePreapprovedOrder() (gas: 58109)
[PASS] test_preapproveOrder_thenMatch() (gas: 628439)
[PASS] test_revert_preapproveOrder_notOperator() (gas: 33107)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 30.03ms (20.50ms CPU time)

Ran 2 tests for src/routers/test/BridgeRouter.t.sol:BridgeRouterTest_bridgePositions
[PASS] test_bridgePositionsTo_emits_explicit_recipient() (gas: 797784)
[PASS] test_emits_initiator_event() (gas: 795995)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 85.67ms (75.98ms CPU time)

Ran 15 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_edgeCases
[PASS] test_getRequestState_activeForValidCondition() (gas: 427973)
[PASS] test_getRequestState_noneForNonExistent() (gas: 19370)
[PASS] test_resolveResult_alreadyResolved_noop() (gas: 556558)
[PASS] test_revert_atomic_requestRejectsChildConditionId() (gas: 453595)
[PASS] test_revert_getRequestState_nonCanonicalRequestId() (gas: 15150)
[PASS] test_revert_initializeRequest_alreadyExists() (gas: 425827)
[PASS] test_revert_initializeRequest_atomicTypeRejectsWrongResultLength() (gas: 62741)
[PASS] test_revert_initializeRequest_nonCanonicalEventId() (gas: 41262)
[PASS] test_revert_initializeRequest_notOperator() (gas: 72698)
[PASS] test_revert_pauseOracle_notAdmin() (gas: 18151)
[PASS] test_revert_reportResult_invalidConditionIndex() (gas: 453709)
[PASS] test_revert_reportResult_requestNotFound() (gas: 22035)
[PASS] test_revert_resolveResult_notAuthorized() (gas: 430300)
[PASS] test_revert_unpauseOracle_notAdmin() (gas: 47392)
[PASS] test_revert_upgradeToAndCall_notOwner() (gas: 6166970)
Suite result: ok. 15 passed; 0 failed; 0 skipped; finished in 82.79ms (71.12ms CPU time)

Ran 1 test for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_reportResultCoverage
[PASS] test_revert_migratePositions_operator_arrayMismatch() (gas: 31905)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 63.14ms (1.81ms CPU time)

Ran 4 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:PreapprovedTest_reverts_Migratable
[PASS] test_revert_invalidSignature() (gas: 41242)
[PASS] test_revert_invalidSignatureNotPreapproved() (gas: 466076)
[PASS] test_revert_invalidatedPreapproval() (gas: 513787)
[PASS] test_revert_notOperator() (gas: 31011)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 61.29ms (10.87ms CPU time)

Ran 3 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_finalize
[PASS] test_finalize_targetAlreadyResolvedWithMatchingPayout() (gas: 712296)
[PASS] test_incremental_finalize_reportsTranslatedPayout() (gas: 740721)
[PASS] test_revert_finalize_targetPayoutMismatch() (gas: 715338)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 19.77ms (6.78ms CPU time)

Ran 5 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_resolverRole
[PASS] test_addResolver_thenReport() (gas: 165686)
[PASS] test_reportResult_replayCallerSemantics() (gas: 166150)
[PASS] test_revert_reportResult_resolverReplayPayoutMismatch() (gas: 141331)
[PASS] test_revert_reportResult_wrongResolutionChain() (gas: 27331)
[PASS] test_revert_unauthorized() (gas: 26750)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 67.40ms (48.14ms CPU time)

Ran 6 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_finalizer
[PASS] test_finalize_byConfiguredFinalizer() (gas: 714732)
[PASS] test_finalize_permissionlessWhenFinalizerZero() (gas: 694667)
[PASS] test_revert_finalize_byNonFinalizer() (gas: 642568)
[PASS] test_revert_setFinalizer_requestNotFound() (gas: 58677)
[PASS] test_revert_setFinalizer_unauthorized() (gas: 428205)
[PASS] test_setFinalizer_togglesGate() (gas: 720615)

Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 57.26ms (45.66ms CPU time)

Ran 4 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_merge

[PASS] test_merge() (gas: 337428)

[PASS] test_merge_unpreparedLegacyPair() (gas: 277845)

[PASS] test_revert_merge_wrongModuleId() (gas: 18701)

[PASS] test_splitMerge_roundtrip() (gas: 326147)

Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 161.87ms (114.99ms CPU time)

Ran 4 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_upgrade

[PASS] test_revert_upgrade_incompatibleModuleId() (gas: 5693582)

[PASS] test_revert_upgrade_incompatibleUsdce() (gas: 6841041)

[PASS] test_revert_upgrade_unauthorized() (gas: 6836459)

[PASS] test_upgradeToAndCall_preservesState() (gas: 6954826)

Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 12.07ms (5.34ms CPU time)

Ran 2 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_viaRouter

[PASS] test_horizontalMerge_viaRouter() (gas: 491999)

[PASS] test_horizontalSplit_viaRouter() (gas: 334436)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 17.18ms (1.71ms CPU time)

Ran 10 tests for src/modules/test/MigrationSnapshots.t.sol:MigrationSnapshots_Test

[PASS] test_MigrationSnapshots_binary_migrate_batch16() (gas: 675007)

[PASS] test_MigrationSnapshots_binary_migrate_batch2() (gas: 449042)

[PASS] test_MigrationSnapshots_binary_migrate_batch4() (gas: 479604)

[PASS] test_MigrationSnapshots_binary_migrate_batch8() (gas: 540803)

[PASS] test_MigrationSnapshots_binary_migrate_operator_batch4() (gas: 483240)

[PASS] test_MigrationSnapshots_negRisk_migrate_batch16() (gas: 5911036)

[PASS] test_MigrationSnapshots_negRisk_migrate_batch2() (gas: 1075537)

[PASS] test_MigrationSnapshots_negRisk_migrate_batch4() (gas: 1751774)

[PASS] test_MigrationSnapshots_negRisk_migrate_batch8() (gas: 3151971)

[PASS] test_MigrationSnapshots_negRisk_migrate_operator_batch4() (gas: 1755305)

Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 116.26ms (104.60ms CPU time)

Ran 1 test for src/modules/test/NegRiskMigration.t.sol:NegRiskMigrationTest_aliasKeyRegression

[PASS] test_revert_resolveMigrationCondition_aliasOutcomeByte() (gas: 959919)

Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 15.16ms (8.87ms CPU time)

Ran 1 test for src/routers/test/CtfRouter.t.sol:CtfRouterTest_redeemPositions

[PASS] test_redeem(bool,uint256) (runs: 256, μ : 483281, $\sim$: 498915)

Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 463.81ms (455.50ms CPU time)

Ran 13 tests for

src/oracle/test/modules/reporters/ChainlinkReporterModule.t.sol:ChainlinkReporterModuleTest_report

[PASS] testFuzz_signedPriceComparison(int192,int192) (runs: 256, μ : 568737, $\sim$: 569912)

[PASS] test_priceDown() (gas: 570363)

[PASS] test_priceDownAcrossZero() (gas: 569424)

[PASS] test_priceUp() (gas: 652783)

[PASS] test_priceUpAcrossZero() (gas: 576309)

[PASS] test_priceUpWhileNegative() (gas: 569460)

[PASS] test_revert_eventNotInitialized() (gas: 18322)

[PASS] test_revert_feedIdNotSet() (gas: 350844)

[PASS] test_revert_priceMissingInDataStore() (gas: 413133)

[PASS] test_revert_report_nonBinaryRequestId() (gas: 348030)

[PASS] test_revert_whenPaused() (gas: 475002)

[PASS] test_succeedsBeforeWindowEndWhenPricesExist() (gas: 569297)

[PASS] test_zeroLiveness_reportAndFinalizeInSameBlock() (gas: 660869)

Suite result: ok. 13 passed; 0 failed; 0 skipped; finished in 300.92ms (292.76ms CPU time)

Ran 3 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_mergeFromYesBasket

[PASS] test_mergeFromYesBasket() (gas: 677473)

[PASS] test_revert_mergeFromYesBasket_requiresNoSide() (gas: 231453)

[PASS] test_revert_mergeFromYesBasket_unprepared() (gas: 150079)

Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 87.98ms (51.73ms CPU time)

Ran 2 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_redeem

[PASS] test_redeem(bool) (runs: 256, μ : 440010, $\sim$: 439996)

[PASS] test_redeem_emitsPositionRedeemed() (gas: 417067)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 378.80ms (372.17ms CPU time)

Ran 6 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_realContract

[PASS]

```
testFuzz_hashOrderLegacy_matchesRealV2(uint256,address,address,uint256,uint256,uint8,uint8,uint256,bytes3
2,bytes32) (runs: 256, μ: 72179, ~: 72179)
[PASS] test_realContract_hashOrderLegacy_matchesRealContract() (gas: 158840)
[PASS] test_realContract_matchOrders_fill() (gas: 990000)
[PASS] test_realContract_matchOrders_partialResume() (gas: 921332)
[PASS] test_realContract_validateOrder() (gas: 146645)
[PASS] test_revert_realContract_orderConsumedOnLegacy() (gas: 145771)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 482.62ms (468.72ms CPU time)

Ran 1 test for src/routers/test/CtfRouter.t.sol:CtfRouterTest_splitPosition
[PASS] test_split() (gas: 314685)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 23.59ms (937.29μs CPU time)

Ran 1 test for src/test/integration/DeployPipelineFork.t.sol:DeployPipelineForkTest
[SKIP: skipped] setUp() (gas: 0)
Suite result: ok. 0 passed; 0 failed; 1 skipped; finished in 442.42μs (0.00ns CPU time)

Ran 2 tests for
src/oracle/test/modules/reporters/ChainlinkReporterModule.t.sol:ChainlinkReporterModuleTest_upgrade
[PASS] test_revert_upgrade_unauthorized() (gas: 2212325)
[PASS] test_upgradeToAndCall() (gas: 2216652)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 18.43ms (9.94ms CPU time)

Ran 4 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_mergeOnCondition
[PASS] test_mergeOnCondition() (gas: 584141)
[PASS] test_revert_mergeOnCondition_parentMustBeYes() (gas: 208205)
[PASS] test_revert_mergeOnCondition_parentUnprepared() (gas: 148700)
[PASS] test_revert_mergeOnCondition_referenceMustBeConditionId() (gas: 7248)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 19.87ms (12.22ms CPU time)

Ran 1 test for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_view
[PASS] test_getPositionId(bytes32,uint256) (runs: 256, μ: 917, ~: 917)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 106.99ms (64.70ms CPU time)

Ran 2 tests for
src/oracle/test/integration/OOReporterIntegration.t.sol:OOReporterIntegrationTest_registration
[PASS] test_aggregatorInitDataRegistersWithoutInitializingUMA() (gas: 439825)
[PASS] test_operatorCanRegisterAfterAggregatorInitialization() (gas: 435310)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 18.16ms (2.84ms CPU time)

Ran 6 tests for src/modules/test/BinaryModule.t.sol:BinaryModuleTest_reportResult
[PASS] test_reportResult() (gas: 84173)
[PASS] test_reportResult_replayCallerSemantics() (gas: 106958)
[PASS] test_revert_invalidSum() (gas: 33283)
[PASS] test_revert_reportResult_bridgeReplayPayoutMismatch() (gas: 96066)
[PASS] test_revert_reportResult_resolverReplayPayoutMismatch() (gas: 90184)
[PASS] test_revert_unauthorized() (gas: 23239)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 58.82ms (48.86ms CPU time)

Ran 15 tests for
src/oracle/test/modules/reporters/ChainlinkReporterModule.t.sol:ChainlinkReporterModuleTest_view
[PASS] test_computeQuestionId() (gas: 14014)
[PASS] test_getEventWindow() (gas: 346554)
[PASS] test_getPrice() (gas: 71876)
[PASS] test_getPrice_int192Bounds() (gas: 123994)
[PASS] test_getPrice_negative() (gas: 71919)
[PASS] test_isPriceAvailable_false() (gas: 22609)
[PASS] test_isPriceAvailable_falseForNonCanonicalNegativeEncoding() (gas: 74033)
[PASS] test_isPriceAvailable_falseForNonCanonicalPositiveEncoding() (gas: 74002)
[PASS] test_isPriceAvailable_noFeedId() (gas: 13597)
[PASS] test_isPriceAvailable_true() (gas: 73796)
[PASS] test_isPriceAvailable_trueForNegativePrice() (gas: 73818)
[PASS] test_revert_getEventWindow_notInitialized() (gas: 16850)
[PASS] test_revert_getPrice_feedIdNotSet() (gas: 16413)
[PASS] test_revert_getPrice_nonCanonicalNegativeEncoding() (gas: 74803)
[PASS] test_revert_getPrice_nonCanonicalPositiveEncoding() (gas: 74752)
Suite result: ok. 15 passed; 0 failed; 0 skipped; finished in 46.06ms (26.87ms CPU time)

Ran 3 tests for src/modules/test/NegRiskMigration.t.sol:NegRiskMigrationTest_horizontalMerge
[PASS] test_horizontalMerge() (gas: 23964159)
[PASS] test_horizontalMerge_beforeResolution() (gas: 1375916)
[PASS] test_horizontalMerge_nonMigratedEvent() (gas: 447389)
```

Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 101.23ms (78.54ms CPU time)

Ran 9 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_initialize

[PASS] testFuzz_initializeRequest_emitsCompleteConfiguration(uint8,uint32,address) (runs: 256, μ : 219700, ~: 220122)

[PASS] test_initializeAtomicNegRiskEvent() (gas: 424963)

[PASS] test_initializeBinaryEvent() (gas: 433824)

[PASS] test_initializeIncrementalNegRiskEventWithArbitraryTarget() (gas: 426984)

[PASS] test_revert_initializeAtomicNegRiskTypeWithArityOne() (gas: 73769)

[PASS] test_revert_initializeAtomicNegRiskTypeWithZeroArity() (gas: 85140)

[PASS] test_revert_initializeBinaryTypeWithNonzeroArity() (gas: 85456)

[PASS] test_revert_initializeIncrementalNegRiskTypeWithArityOne() (gas: 73836)

[PASS] test_revert_initializeIncrementalNegRiskTypeWithZeroArity() (gas: 85097)

Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 171.09ms (160.62ms CPU time)

Ran 2 tests for

src/oracle/test/integration/00ReporterIntegration.t.sol:00ReporterIntegrationTest_ruleUpdates

[PASS] test_revert_updateRules_whenProduction00ReporterRequesterRoleIsRevoked() (gas: 505513)

[PASS] test_updateRules_ignoresResolvedProduction00ReporterRequest() (gas: 737040)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 15.78ms (3.10ms CPU time)

Ran 2 tests for src/exchange/test/EIP712Digest.t.sol:Exchange_EIP712DigestStability

[PASS] test_orderHash_matchesIndependentComputation() (gas: 18600)

[PASS] test_orderStructHash_baseline() (gas: 3075)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 64.65ms (28.96ms CPU time)

Ran 5 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_marketDataRegistry

[PASS] test_revert_getLatestRule_none() (gas: 426058)

[PASS] test_revert_getRuleAt_outOfBounds() (gas: 509109)

[PASS] test_revert_setProductSpecification_emptyName() (gas: 21562)

[PASS] test_setProductSpecification_caseInsensitiveUpdate() (gas: 137855)

[PASS] test_setProductSpecification_firstWrite() (gas: 132227)

Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 20.36ms (11.91ms CPU time)

Ran 5 tests for src/collateral/test/CollateralOfframp.t.sol:CollateralOfframpTest_admin

[PASS] test_addAdmin() (gas: 48292)

[PASS] test_removeAdmin() (gas: 37732)

[PASS] test_revert_addAdmin_unauthorized() (gas: 20087)

[PASS] test_revert_pause_unauthorized() (gas: 15917)

[PASS] test_revert_removeAdmin_unauthorized() (gas: 20208)

Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 34.01ms (23.97ms CPU time)

Ran 2 tests for

src/oracle/test/integration/00ReporterIntegration.t.sol:00ReporterIntegrationTest_settlement

[PASS] test_atomicSettlement_reportsThenFinalizesAfterPolymarketLiveness() (gas: 914256)

[PASS] test_binarySettlement_reportsThenFinalizesAfterPolymarketLiveness() (gas: 864620)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 24.68ms (9.59ms CPU time)

Ran 12 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_split

[PASS] test_prepareCondition_maxLegs() (gas: 1948576)

[PASS] test_revert_conflictingConditions() (gas: 21524)

[PASS] test_revert_duplicateCondition() (gas: 18960)

[PASS] test_revert_emptyConditions() (gas: 14413)

[PASS] test_revert_nonCanonicalOrder() (gas: 21143)

[PASS] test_revert_prepareCondition_tooManyLegs() (gas: 671907)

[PASS] test_revert_splitOnCondition_tooManyLegs() (gas: 1963018)

[PASS] test_revert_split_unprepared() (gas: 35381)

[PASS] test_revert_split_wrongModuleId() (gas: 20112)

[PASS] test_revert_wrongToLength() (gas: 25183)

[PASS] test_split() (gas: 221344)

[PASS] test_split_singleCondition() (gas: 187540)

Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 46.10ms (34.79ms CPU time)

Ran 9 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_marketPause

[PASS] test_pause_adminResolveStillWorks() (gas: 580351)

[PASS] test_pause_arbitratorResolveStillWorks() (gas: 583420)

[PASS] test_pause_batch() (gas: 817392)

[PASS] test_pause_blocksFinalize() (gas: 674766)

[PASS] test_pause_independentOfGlobalPause() (gas: 452136)

[PASS] test_pause_reportAndDisputeUnaffected() (gas: 734300)

[PASS] test_revert_pause_requestNotFound() (gas: 57424)

[PASS] test_revert_pause_unauthorized() (gas: 432234)


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[PASS] test_unpause_reenablesFinalize() (gas: 712530)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 30.35ms (23.35ms CPU time)

Ran 7 tests for
src/oracle/test/integration/00ReporterIntegration.t.sol:00ReporterIntegrationTest_trustBoundaries
[PASS] test_fixtureSeparatesPolymarketAndUMARoles() (gas: 53066)
[PASS] test_revert_moduleCannotInitializeUMAResponse() (gas: 416774)
[PASS] test_revert_nonUMAResolverCannotSettleM00V2Request() (gas: 590709)
[PASS] test_revert_polymarketOwnerCannotManageUMAResponderRoles() (gas: 22007)
[PASS] test_revert_umaInitializationWhen00ReporterIsNotM00V2Requester() (gas: 531303)
[PASS] test_revert_umaOwnerCannotRegisterPolymarketRequest() (gas: 219233)
[PASS] test_revert_umaRolesCannotResolvePolymarketRequest() (gas: 256565)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 34.92ms (17.63ms CPU time)

Ran 4 tests for src/collateral/test/CollateralOfframp.t.sol:CollateralOfframpTest_pause
[PASS] test_pause_emitsEvent() (gas: 43254)
[PASS] test_revert_unwrapUSDC_paused() (gas: 212207)
[PASS] test_revert_unwrapUSDCe_paused() (gas: 212260)
[PASS] test_unpause() (gas: 273517)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 39.63ms (38.33ms CPU time)

Ran 2 tests for src/collateral/test/CollateralOfframp.t.sol:CollateralOfframpTest_unwrap
[PASS] test_unwrapUSDC() (gas: 245176)
[PASS] test_unwrapUSDCe() (gas: 245249)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 8.42ms (837.96µs CPU time)

Ran 6 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_splitOnCondition
[PASS] test_revert_conditionAlreadyPresent() (gas: 210363)
[PASS] test_revert_splitOnCondition_parentMustBeYes() (gas: 209259)
[PASS] test_revert_splitOnCondition_parentUnprepared() (gas: 149777)
[PASS] test_revert_splitOnCondition_referenceMustBeConditionId() (gas: 7269)
[PASS] test_revert_splitOnCondition_wrongToLength() (gas: 256270)
[PASS] test_splitOnCondition() (gas: 519022)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 35.15ms (18.09ms CPU time)

Ran 6 tests for src/oracle/test/modules/reporters/E0AResponderModule.t.sol:E0AResponderModuleTest_admin
[PASS] test_removeAdmin() (gas: 23094)
[PASS] test_removeOperator() (gas: 38148)
[PASS] test_revert_removeAdmin_unauthorized() (gas: 20720)
[PASS] test_revert_removeOperator_unauthorized() (gas: 50809)
[PASS] test_revert_setAggregator_onlyAdmin() (gas: 20710)
[PASS] test_setAggregator() (gas: 28080)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 69.29ms (49.95ms CPU time)

Ran 1 test for
src/oracle/test/integration/00ReporterIntegration.t.sol:00ReporterIntegrationTest_umaDispute
[PASS] test_firstDispute_usesRealReporterAutomaticRerequest() (gas: 686898)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 16.57ms (2.68ms CPU time)

Ran 14 tests for src/bridge/test/CcipBridgeCollateral.t.sol:CcipBridgeCollateralTest
[PASS] test_bridge_collateral_emitsCollateralBridged() (gas: 403560)
[PASS] test_bridge_collateral_emitsCollateralReceived() (gas: 532716)
[PASS] test_bridge_collateral_hubToSpoke() (gas: 538646)
[PASS] test_bridge_collateral_insufficientApproval_reverts() (gas: 432288)
[PASS] test_bridge_collateral_multipleBridges() (gas: 973510)
[PASS] test_bridge_collateral_partialAmount() (gas: 576375)
[PASS] test_bridge_collateral_quote_amountDoesNotAffectFee() (gas: 45279)
[PASS] test_bridge_collateral_quote_hubToSpoke() (gas: 32727)
[PASS] test_bridge_collateral_quote_spokeToHub() (gas: 32704)
[PASS] test_bridge_collateral_roundTrip() (gas: 910627)
[PASS] test_bridge_collateral_spokeToHub() (gas: 536365)
[PASS] test_bridge_collateral_spokeToSpoke() (gas: 536408)
[PASS] test_bridge_collateral_toSelf() (gas: 532124)
[PASS] test_revert_CcipBridge_receiveCollateral_noMinterRole() (gas: 481325)
Suite result: ok. 14 passed; 0 failed; 0 skipped; finished in 355.35ms (323.17ms CPU time)

Ran 1 test for src/modules/test/NegRiskMigration.t.sol:NegRiskMigrationTest_horizontalSplit
[PASS] test_horizontalSplit() (gas: 22118916)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 107.44ms (96.35ms CPU time)

Ran 9 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_pausable
[PASS] test_globalPauseBeforeProposalDoesNotExtendWindow() (gas: 626478)
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[PASS] test_globalPauseDoesNotExtendDisputeWindow() (gas: 715227)
[PASS] test_globalPauseDoesNotPreserveRemainingDisputeTime() (gas: 629906)
[PASS] test_resolvedDisputeWindowDoesNotDriftWithLaterPauses() (gas: 723810)
[PASS] test_revert_disputeResult_whenPaused() (gas: 643426)
[PASS] test_revert_finalize_whenPaused() (gas: 641790)
[PASS] test_revert_reportResult_whenPaused() (gas: 479586)
[PASS] test_revert_resolveResult_whenPaused() (gas: 451950)
[PASS] test_unpauseOracle() (gas: 35649)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 46.95ms (33.55ms CPU time)

Ran 4 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_unwrap
[PASS] test_revert_unwrap_multiCondition() (gas: 254440)
[PASS] test_unwrap_noPosition() (gas: 269230)
[PASS] test_unwrap_yesPosition() (gas: 267410)
[PASS] test_unwrap_yesPosition_ofNCondition() (gas: 265513)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 33.11ms (12.35ms CPU time)

Ran 1 test for
src/oracle/test/modules/reporters/EOAReporterModule.t.sol:EOAReporterModuleTest_constructor
[PASS] test_constructor() (gas: 20706)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 30.92ms (15.67ms CPU time)

Ran 1 test for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_report
[PASS] test_revert_invalidResultLength() (gas: 479553)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 18.26ms (1.13ms CPU time)

Ran 9 tests for src/oracle/test/common/OOReporterModule.t.sol:OOReporterModuleTest_finalize
[PASS] test_arbitrationBlocksFinalizeAndSeparateArbitratorResolves() (gas: 769109)
[PASS] test_finalize_afterLivenessResolvesBinaryResult() (gas: 655297)
[PASS] test_finalize_atomicNumericalResult() (gas: 720319)
[PASS] test_finalize_zeroFinalizerAllowsDirectPermissionlessFinalization() (gas: 619723)
[PASS] test_finalize_zeroLivenessResolvesInReportingBlock() (gas: 647051)
[PASS] test_moduleHasNoResolveAuthority() (gas: 400235)
[PASS] test_revert_finalize_beforeLiveness() (gas: 567976)
[PASS] test_revert_finalize_beforeThreshold() (gas: 552889)
[PASS] test_revert_finalize_whenAnotherFinalizerConfigured() (gas: 574030)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 49.20ms (32.55ms CPU time)

Ran 8 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_tokenIdMapping
[PASS] test_legacyOrder_markerStripped_failsAsNative() (gas: 156517)
[PASS] test_legacyOrder_nonMigrationModule_rejected() (gas: 39121)
[PASS] test_mixedBatch_legacyTaker_nativeMaker() (gas: 911380)
[PASS] test_mixedBatch_nativeTaker_legacyMaker() (gas: 891902)
[PASS] test_nativeOrder_freshMarket_validates() (gas: 66632)
[PASS] test_nativeOrder_makesNoLegacyCalls() (gas: 828493)
[PASS] test_nativeOrder_nonMigrationModule_validates() (gas: 39729)
[PASS] test_revert_legacySig_wrongModernId() (gas: 229599)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 194.81ms (170.49ms CPU time)

Ran 10 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_redeem
[PASS] test_redeem_fractionalYesNoPairCannotExtractDust() (gas: 502923)
[PASS] test_redeem_noPosition_allTrue() (gas: 394771)
[PASS] test_redeem_noPosition_anyFalse() (gas: 438324)
[PASS] test_redeem_noPosition_fractionalComplement() (gas: 483287)
[PASS] test_redeem_noPosition_laterFalseBeforeAllResolved() (gas: 368386)
[PASS] test_redeem_yesPosition_allTrue() (gas: 442562)
[PASS] test_redeem_yesPosition_anyFalse() (gas: 387804)
[PASS] test_redeem_yesPosition_fractionalProduct() (gas: 483052)
[PASS] test_redeem_yesPosition_laterFalseBeforeAllResolved() (gas: 333259)
[PASS] test_revert_notFullyResolved() (gas: 339968)
Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 66.22ms (51.55ms CPU time)

Ran 4 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_upgrade
[PASS] test_revert_upgrade_incompatibleModuleId() (gas: 5702485)
[PASS] test_revert_upgrade_incompatiblePositionManager() (gas: 36583687)
[PASS] test_revert_upgrade_unauthorized() (gas: 6116023)
[PASS] test_upgradeToAndCall_preservesState() (gas: 6212517)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 35.43ms (29.41ms CPU time)

Ran 6 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_reporterConflict
[PASS] test_additionalSupportForProposedResultDoesNotTriggerArbitration() (gas: 536519)
[PASS] test_conflictingResultAtThresholdTriggersArbitration() (gas: 673647)
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[PASS] test_conflictingResultBelowThresholdDoesNotTriggerArbitration() (gas: 558555)
[PASS] test_globalPauseDoesNotPreserveReporterConflictWindow() (gas: 532657)
[PASS] test_revert_reportAfterDisputeWindowExpired() (gas: 521326)
[PASS] test_revert_report_zeroLivenessClosesImmediately() (gas: 529694)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 26.05ms (7.93ms CPU time)

Ran 1 test for
src/oracle/test/modules/reporters/EOAReporterModule.t.sol:EOAReporterModuleTest_incrementalNegRisk
[PASS] test_reportOnSubCondition() (gas: 494939)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 39.04ms (24.70ms CPU time)

Ran 4 tests for src/oracle/test/common/OOReporterModule.t.sol:OOReporterModuleTest_initializer
[PASS] test_constructor() (gas: 24434)
[PASS] test_revert_constructor_zeroOOReporter() (gas: 62614)
[PASS] test_revert_initialize_onImplementation() (gas: 2300719)
[PASS] test_revert_initialize_twice() (gas: 26177)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 24.23ms (15.02ms CPU time)

Ran 2 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_wrap
[PASS] test_wrap() (gas: 255331)
[PASS] test_wrapUnwrap_roundtrip() (gas: 306764)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 17.15ms (5.68ms CPU time)

Ran 4 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_resolve
[PASS] test_adminResolve_duringArbitration_arbitratorHookReverts_stillResolves() (gas: 945667)
[PASS] test_adminResolve_duringArbitration_clearsArbitratorState() (gas: 763342)
[PASS] test_adminResolve_outsideArbitration_skipsArbitratorHook() (gas: 557245)
[PASS] test_revert_atomic_winnerIndexOutOfRange() (gas: 433425)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 16.33ms (11.29ms CPU time)

Ran 4 tests for src/routers/test/CtfRouter.t.sol:CtfRouterTest_initialize
[PASS] test_initialize_setsOwner() (gas: 14972)
[PASS] test_revert_initialize_onImplementation() (gas: 1565509)
[PASS] test_revert_initialize_twice() (gas: 21093)
[PASS] test_revert_initialize_zeroOwner() (gas: 1652458)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 7.33ms (2.82ms CPU time)

Ran 7 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_ruleManagerRole
[PASS] test_addRuleManager_grantsRole() (gas: 132433)
[PASS] test_admin_canPerformConfigActions() (gas: 479610)
[PASS] test_admin_canWriteMarketData() (gas: 104256)
[PASS] test_removeRuleManager_revokesRole() (gas: 37134)
[PASS] test_revert_addRuleManager_notAdmin() (gas: 18726)
[PASS] test_revert_removeRuleManager_notAdmin() (gas: 20784)
[PASS] test_revert_setProductSpecification_operatorNotAllowed() (gas: 19206)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 17.48ms (12.50ms CPU time)

Ran 1 test for src/routers/test/CtfRouter.t.sol:CtfRouterTest_mergePositions
[PASS] test_merge() (gas: 439272)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 12.78ms (1.19ms CPU time)

Ran 12 tests for src/oracle/test/common/OOReporterModule.t.sol:OOReporterModuleTest_registration
[PASS] test_createRequest_derivesNumericalIdentifier() (gas: 392484)
[PASS] test_createRequest_registersBinaryRequestAndEmitsEvents() (gas: 408406)
[PASS] test_initializeReporterModule_registersFromAggregatorInitData() (gas: 400676)
[PASS] test_initializeReporterModule_registersMultipleIncrementalRequests() (gas: 459487)
[PASS] test_revert_createRequest_beforeAggregatorInitialization() (gas: 92635)
[PASS] test_revert_createRequest_invalidLivenessRange() (gas: 248577)
[PASS] test_revert_createRequest_nonCanonicalRequestId() (gas: 220135)
[PASS] test_revert_createRequest_unauthorized() (gas: 219283)
[PASS] test_revert_duplicateRequestIdAcrossRegistrationPaths() (gas: 398497)
[PASS] test_revert_initializeReporterModule_emptyRegistrations() (gas: 56191)
[PASS] test_revert_initializeReporterModule_eventIdMismatch() (gas: 66909)
[PASS] test_revert_initializeReporterModule_onlyAggregator() (gas: 57080)
Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 36.04ms (20.75ms CPU time)

Ran 4 tests for
src/oracle/test/modules/reporters/EOAReporterModule.t.sol:EOAReporterModuleTest_initializeRequest
[PASS] test_initializeRequest() (gas: 367581)
[PASS] test_revert_cannotReinitialize() (gas: 361629)
[PASS] test_revert_emptyReporters() (gas: 51629)
[PASS] test_revert_onlyAggregator() (gas: 22227)
```

Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 47.55ms (33.83ms CPU time)

Ran 3 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_walletTypes

[PASS] test_replay_1271LegacySignature() (gas: 150069)

[PASS] test_replay_proxyLegacySignature() (gas: 195273)

[PASS] test_replay_safeLegacySignature() (gas: 232056)

Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 66.73ms (45.31ms CPU time)

Ran 3 tests for src/routers/test/CtfRouter.t.sol:CtfRouterTest_redeemInvalidIndexSet

[PASS] test_revert_redeemPositions_indexSetInvalidAmongValid() (gas: 442720)

[PASS] test_revert_redeemPositions_indexSetThree() (gas: 41422)

[PASS] test_revert_redeemPositions_indexSetZero() (gas: 41443)

Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 18.80ms (4.40ms CPU time)

Ran 1 test for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_targetFinalization

[PASS] test_revert_finalizeConditions_targetReverts() (gas: 751465)

Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 22.20ms (7.43ms CPU time)

Ran 1 test for src/modules/test/CombinatorialModuleInvariant.t.sol:CombinatorialModuleInvariantTest

[PASS] test_handlerResolvePath() (gas: 167019)

Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 47.16ms (861.29µs CPU time)

Ran 5 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_updateRequestRules

[PASS] test_revert_updateRequestRules_resolvedRequest() (gas: 692917)

[PASS] test_revert_updateRequestRules_unauthorized() (gas: 426262)

[PASS] test_revert_updateRequestRules_unknownRequest() (gas: 56948)

[PASS] test_revert_updateRules_notAggregator() (gas: 19009)

[PASS] test_updateRequestRules_appendsRuleAndNotifiesReporters() (gas: 611655)

Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 41.19ms (20.14ms CPU time)

Ran 2 tests for src/routers/test/CtfRouter.t.sol:CtfRouterTest_authorizeUpgrade

[PASS] test_revert_upgrade_unauthorized() (gas: 1578046)

[PASS] test_upgrade_ownerSucceeds() (gas: 32056365)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 44.47ms (25.92ms CPU time)

Ran 16 tests for src/oracle/test/common/OOReporterModule.t.sol:OOReporterModuleTest_report

[PASS] test_report_atomicNumericalResult() (gas: 558034)

[PASS] test_report_belowThresholdDoesNotCreateProposal() (gas: 545248)

[PASS] test_report_binaryP3TranslatesToHalf() (gas: 557507)

[PASS] test_report_incrementalNegRiskNo() (gas: 538576)

[PASS] test_report_incrementalNegRiskYes() (gas: 558528)

[PASS] test_report_noTranslatesToZero() (gas: 537121)

[PASS] test_report_yesCreatesProposalWithoutResolvingTarget() (gas: 576111)

[PASS] test_revert_report_afterModuleRemoval() (gas: 476488)

[PASS] test_revert_report_atomicNumericalNegativeResult() (gas: 446722)

[PASS] test_revert_report_atomicNumericalNonMultipleResult() (gas: 446892)

[PASS] test_revert_report_atomicNumericalWinnerOutOfRange() (gas: 447288)

[PASS] test_revert_report_beforeUMAResultAvailable() (gas: 394913)

[PASS] test_revert_report_incrementalNegRiskP3() (gas: 453855)

[PASS] test_revert_report_malformedBinaryPrice() (gas: 446415)

[PASS] test_revert_report_twice() (gas: 568161)

[PASS] test_revert_report_unregisteredRequest() (gas: 213612)

Suite result: ok. 16 passed; 0 failed; 0 skipped; finished in 70.13ms (57.08ms CPU time)

Ran 5 tests for src/modules/test/NegRiskMigration.t.sol:NegRiskMigrationTest_migrate

[PASS] test_getLegacyPositionId_matchesAdapter() (gas: 1090186)

[PASS] test_migrate() (gas: 20991906)

[PASS] test_revert_amountsLengthMismatch() (gas: 1567168)

[PASS] test_revert_conditionNotPrepared() (gas: 1464711)

[PASS] test_revert_invalidArrayLength() (gas: 1567169)

Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 161.57ms (147.47ms CPU time)

Ran 2 tests for

src/oracle/test/modules/reporters/EOReporterModule.t.sol:EOReporterModuleTest_initializer

[PASS] test_revert_initialize_onImplementation() (gas: 1660544)

[PASS] test_revert_initialize_twice() (gas: 26155)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 70.41ms (60.53ms CPU time)

Ran 3 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_voteOnce

[PASS] test_reportResult_differentModulesSameRequest() (gas: 619751)

[PASS] test_reportResult_sameModuleDifferentRequests() (gas: 601808)

[PASS] test_revert_reportResult_alreadyVoted() (gas: 540139)

Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 23.98ms (3.18ms CPU time)

Ran 3 tests for src/routers/test/CtfRouter.t.sol:CtfRouterTest_events

[PASS] test_mergePositions_emitsPositionsMerged() (gas: 436100)

[PASS] test_redeemPositions_emitsPositionRedeemedPerIndex() (gas: 509196)

[PASS] test_splitPosition_emitsPositionSplit() (gas: 291800)

Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 21.83ms (7.15ms CPU time)

Ran 8 tests for

src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersBalanceFlowsTest_complementary_Migratable

[PASS] test_complementaryBuyMakerFeeOnly() (gas: 609139)

[PASS] test_complementaryBuyTakerFeeOnly() (gas: 608544)

[PASS] test_complementarySellMakerFeeOnly() (gas: 643384)

[PASS] test_complementarySellMixedMakerFees() (gas: 825436)

[PASS] test_takerBuyTwoMakers() (gas: 859371)

[PASS] test_takerBuyTwoMakersWithFees() (gas: 890479)

[PASS] test_takerSellTwoMakers() (gas: 779415)

[PASS] test_takerSellTwoMakersWithFees() (gas: 828901)

Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 79.06ms (42.84ms CPU time)

Ran 1 test for src/libraries/test/Ids.t.sol:EventIdLibTest_conditionRoundtrip

[PASS] test_eventConditionRoundtrip(bytes32,uint256) (runs: 256, μ : 1730, $\sim$: 1730)

Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 11.65ms (11.04ms CPU time)

Ran 6 tests for src/test/integration/OracleDeployScripts.t.sol:OracleDeployScriptsTest

[PASS] test_checkOracleProxyAddresses_revertsForStaleReporterConfigs() (gas: 124010)

[PASS] test_checkedOracleAggregatorAddress_revertsForStaleConfig() (gas: 61985)

[PASS] test_initcodeScriptsComputeCanonicalCreate2Addresses() (gas: 21191401)

[PASS] test_oracleAggregatorVanitySaltAndAddressArePinned() (gas: 28248)

[PASS] test_postDeployWiresRolesAndHandsOffUpgradeAuthority() (gas: 18000315)

[PASS] test_proxyScriptsComputeDistinctCreate3Addresses() (gas: 8121498)

Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 23.73ms (37.56ms CPU time)

Ran 1 test for

src/test/integration/OracleResolutionModulesUpgrade.t.sol:OracleResolutionModulesUpgradeConfigTest

[SKIP: skipped] setUp() (gas: 0)

Suite result: ok. 0 passed; 0 failed; 1 skipped; finished in 1.46ms (0.00ns CPU time)

Ran 4 tests for src/exchange/test/MigratableExchangeHarness.t.sol:MigratableExchangeTest_guards

[PASS] test_hashOrderLegacy_unregisteredModule_defaultsBinaryDomain() (gas: 55388)

[PASS] test_hashTypedDataLegacy_unknownVenue_reverts() (gas: 11534)

[PASS] test_revert_legacyOrder_unmigratedMarket() (gas: 81586)

[PASS] test_updateOrderStatus_revertsWhenFillExceedsRemaining() (gas: 32897)

Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 97.33ms (85.02ms CPU time)

Ran 1 test for

src/test/integration/OracleResolutionModulesUpgrade.t.sol:OracleResolutionModulesUpgradeForkTest

[SKIP: skipped] setUp() (gas: 0)

Suite result: ok. 0 passed; 0 failed; 1 skipped; finished in 1.26ms (0.00ns CPU time)

Ran 7 tests for src/oracle/test/common/OOReporterModule.t.sol:OOReporterModuleTest_updateRules

[PASS] test_revert_updateRules_emptyOOReporterError() (gas: 486695)

[PASS] test_revert_updateRules_notAggregator() (gas: 394112)

[PASS] test_revert_updateRules_unexpectedOOReporterError() (gas: 487275)

[PASS] test_updateRules_forwardsToOOReporterForRegisteredRequest() (gas: 479107)

[PASS] test_updateRules_ignoresRequestAlreadyResolved() (gas: 534553)

[PASS] test_updateRules_ignoresRequestNotRegistered() (gas: 470082)

[PASS] test_updateRules_skipsUnregisteredRequest() (gas: 63882)

Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 35.87ms (14.44ms CPU time)

Ran 2 tests for

src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersBalanceFlowsTest_merge_Migratable

[PASS] test_twoMakers() (gas: 1083252)

[PASS] test_twoMakersWithFees() (gas: 1113503)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 17.41ms (5.15ms CPU time)

Ran 1 test for src/libraries/test/Ids.t.sol:EventIdLibTest_encodeFromData

[PASS] test_negRiskEventId(uint256,bytes) (runs: 256, μ : 2575, $\sim$: 2561)

Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 11.96ms (11.30ms CPU time)

Ran 4 tests for src/libraries/test/Ids.t.sol:IdsTest_canonicalBoundaries

[PASS] test_conditionIdFrom_canonicalMax() (gas: 389)

```
[PASS] test_conditionIdFrom_zero() (gas: 371)
[PASS] test_eventIdFrom_canonicalMax() (gas: 985)
[PASS] test_eventIdFrom_zero() (gas: 729)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 1.63ms (75.17µs CPU time)
```

```
Ran 9 tests for src/modules/test/NegRiskMigration.t.sol:NegRiskMigrationTest_prepareEvent
[PASS] test_maxLegacyConditionCount() (gas: 6491352)
[PASS] test_minLegacyConditionCount() (gas: 100055)
[PASS] test_prepareMigrationEvent_emitsLegacyEventIdInEventPrepared() (gas: 1564921)
[PASS] test_revert_alreadyPrepared() (gas: 1565406)
[PASS] test_revert_eventNotPrepared() (gas: 24287)
[PASS] test_revert_insufficientConditionCount() (gas: 23086)
[PASS] test_revert_legacyConditionCountTooLarge() (gas: 23099)
[PASS] test_revert_migrationNotSupported() (gas: 6986327)
[PASS] test_revert_prepareAgainAfterAddedQuestion() (gas: 1009905)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 50.28ms (28.17ms CPU time)
```

```
Ran 8 tests for
src/test/integration/OracleResolutionModulesUpgrade.t.sol:OracleResolutionModulesUpgradeTest
[PASS] test_buildCallsAndExecute_preservesRolesAndOwnership() (gas: 462308)
[PASS] test_buildCalls_doesNotRequireResolutionChainGetterOnOldImplementations() (gas: 247770)
[PASS] test_buildCalls_revertsForBinaryImmutableMismatch() (gas: 5853233)
[PASS] test_buildCalls_revertsForNegRiskImmutableMismatch() (gas: 7117212)
[PASS] test_buildCalls_revertsForPositionManagerImmutableMismatch() (gas: 3104401)
[PASS] test_buildCalls_revertsWhenBinaryConfigTargetIsStale() (gas: 142254)
[PASS] test_buildCalls_revertsWhenPositionManagerConfigTargetIsStale() (gas: 142145)
[PASS] test_buildCalls_revertsWhenProxyAlreadyUsesTarget() (gas: 315129)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 15.79ms (10.48ms CPU time)
```

```
Ran 2 tests for src/oracle/test/common/00ReporterModule.t.sol:00ReporterModuleTest_upgrade
[PASS] test_revert_upgrade_unauthorized() (gas: 2307763)
[PASS] test_upgradeToAndCall_preservesStateAndRoles() (gas: 2709376)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 15.12ms (1.91ms CPU time)
```

```
Ran 2 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersBalanceFlowsTest_mint_Migratable
[PASS] test_twoMakers() (gas: 760746)
[PASS] test_twoMakersWithFees() (gas: 790287)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 17.21ms (8.60ms CPU time)
```

```
Ran 7 tests for
src/oracle/test/libraries/OptimisticOraclePayoutLib.t.sol:OptimisticOraclePayoutLibTest_priceToPayouts
[PASS] test_noPrice_singleOutcome() (gas: 8327)
[PASS] test_numerical_winnerIndex0() (gas: 9762)
[PASS] test_numerical_winnerIndex2() (gas: 9696)
[PASS] test_p3Price_singleOutcome() (gas: 8550)
[PASS] test_revert_zeroOutcomes() (gas: 11694)
[PASS] test_revert_zeroOutcomes_nonZeroPrice() (gas: 11694)
[PASS] test_yesPrice_singleOutcome() (gas: 8423)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 2.85ms (996.67µs CPU time)
```

```
Ran 6 tests for src/oracle/test/modules/reporters/E0ReporterModule.t.sol:E0ReporterModuleTest_report
[PASS] test_revert_alreadyReported() (gas: 492394)
[PASS] test_revert_eventNotInitialized() (gas: 19805)
[PASS] test_revert_invalidResultLength() (gas: 417258)
[PASS] test_revert_invalidResultSum() (gas: 417076)
[PASS] test_revert_unauthorizedReporter() (gas: 364680)
[PASS] test_updatesAggregatorState() (gas: 497142)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 62.14ms (46.57ms CPU time)
```

```
Ran 1 test for
src/exchange/test/MigratableExchangeMatrix.t.sol:ERC1271SignatureTest_gasGriefing_Migratable
[PASS] test_validSignature_boundsReturndata() (gas: 3101260)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 51.33ms (33.58ms CPU time)
```

```
Ran 4 tests for src/oracle/test/integration/OracleSnapshots.t.sol:OracleSnapshotsTest_binary
[PASS] test_binary_createRequest() (gas: 259842)
[PASS] test_binary_finalize() (gas: 512073)
[PASS] test_binary_initializeRequest() (gas: 180708)
[PASS] test_binary_report() (gas: 421714)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 35.62ms (13.36ms CPU time)
```

```
Ran 23 tests for src/bridge/test/CcipBridgeStandard.t.sol:CcipBridgeStandardTest
[PASS] test_bridge_binaryPositionId_matchesAcrossChains(bytes32,uint256) (runs: 256, μ: 10110, ~: 10303)
[PASS] test_bridge_binary_positions_hubToSpoke() (gas: 901355)
[PASS] test_bridge_binary_positions_idempotentConditionPrep() (gas: 1448279)
[PASS] test_bridge_binary_positions_partialAmount() (gas: 918165)
[PASS] test_bridge_binary_positions_spokeToHub() (gas: 1435098)
[PASS] test_bridge_binary_positions_spokeToSpoke() (gas: 1482700)
[PASS] test_bridge_binary_resultBridge_hubToSpoke() (gas: 1318031)
[PASS] test_bridge_binary_resultBridge_spokeToSpoke() (gas: 2267405)
[PASS] test_bridge_binary_roundTrip() (gas: 1792970)
[PASS] test_bridge_moduleId_matches() (gas: 44479)
[PASS] test_bridge_negRiskDerivation_matchesAcrossChains(bytes32,uint8,uint256) (runs: 256, μ: 10779, ~: 10955)
[PASS] test_bridge_negRisk_positions_hubToSpoke() (gas: 904778)
[PASS] test_bridge_negRisk_positions_idempotentConditionPrep() (gas: 1452230)
[PASS] test_bridge_negRisk_positions_partialAmount() (gas: 922500)
[PASS] test_bridge_negRisk_positions_spokeToHub() (gas: 1438714)
[PASS] test_bridge_negRisk_positions_spokeToSpoke() (gas: 1486280)
[PASS] test_bridge_negRisk_resultBridge_hubToSpoke() (gas: 1418476)
[PASS] test_bridge_negRisk_resultBridge_spokeToSpoke() (gas: 2416616)
[PASS] test_bridge_negRisk_roundTrip() (gas: 1798308)
[PASS] test_bridge_router_bridgeCollateral() (gas: 639914)
[PASS] test_bridge_router_bridgePositions_binary() (gas: 933240)
[PASS] test_bridge_router_bridgePositions_negRisk() (gas: 936268)
[PASS] test_revert_CcipBridge_receivePositions_noBridgeRole() (gas: 845471)
Suite result: ok. 23 passed; 0 failed; 0 skipped; finished in 908.88ms (899.41ms CPU time)
```

```
Ran 4 tests for src/oracle/test/integration/OracleSnapshots.t.sol:OracleSnapshotsTest_negRisk
[PASS] test_negRisk_atomicReportAndFinalize() (gas: 563787)
[PASS] test_negRisk_createRequest() (gas: 260615)
[PASS] test_negRisk_initializeRequest() (gas: 181230)
[PASS] test_negRisk_reportAndFinalize() (gas: 563679)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 21.38ms (12.88ms CPU time)
```

```
Ran 6 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersBalanceFlowsTest_mixedPath_Migratable
[PASS] test_comboComplementaryMerge() (gas: 955664)
[PASS] test_comboComplementaryMergeWithFees() (gas: 985571)
[PASS] test_comboComplementaryMintNoFees() (gas: 862542)
[PASS] test_comboComplementaryMintWithFees() (gas: 18471054)
[PASS] test_threeMakersCompMergeWithFees() (gas: 1267809)
[PASS] test_threeMakersCompMintWithFees() (gas: 1078644)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 60.94ms (45.09ms CPU time)
```

```
Ran 4 tests for src/libraries/test/Ids.t.sol:IdsTest_canonicalization
[PASS] test_revert_conditionIdNonCanonical_outcomeByte(bytes32,uint8) (runs: 256, μ: 10619, ~: 10619)
[PASS] test_revert_eventIdNonCanonical_bothDirty(bytes32,uint8,uint16) (runs: 256, μ: 11928, ~: 11928)
[PASS] test_revert_eventIdNonCanonical_conditionIndex(bytes32,uint16) (runs: 256, μ: 13645, ~: 13645)
[PASS] test_revert_eventIdNonCanonical_outcomeByte(bytes32,uint8) (runs: 256, μ: 11215, ~: 11215)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 76.54ms (75.90ms CPU time)
```

```
Ran 18 tests for src/oracle/test/OracleAggregator.t.sol:OracleAggregatorTest_configMutation
[PASS] test_addDisputerModules_newModuleCanDispute() (gas: 1179580)
[PASS] test_addReporterModules_newModuleCanVote() (gas: 2545753)
[PASS] test_dispute_codelessArbitratorParksForAdmin() (gas: 649809)
[PASS] test_dispute_revertingArbitratorParksForAdmin() (gas: 739399)
[PASS] test_removeDisputerModules_blocksDisputing() (gas: 1091370)
[PASS] test_removeReporterModules_blocksReporting() (gas: 452240)
[PASS] test_revert_configMutation_requestNotFound() (gas: 122764)
[PASS] test_revert_configMutation_unauthorized() (gas: 452100)
[PASS] test_revert_removeDisputerModules_belowThreshold() (gas: 433385)
[PASS] test_revert_removeReporterModules_belowThreshold() (gas: 431519)
[PASS] test_revert_setArbitratorModule_zeroOnDisputableRequest() (gas: 426907)
[PASS] test_setArbitratorModule_codelessWithInitData_reverts() (gas: 438566)
[PASS] test_setArbitratorModule_midArbitration() (gas: 1528926)
[PASS] test_setArbitratorModule_noInitData_skipsInitializer() (gas: 1171363)
[PASS] test_setArbitratorModule_swapsAuthority() (gas: 1310657)
[PASS] test_setArbitratorModule_withInitData() (gas: 1270963)
[PASS] test_setLivenessWindow_appliesToFutureProposals() (gas: 626776)
[PASS] test_setLivenessWindow_zeroAllowsSameBlockFinalization() (gas: 707471)
Suite result: ok. 18 passed; 0 failed; 0 skipped; finished in 62.97ms (52.75ms CPU time)
```


Ran 6 tests for src/collateral/test/PermissionedRamp.t.sol:PermissionedRampTest_admin
[PASS] test_addAdmin() (gas: 48314)
[PASS] test_addWitness() (gas: 218280)
[PASS] test_removeAdmin() (gas: 37767)
[PASS] test_removeWitness() (gas: 144560)
[PASS] test_revert_addAdmin_unauthorized() (gas: 20109)
[PASS] test_revert_removeAdmin_unauthorized() (gas: 20208)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 10.44ms (4.13ms CPU time)

Ran 2 tests for src/oracle/test/modules/reporters/EOAReporterModule.t.sol:EOAReporterModuleTest_upgrade
[PASS] test_revert_upgrade_unauthorized() (gas: 1667610)
[PASS] test_upgradeToAndCall() (gas: 1671827)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 58.07ms (35.03ms CPU time)

Ran 1 test for src/libraries/test/Ids.t.sol:IdsTest_computeBaseHash
[PASS] test_computeBaseHash(uint256,bytes) (runs: 256, μ : 2168, $\sim$: 2144)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 9.61ms (9.01ms CPU time)

Ran 2 tests for src/collateral/test/PermissionedRamp.t.sol:PermissionedRampTest_pause
[PASS] test_revert_unwrap_paused() (gas: 258872)
[PASS] test_revert_wrap_paused() (gas: 134683)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 4.78ms (1.13ms CPU time)

Ran 3 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:ERC1271SignatureTest_reverts_Migratable
[PASS] test_revert_incorrectSigner() (gas: 471463)
[PASS] test_revert_invalidSignerMaker() (gas: 466800)
[PASS] test_revert_nonContract() (gas: 465304)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 43.14ms (22.47ms CPU time)

Ran 7 tests for src/collateral/test/PermissionedRamp.t.sol:PermissionedRampTest_signature
[PASS] test_revert_unwrap_expiredDeadline() (gas: 230413)
[PASS] test_revert_unwrap_invalidWitness() (gas: 237496)
[PASS] test_revert_wrap_expiredDeadline() (gas: 107086)
[PASS] test_revert_wrap_invalidNonce() (gas: 129426)
[PASS] test_revert_wrap_invalidWitness() (gas: 134245)
[PASS] test_revert_wrap_replaySignature() (gas: 232486)
[PASS] test_revert_wrap_signerNotWitness() (gas: 132224)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 10.55ms (4.50ms CPU time)

Ran 2 tests for src/libraries/test/Ids.t.sol:IdsTest_conditionEq
[PASS] test_conditionEq_distinct(bytes32,bytes32) (runs: 256, μ : 4275, $\sim$: 4275)
[PASS] test_conditionEq_reflexive(bytes32) (runs: 256, μ : 856, $\sim$: 856)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 13.45ms (12.86ms CPU time)

Ran 2 tests for src/collateral/test/PermissionedRamp.t.sol:PermissionedRampTest_unwrap
[PASS] test_unwrapUSDC() (gas: 285581)
[PASS] test_unwrapUSDCe() (gas: 285654)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 4.81ms (1.67ms CPU time)

Ran 3 tests for src/collateral/test/PermissionedRamp.t.sol:PermissionedRampTest_wrap
[PASS] test_incrementsNonce() (gas: 243948)
[PASS] test_wrapUSDC() (gas: 201599)
[PASS] test_wrapUSDCe() (gas: 201617)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 6.37ms (1.96ms CPU time)

Ran 6 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersBalanceFlowsTest_surplusRefund_Migratable
[PASS] test_batchBuyCollateralSurplusRefunded() (gas: 595539)
[PASS] test_batchBuyPositionSurplusRefunded() (gas: 887444)
[PASS] test_batchBuyPositionSurplusWithFees() (gas: 911713)
[PASS] test_batchMintOverfundedRefundedToTaker() (gas: 595572)
[PASS] test_batchSellPositiveSlippageRefundedToTaker() (gas: 958692)
[PASS] test_batchSellPositiveSlippageWithFees() (gas: 982220)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 35.31ms (22.59ms CPU time)

Ran 2 tests for src/oracle/test/modules/reporters/EOAReporterModule.t.sol:EOAReporterModuleTest_view
[PASS] test_hasReporterReported() (gas: 500621)
[PASS] test_isReporter() (gas: 372582)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 27.43ms (22.06ms CPU time)

Ran 1 test for
src/abstract/test/ERC1155TokenReceiver.t.sol:ERC1155TokenReceiverTest_onERC1155BatchReceived

```
[PASS] test_returnsAcceptanceSelector() (gas: 10081)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 208.83µs (57.08µs CPU time)

Ran 1 test for src/abstract/test/ERC1155TokenReceiver.t.sol:ERC1155TokenReceiverTest_onERC1155Received
[PASS] test_returnsAcceptanceSelector() (gas: 7893)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 173.79µs (41.58µs CPU time)

Ran 3 tests for src/abstract/test/ERC1155TokenReceiver.t.sol:ERC1155TokenReceiverTest_supportsInterface
[PASS] test_rejectsUnknown() (gas: 7897)
[PASS] test_supportsERC1155TokenReceiver() (gas: 6351)
[PASS] test_supportsERC165() (gas: 6400)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 249.13µs (105.71µs CPU time)

Ran 3 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_addModule
[PASS] test_revert_addModule_alreadyRegistered() (gas: 26426)
[PASS] test_revert_addModule_invalidModuleId() (gas: 368795)
[PASS] test_revert_removeModule_notRegistered() (gas: 20844)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 5.25ms (900.46µs CPU time)

Ran 3 tests for
src/modules/test/NegRiskMigration.t.sol:NegRiskMigrationTest_resolveMigrationCondition_sideEffects
[PASS] test_reportResult_migrationConditionSettlesCollateralToVault() (gas: 1913424)
[PASS] test_resolveMigrationCondition_migratedLoser_redeemsLegacy() (gas: 1982371)
[PASS] test_revert_resolveMigrationCondition_migratedLoser_legacyUnresolved() (gas: 1469540)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 41.20ms (30.42ms CPU time)

Ran 1 test for src/exchange/test/MigratableExchangeMatrix.t.sol:ERC1271SignatureTest_valid_Migratable
[PASS] test_validSignature() (gas: 572433)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 33.33ms (9.52ms CPU time)

Ran 5 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_crossModuleBurn
[PASS] test_crossModuleBatchBurn() (gas: 103531)
[PASS] test_crossModuleBurn() (gas: 78896)
[PASS] test_crossModuleMint_revoked() (gas: 52500)
[PASS] test_revert_crossModuleBatchBurn_unauthorized() (gas: 90803)
[PASS] test_revert_crossModuleBurn_unauthorized() (gas: 57914)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 12.63ms (8.02ms CPU time)

Ran 1 test for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersBalanceFlowsTest_takerPositionRefund_Migratable
[PASS] test_batchSellTakerExcessPositionsRefunded() (gas: 604250)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 23.42ms (2.19ms CPU time)

Ran 1 test for src/exchange/test/ERC1271Signature.t.sol:ERC1271SignatureTest_gasGriefing
[PASS] test_validSignature_boundsReturndata() (gas: 3098388)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 23.00ms (11.51ms CPU time)

Ran 4 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_crossModuleMint
[PASS] test_crossModuleBatchMint() (gas: 124727)
[PASS] test_crossModuleMint() (gas: 93149)
[PASS] test_revert_crossModuleBatchMint_unauthorized() (gas: 35525)
[PASS] test_revert_crossModuleMint_unauthorized() (gas: 31809)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 10.55ms (4.84ms CPU time)

Ran 1 test for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_deployment
[PASS] test_deployment() (gas: 25654)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 2.07ms (315.50µs CPU time)

Ran 1 test for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_getPayout
[PASS] test_revert_getPayout_moduleNotRegistered() (gas: 17254)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 2.37ms (640.00µs CPU time)

Ran 1 test for
src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_domainSeparator_Migratable
[PASS] test_domainSeparator() (gas: 11094)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 20.22ms (723.67µs CPU time)

Ran 4 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_initRoles
[PASS] test_creatorAction_success() (gas: 36300)
[PASS] test_operatorAction_success() (gas: 36166)
[PASS] test_revert_creatorAction_unauthorized() (gas: 13294)
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[PASS] test_revert_operatorAction_unauthorized() (gas: 13228)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 1.24ms (244.21µs CPU time)

Ran 4 tests for src/libraries/test/Ids.t.sol:IdsTest_encodeIdempotence
[PASS] test_conditionIdEncodeFromDataIdempotent(uint256,uint256,bytes) (runs: 256, µ: 2397, ~: 2381)
[PASS] test_conditionIdEncodeFromIdempotent(uint256,bytes32,uint256,uint256) (runs: 256, µ: 1615, ~: 1615)
[PASS] test_eventIdEncodeFromDataIdempotent(uint256,uint256,bytes) (runs: 256, µ: 2730, ~: 2714)
[PASS] test_eventIdEncodeFromIdempotent(uint256,bytes32,uint256) (runs: 256, µ: 1813, ~: 1813)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 43.06ms (41.88ms CPU time)

Ran 1 test for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_merge
[PASS] test_merge() (gas: 123302)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 2.29ms (455.79µs CPU time)

Ran 2 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_hashOrder_Migratable
[PASS] test_deterministic() (gas: 23937)
[PASS] test_differentSaltDifferentHash() (gas: 24931)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 7.93ms (1.57ms CPU time)

Ran 5 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_module
[PASS] test_prepareCondition() (gas: 27500)
[PASS] test_removeModule() (gas: 32271)
[PASS] test_resolveCondition() (gas: 16632)
[PASS] test_revert_removeModule_unauthorized() (gas: 23995)
[PASS] test_revert_split_moduleNotRegistered() (gas: 27241)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 5.85ms (2.47ms CPU time)

Ran 6 tests for src/modules/test/NegRiskMigration.t.sol:NegRiskMigrationTest_resolveMigrationCondition
[PASS] test_resolveMigrationCondition() (gas: 20986071)
[PASS] test_resolveMigrationCondition_alreadyResolvedNoop() (gas: 2044236)
[PASS] test_revert_nonMigratedCondition() (gas: 1461538)
[PASS] test_revert_notResolvedOnLegacy() (gas: 1567724)
[PASS] test_revert_resolutionPaused() (gas: 1637521)
[PASS] test_unpauseAllowsResolution() (gas: 1817164)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 142.75ms (135.68ms CPU time)

Ran 4 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_moduleAuth
[PASS] test_revert_batchBurn_unauthorized() (gas: 30211)
[PASS] test_revert_batchMint_unauthorized() (gas: 30764)
[PASS] test_revert_burn_unauthorized() (gas: 27835)
[PASS] test_revert_mint_unauthorized() (gas: 28451)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 4.26ms (1.25ms CPU time)

Ran 1 test for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_redeem
[PASS] test_redeem() (gas: 96776)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 2.63ms (613.92µs CPU time)

Ran 2 tests for src/libraries/test/Ids.t.sol:IdsTest_eventEq
[PASS] test_eventEq_distinct(bytes32,bytes32) (runs: 256, µ: 5855, ~: 5855)
[PASS] test_eventEq_reflexive(bytes32) (runs: 256, µ: 1406, ~: 1406)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 18.24ms (17.65ms CPU time)

Ran 3 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_initializer_Migratable
[PASS] test_revert_constructor_maxFeeRateExceedsCeiling() (gas: 66330)
[PASS] test_revert_initialize_onImplementation() (gas: 5591691)
[PASS] test_revert_initialize_twice() (gas: 23556)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 12.28ms (4.86ms CPU time)

Ran 2 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_batchMatchTakerBuy_Migratable
[PASS] test_revert_batchTakerBuy_makerBuyComplement_notCrossing() (gas: 636850)
[PASS] test_revert_batchTakerBuy_makerSell_notCrossing() (gas: 635832)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 28.42ms (6.48ms CPU time)

Ran 10 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_roles
[PASS] test_addAdmin() (gas: 53908)
[PASS] test_addCreator() (gas: 54008)
[PASS] test_addOperator() (gas: 53984)
[PASS] test_removeAdmin() (gas: 45604)
[PASS] test_removeCreator() (gas: 42464)
[PASS] test_removeOperator() (gas: 42464)
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[PASS] test_revert_addAdmin_byAdmin() (gas: 29642)
[PASS] test_revert_addAdmin_unauthorized() (gas: 29597)
[PASS] test_revert_addCreator_unauthorized() (gas: 29766)
[PASS] test_revert_addOperator_unauthorized() (gas: 29700)
Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 11.69ms (7.27ms CPU time)

Ran 2 tests for src/libraries/test/Ids.t.sol:IdsTest_positionEq
[PASS] test_positionEq_distinct(uint256,uint256) (runs: 256, μ: 3847, ~: 3847)
[PASS] test_positionEq_reflexive(uint256) (runs: 256, μ: 729, ~: 729)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 15.13ms (13.95ms CPU time)

Ran 8 tests for
src/oracle/test/modules/reporters/ChainlinkReporterModule.t.sol:ChainlinkReporterModuleTest_admin
[PASS] test_pause() (gas: 44769)
[PASS] test_revert_pause_notAdmin() (gas: 18100)
[PASS] test_revert_setFeedId_notAdmin() (gas: 18714)
[PASS] test_revert_setPriceSource_notAdmin() (gas: 20643)
[PASS] test_revert_unpause_notAdmin() (gas: 47332)
[PASS] test_setFeedId() (gas: 46677)
[PASS] test_setPriceSource() (gas: 33247)
[PASS] test_unpause() (gas: 35518)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 23.66ms (15.39ms CPU time)

Ran 1 test for src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_operators_Migratable
[PASS] test_addRemoveOperator() (gas: 38437)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 15.49ms (3.72ms CPU time)

Ran 7 tests for src/modules/test/NegRiskMigration.t.sol:NegRiskMigrationTest_resolveMigrationCounters
[PASS] test_countersUpdateOnNo() (gas: 1766433)
[PASS] test_countersUpdateOnYes() (gas: 1787293)
[PASS] test_finalizeMigration_allNo() (gas: 1759238)
[PASS] test_migrationThenResolveLosers() (gas: 1767929)
[PASS] test_revert_lastConditionMustBeNo() (gas: 1678194)
[PASS] test_revert_resolveMigrationCondition_syntheticOther() (gas: 1247333)
[PASS] test_revert_twoYesLegacyResolves() (gas: 1834022)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 65.86ms (50.24ms CPU time)

Ran 5 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_setCrossModuleAuth
[PASS] test_revert_setCrossModuleAuth_moduleIdZero() (gas: 371234)
[PASS] test_revert_setCrossModuleAuth_notRegistered() (gas: 774754)
[PASS] test_revert_setCrossModuleAuth_unauthorized() (gas: 20940)
[PASS] test_setCrossModuleAuth() (gas: 57241)
[PASS] test_setCrossModuleAuth_revoke() (gas: 46160)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 16.34ms (11.46ms CPU time)

Ran 3 tests for src/exchange/test/ERC1271Signature.t.sol:ERC1271SignatureTest_reverts
[PASS] test_revert_incorrectSigner() (gas: 470470)
[PASS] test_revert_invalidSignerMaker() (gas: 465807)
[PASS] test_revert_nonContract() (gas: 464311)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 56.21ms (39.11ms CPU time)

Ran 1 test for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_split
[PASS] test_split() (gas: 92452)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 5.07ms (1.23ms CPU time)

Ran 3 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_batchMatchTakerSell_Migratable
[PASS] test_revert_batchTakerSell_makerBuy_notCrossing() (gas: 727327)
[PASS] test_revert_batchTakerSell_makerSellComplement_notCrossing() (gas: 743568)
[PASS] test_revert_batchTakerSell_takerFillMismatch() (gas: 836609)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 30.29ms (18.46ms CPU time)

Ran 3 tests for
src/oracle/test/modules/reporters/ChainlinkReporterModule.t.sol:ChainlinkReporterModuleTest_initializeReq
uest
[PASS] test_initializeRequest() (gas: 345265)
[PASS] test_revert_initializeRequest_notBinaryEvent() (gas: 56168)
[PASS] test_revert_initializeRequest_windowAlreadyEnded() (gas: 57093)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 23.59ms (12.04ms CPU time)

Ran 2 tests for src/libraries/test/Ids.t.sol:PositionIdLibTest_pathEquivalence
[PASS] test_conditionIdPaths(uint256,uint256,bytes) (runs: 256, μ: 3524, ~: 3490)
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[PASS] test_eventIdPaths(uint256,uint256,bytes) (runs: 256, μ: 3445, ~: 3411)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 25.21ms (24.22ms CPU time)

Ran 4 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_pause_Migratable
[PASS] test_pauseThenMatchReverts() (gas: 595730)
[PASS] test_pauseUnpause() (gas: 33412)
[PASS] test_revert_pauseUser_alreadyPaused() (gas: 43963)
[PASS] test_selfServicePauseUser() (gas: 41847)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 28.36ms (16.53ms CPU time)

Ran 5 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_unsafeMint
[PASS] test_batchMintToNonReceiver() (gas: 127894)
[PASS] test_mintToNonReceiver() (gas: 95311)
[PASS] test_revert_batchMint_lengthMismatch() (gas: 32660)
[PASS] test_revert_batchMint_toZeroAddress() (gas: 30931)
[PASS] test_revert_mint_toZeroAddress() (gas: 24747)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 16.04ms (8.73ms CPU time)

Ran 1 test for src/exchange/test/ERC1271Signature.t.sol:ERC1271SignatureTest_valid
[PASS] test_validSignature() (gas: 570136)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 22.99ms (9.08ms CPU time)

Ran 2 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_upgrade
[PASS] test_revert_upgrade_unauthorized() (gas: 2970744)
[PASS] test_upgradeToAndCall() (gas: 2975122)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 12.59ms (8.44ms CPU time)

Ran 2 tests for
src/oracle/test/modules/reporters/ChainlinkReporterModule.t.sol:ChainlinkReporterModuleTest_initializer
[PASS] test_revert_initialize_onImplementation() (gas: 2205781)
[PASS] test_revert_initialize_twice() (gas: 26597)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 24.31ms (2.60ms CPU time)

Ran 1 test for src/libraries/test/Ids.t.sol:PositionIdLibTest_roundtrip
[PASS] test_positionId(uint256,uint256,uint256,uint256,bytes) (runs: 256, μ: 3348, ~: 3332)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 30.53ms (28.93ms CPU time)

Ran 1 test for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_domainSeparator
[PASS] test_domainSeparator() (gas: 11094)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 23.28ms (701.88μs CPU time)

Ran 2 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_uri
[PASS] test_revert_uri_moduleNotRegistered() (gas: 18848)
[PASS] test_uri_returnsTemplate() (gas: 24395)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 18.08ms (5.53ms CPU time)

Ran 3 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_removeAdminAndOperator_Migratable
[PASS] test_removeAdmin() (gas: 38368)
[PASS] test_revert_removeAdmin_unauthorized() (gas: 48821)
[PASS] test_revert_removeOperator_unauthorized() (gas: 50839)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 36.52ms (10.60ms CPU time)

Ran 3 tests for src/positionManager/test/PositionManager.t.sol:PositionManagerTest_view
[PASS] test_balanceOf_conditionId() (gas: 88544)
[PASS] test_getPayout() (gas: 29591)
[PASS] test_getPositionId() (gas: 14352)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 13.20ms (6.87ms CPU time)

Ran 5 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_complementValidation_Migratable
[PASS] test_revert_invalidComplement_sameOutcome() (gas: 378435)
[PASS] test_revert_mismatchedTokenIds_batchSellBuyMaker() (gas: 488638)
[PASS] test_revert_notCrossing_batchBuySellMaker() (gas: 469309)
[PASS] test_revert_notCrossing_mergePath() (gas: 587380)
[PASS] test_revert_notCrossing_mintPath() (gas: 380876)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 62.43ms (39.61ms CPU time)

Ran 2 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_setup_Migratable
[PASS] test_initialState() (gas: 44973)
[PASS] test_initializerDefaults() (gas: 17752021)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 42.80ms (15.27ms CPU time)
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Ran 2 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_renounceOperatorRole_Migratable
[PASS] test_renounceOperatorRole() (gas: 22072)
[PASS] test_revert_renounceOperatorRole_notOperator() (gas: 18102)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 23.31ms (8.90ms CPU time)

Ran 2 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_hashOrder
[PASS] test_deterministic() (gas: 23893)
[PASS] test_differentSaltDifferentHash() (gas: 24887)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 33.08ms (12.14ms CPU time)

Ran 3 tests for src/libraries/test/Ids.t.sol:PositionIdLibTest_structuralInvariants
[PASS] test_conditionIdOutcomeByteZero(uint256,uint256,bytes) (runs: 256, μ: 2270, ~: 2254)
[PASS] test_reservedBitsZero(uint256,uint256,uint256,uint256,bytes) (runs: 256, μ: 2746, ~: 2730)
[PASS] test_resolutionChainBitsAreRightAligned(bytes32) (runs: 256, μ: 768, ~: 768)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 41.05ms (38.98ms CPU time)

Ran 1 test for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_complementaryNotCrossing_Migratable
[PASS] test_revert_complementaryNotCrossing() (gas: 469076)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 26.29ms (6.82ms CPU time)

Ran 1 test for src/exchange/test/Preapproved.t.sol:PreapprovedTest_erc1271
[PASS] test_signerInvalidated() (gas: 839608)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 42.66ms (10.84ms CPU time)

Ran 2 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_roleViews_Migratable
[PASS] test_isAdmin() (gas: 21873)
[PASS] test_isOperator() (gas: 30570)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 36.11ms (3.91ms CPU time)

Ran 3 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_initializer
[PASS] test_revert_constructor_maxFeeRateExceedsCeiling() (gas: 66330)
[PASS] test_revert_initialize_onImplementation() (gas: 5591691)
[PASS] test_revert_initialize_twice() (gas: 23534)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 38.34ms (13.03ms CPU time)

Ran 2 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_tradingPausedEvents_Migratable
[PASS] test_pauseTrading_emitsEvent() (gas: 43246)
[PASS] test_unpauseTrading_emitsEvent() (gas: 33042)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 41.69ms (10.19ms CPU time)

Ran 2 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_complementarySellNotCrossing_Migratable
[PASS] test_revert_complementarySellNotCrossing_withFees() (gas: 485908)
[PASS] test_revert_complementarySellNotCrossing_zeroFee() (gas: 486023)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 29.54ms (6.30ms CPU time)

Ran 1 test for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_operators
[PASS] test_addRemoveOperator() (gas: 38402)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 12.19ms (2.97ms CPU time)

Ran 2 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_batchMatchTakerBuy
[PASS] test_revert_batchTakerBuy_makerBuyComplement_notCrossing() (gas: 635824)
[PASS] test_revert_batchTakerBuy_makerSell_notCrossing() (gas: 634806)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 56.68ms (15.98ms CPU time)

Ran 5 tests for src/exchange/test/Preapproved.t.sol:PreapprovedTest_matching
[PASS] test_partialFill() (gas: 912502)
[PASS] test_preapprovedMakerComplementary() (gas: 594443)
[PASS] test_preapprovedTakerComplementary() (gas: 594456)
[PASS] test_respectsFilledStatus() (gas: 621471)
[PASS] test_respectsUserPause() (gas: 513144)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 40.04ms (20.88ms CPU time)

Ran 2 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_upgrade_Migratable
[PASS] test_revert_upgrade_unauthorized() (gas: 5601368)
[PASS] test_upgradeToAndCall() (gas: 5605717)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 29.10ms (7.48ms CPU time)
```



```
Ran 4 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_pause
[PASS] test_pauseThenMatchReverts() (gas: 593488)
[PASS] test_pauseUnpause() (gas: 33517)
[PASS] test_revert_pauseUser_alreadyPaused() (gas: 43963)
[PASS] test_selfServicePauseUser() (gas: 41882)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 26.82ms (11.71ms CPU time)

Ran 2 tests for src/modules/test/NegRiskMigration.t.sol:NegRiskMigrationTest_view
[PASS] test_getConditionId(bytes32,uint8) (runs: 256, μ: 4393, ~: 4393)
[PASS] test_positionIds(bytes32,uint8,bool) (runs: 256, μ: 1694, ~: 1705)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 171.34ms (156.49ms CPU time)

Ran 4 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_mint_Migratable
[PASS] test_mint() (gas: 574331)
[PASS] test_mintConsumesLimitFill_whenFundedAtExecutionPrice() (gas: 579848)
[PASS] test_mintFees() (gas: 601842)
[PASS] test_mintRefundedTakerEventReportsNetFill() (gas: 616140)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 58.12ms (33.80ms CPU time)

Ran 3 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_batchMatchTakerSell
[PASS] test_revert_batchTakerSell_makerBuy_notCrossing() (gas: 726316)
[PASS] test_revert_batchTakerSell_makerSellComplement_notCrossing() (gas: 742558)
[PASS] test_revert_batchTakerSell_takerFillMismatch() (gas: 833648)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 38.85ms (11.65ms CPU time)

Ran 3 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_userPauseBlockInterval_Migratable
[PASS] test_revert_exceedsMaxPauseInterval() (gas: 18640)
[PASS] test_setUserPauseBlockInterval() (gas: 29048)
[PASS] test_setUserPauseBlockInterval_maxAllowed() (gas: 26119)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 33.70ms (8.56ms CPU time)

Ran 3 tests for src/exchange/test/Preapproved.t.sol:PreapprovedTest_preapproval
[PASS] test_invalidatePreapprovedOrder() (gas: 57727)
[PASS] test_preapproveOrder_thenMatch() (gas: 626210)
[PASS] test_revert_preapproveOrder_notOperator() (gas: 33063)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 36.60ms (12.10ms CPU time)

Ran 7 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_complementaryTakerSell_Migratable
[PASS] test_complementaryTakerSell_withFees() (gas: 653536)
[PASS] test_complementaryTakerSell_zeroFee() (gas: 589092)
[PASS] test_complementaryTakerSell_zeroMakerFee() (gas: 632285)
[PASS] test_revert_complementaryTakerSell_feePathFillExceedsTakerFill() (gas: 560277)
[PASS] test_revert_complementaryTakerSell_makerFeeOnlyNotCrossing() (gas: 486017)
[PASS] test_revert_complementaryTakerSell_zeroFeeFillExceedsTakerFill() (gas: 558608)
[PASS] test_revert_complementaryTakerSell_zeroFeeNotCrossing() (gas: 486073)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 73.34ms (50.26ms CPU time)

Ran 3 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_removeAdminAndOperator
[PASS] test_removeAdmin() (gas: 38368)
[PASS] test_revert_removeAdmin_unauthorized() (gas: 48865)
[PASS] test_revert_removeOperator_unauthorized() (gas: 50839)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 36.98ms (15.87ms CPU time)

Ran 1 test for src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_partialFill_Migratable
[PASS] test_partialFill() (gas: 575576)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 31.71ms (12.33ms CPU time)

Ran 2 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_userPauseEvents_Migratable
[PASS] test_pauseUser_emitsEvent() (gas: 45908)
[PASS] test_unpauseUser_emitsEvent() (gas: 33296)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 40.37ms (17.21ms CPU time)

Ran 13 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_bridge
[PASS] test_burnFromBridge() (gas: 125259)
[PASS] test_burnFromBridge_batch() (gas: 181120)
[PASS] test_conditionCount() (gas: 16761)
[PASS] test_conditionCount_invalidEventId() (gas: 12270)
[PASS] test_directSyntheticNoDoesNotEmitSyntheticYesDerivableEarly() (gas: 255790)
[PASS] test_mintFromBridge() (gas: 114932)
```

```
[PASS] test_mintFromBridge_thenRedeem() (gas: 330015)
[PASS] test_mintFromBridge_zeroAmount() (gas: 85992)
[PASS] test_onERC1155BatchReceived_normalTransfer() (gas: 175108)
[PASS] test_onERC1155Received_normalTransfer() (gas: 128708)
[PASS] test_reportResult_bridgeDoesNotRequireEventBridge() (gas: 139812)
[PASS] test_reportResult_bridgeIdempotent() (gas: 145640)
[PASS] test_revert_mintFromBridge_unauthorized() (gas: 68705)
Suite result: ok. 13 passed; 0 failed; 0 skipped; finished in 62.82ms (55.77ms CPU time)
```

```
Ran 2 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_renounceOperatorRole
[PASS] test_renounceOperatorRole() (gas: 22094)
[PASS] test_revert_renounceOperatorRole_notOperator() (gas: 18080)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 35.08ms (14.78ms CPU time)
```

```
Ran 5 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_complementValidation
[PASS] test_revert_invalidComplement_sameOutcome() (gas: 377442)
[PASS] test_revert_mismatchedTokenIds_batchSellBuyMaker() (gas: 487645)
[PASS] test_revert_notCrossing_batchBuySellMaker() (gas: 468316)
[PASS] test_revert_notCrossing_mergePath() (gas: 586371)
[PASS] test_revert_notCrossing_mintPath() (gas: 379884)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 59.96ms (45.31ms CPU time)
```

```
Ran 4 tests for src/exchange/test/Preapproved.t.sol:PreapprovedTest_reverts
[PASS] test_revert_invalidSignature() (gas: 40848)
[PASS] test_revert_invalidSignatureNotPreapproved() (gas: 465455)
[PASS] test_revert_invalidatedPreapproval() (gas: 511810)
[PASS] test_revert_notOperator() (gas: 30967)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 54.21ms (29.98ms CPU time)
```

```
Ran 6 tests for src/auth/test/Roles.t.sol:RolesTest_admin
[PASS] test_addAdmin() (gas: 39283)
[PASS] test_adminOnlyFunction() (gas: 13196)
[PASS] test_removeAdmin() (gas: 35412)
[PASS] test_revert_addAdmin_unauthorized() (gas: 13672)
[PASS] test_revert_adminOnlyFunction_unauthorized() (gas: 13401)
[PASS] test_revert_removeAdmin_unauthorized() (gas: 43644)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 4.62ms (999.04µs CPU time)
```

```
Ran 6 tests for src/auth/test/Roles.t.sol:RolesTest_bridge
[PASS] test_addBridge() (gas: 39336)
[PASS] test_bridgeOnlyFunction() (gas: 13127)
[PASS] test_removeBridge() (gas: 22023)
[PASS] test_revert_addBridge_unauthorized() (gas: 13774)
[PASS] test_revert_bridgeOnlyFunction_unauthorized() (gas: 13399)
[PASS] test_revert_removeBridge_unauthorized() (gas: 15888)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 5.33ms (1.87ms CPU time)
```

```
Ran 3 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:ExchangeAdminTest_validateOrder_Migratable
[PASS] test_revert_validateOrder_invalidSignature() (gas: 33952)
[PASS] test_revert_validateOrder_userPaused() (gas: 63974)
[PASS] test_validateOrder_validOrder() (gas: 40657)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 28.71ms (12.75ms CPU time)
```

```
Ran 4 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_proxyAndSafeSignatures_Migratable
[PASS] test_polyGnosisSafeSignature() (gas: 654321)
[PASS] test_polyProxySignature() (gas: 624844)
[PASS] test_revert_polyGnosisSafeSignature_unrelatedMaker() (gas: 130559)
[PASS] test_revert_polyProxySignature_unrelatedMaker() (gas: 93777)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 46.87ms (33.10ms CPU time)
```

```
Ran 2 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_roleViews
[PASS] test_isAdmin() (gas: 21873)
[PASS] test_isOperator() (gas: 30636)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 34.40ms (6.33ms CPU time)
```

```
Ran 4 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_convert
[PASS] test_convert() (gas: 397688)
[PASS] test_convert_emitsPositionConverted() (gas: 399419)
[PASS] test_revert_invalidConditionIndex() (gas: 24317)
[PASS] test_revert_invalidEventId() (gas: 19137)
```

Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 45.97ms (10.35ms CPU time)

Ran 6 tests for src/auth/test/Roles.t.sol:RolesTest_creator

[PASS] test_addCreator() (gas: 39315)

[PASS] test_creatorOnlyFunction() (gas: 13125)

[PASS] test_removeCreator() (gas: 22108)

[PASS] test_revert_addCreator_unauthorized() (gas: 13751)

[PASS] test_revert_creatorOnlyFunction_unauthorized() (gas: 13398)

[PASS] test_revert_removeCreator_unauthorized() (gas: 15910)

Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 25.33ms (9.18ms CPU time)

Ran 6 tests for src/auth/test/Roles.t.sol:RolesTest_operator

[PASS] test_addOperator() (gas: 39380)

[PASS] test_operatorOnlyFunction() (gas: 13216)

[PASS] test_removeOperator() (gas: 22042)

[PASS] test_revert_addOperator_unauthorized() (gas: 13774)

[PASS] test_revert_operatorOnlyFunction_unauthorized() (gas: 13401)

[PASS] test_revert_removeOperator_unauthorized() (gas: 15888)

Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 5.47ms (3.63ms CPU time)

Ran 1 test for src/auth/test/Roles.t.sol:RolesTest_resolver

[PASS] test_addRemoveResolver() (gas: 35415)

Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 1.81ms (855.71µs CPU time)

Ran 8 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_complementary_Migratable

[PASS] test_complementary() (gas: 602380)

[PASS] test_complementaryConsumesLimitFill_whenPriceImproves() (gas: 592782)

[PASS] test_complementaryFees() (gas: 602154)

[PASS] test_matchOrdersAndPrepareCombinatorial() (gas: 594412)

[PASS] test_noExchangeBalance() (gas: 595336)

[PASS] test_revert_complementaryFillExceedsTakerFill_overspend() (gas: 542624)

[PASS] test_revert_matchOrdersAndPrepareCombinatorial_notOperator() (gas: 24150)

[PASS] test_revert_matchOrdersAndPrepareCombinatorial_paused() (gas: 49457)

Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 91.00ms (73.73ms CPU time)

Ran 2 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_setup

[PASS] test_initialState() (gas: 45039)

[PASS] test_initializerDefaults() (gas: 5756178)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 26.15ms (12.20ms CPU time)

Ran 3 tests for src/routers/test/RouterNegRiskModule.t.sol:RouterNegRiskModule_Test_events

[PASS] test_convert_emitsPositionConverted() (gas: 7902126)

[PASS] test_horizontalMerge_emitsHorizontalMerge() (gas: 10805663)

[PASS] test_horizontalSplit_emitsHorizontalSplit() (gas: 6412856)

Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 53.95ms (36.19ms CPU time)

Ran 2 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_tradingPausedEvents

[PASS] test_pauseTrading_emitsEvent() (gas: 43268)

[PASS] test_unpauseTrading_emitsEvent() (gas: 33112)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 25.24ms (7.36ms CPU time)

Ran 5 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_directHorizontal

[PASS] test_horizontalMerge_direct() (gas: 467320)

[PASS] test_horizontalSplit_direct() (gas: 279015)

[PASS] test_revert_convert_noPreTransfer() (gas: 119764)

[PASS] test_revert_horizontalMerge_noPreTransfer() (gas: 100566)

[PASS] test_revert_horizontalSplit_noPreTransfer() (gas: 148197)

Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 50.90ms (31.79ms CPU time)

Ran 2 tests for src/routers/test/Router.t.sol:Router_Test_authorizeUpgrade

[PASS] test_revert_upgrade_unauthorized() (gas: 4233625)

[PASS] test_upgrade_ownerSucceeds() (gas: 34705132)

Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 42.04ms (23.79ms CPU time)

Ran 8 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_complementary

[PASS] test_complementary() (gas: 599980)

[PASS] test_complementaryConsumesLimitFill_whenPriceImproves() (gas: 590416)

[PASS] test_complementaryFees() (gas: 599824)

[PASS] test_matchOrdersAndPrepareCombinatorial() (gas: 592064)

[PASS] test_noExchangeBalance() (gas: 592972)

[PASS] test_revert_complementaryFillExceedsTakerFill_overspend() (gas: 540672)

[PASS] test_revert_matchOrdersAndPrepareCombinatorial_notOperator() (gas: 24128)


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[PASS] test_revert_matchOrdersAndPrepareCombinatorial_paused() (gas: 49457)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 89.06ms (58.76ms CPU time)

Ran 10 tests for src/modules/test/CombinatorialModule.t.sol:CombinatorialModuleTest_pathIndependence
[PASS] testFuzz_extractPayoutProperty(uint96,uint256,uint256,uint256) (runs: 256, μ: 665170, ~: 667863)
[PASS] testFuzz_yesBasketPayoutProperty(uint96,uint256,uint256,uint256) (runs: 256, μ: 751300, ~: 753925)
[PASS] test_collateralReturnViaPartialPeel() (gas: 1309962)
[PASS] test_convertToYesBasket_matches_repeatedExtract() (gas: 866701)
[PASS] test_extractInjectRoundtrip() (gas: 577314)
[PASS] test_extractPayoutIdentity_boolean() (gas: 587279)
[PASS] test_extractPayoutWeakInequality_fractional() (gas: 655551)
[PASS] test_splitMergeValuePreservation() (gas: 671553)
[PASS] test_yesBasketPayoutIdentity_boolean() (gas: 666120)
[PASS] test_yesBasketPayoutWeakInequality_fractional() (gas: 741520)
Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 879.63ms (892.92ms CPU time)

Ran 2 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_getEventId
[PASS] test_getEventId() (gas: 15652)
[PASS] test_revert_invalidConditionCount() (gas: 31361)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 19.34ms (1.81ms CPU time)

Ran 14 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_reverts_Migratable
[PASS] test_revert_invalidComplement_makerWrongCondition() (gas: 390783)
[PASS] test_revert_invalidSignature() (gas: 478251)
[PASS] test_revert_invalidSignature_eoaSignerMakerMismatch() (gas: 463300)
[PASS] test_revert_invalidTokenId_takerNonBinaryPosition() (gas: 455821)
[PASS] test_revert_makingGtRemaining() (gas: 464072)
[PASS] test_revert_mismatchedArrayLengths() (gas: 456357)
[PASS] test_revert_noMakerOrders() (gas: 211640)
[PASS] test_revert_notOperator() (gas: 38071)
[PASS] test_revert_orderAlreadyFilled() (gas: 577865)
[PASS] test_revert_paused() (gas: 475685)
[PASS] test_revert_takerReceiveAmountMismatch() (gas: 542329)
[PASS] test_revert_userIsPaused() (gas: 485573)
[PASS] test_revert_zeroMakerAmount() (gas: 77472)
[PASS] test_revert_zeroMakerAmount_taker() (gas: 351790)
Suite result: ok. 14 passed; 0 failed; 0 skipped; finished in 97.62ms (70.95ms CPU time)

Ran 2 tests for src/routers/test/RouterNegRiskModule.t.sol:RouterNegRiskModule_Test_horizontal
[PASS] test_horizontalMerge() (gas: 17441784)
[PASS] test_horizontalSplit() (gas: 8371253)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 48.20ms (36.62ms CPU time)

Ran 4 tests for src/routers/test/Router.t.sol:Router_Test_initialize
[PASS] test_initialize_setsOwner() (gas: 14972)
[PASS] test_revert_initialize_onImplementation() (gas: 4221177)
[PASS] test_revert_initialize_twice() (gas: 21160)
[PASS] test_revert_initialize_zeroOwner() (gas: 4308126)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 33.73ms (4.88ms CPU time)

Ran 118 tests for src/bridge/test/CcipBridgeUnit.t.sol:CcipBridgeUnitTest
[PASS] testFuzz_revert_bridge_bridgeCollateral_dirtyRecipient(uint256) (runs: 256, μ: 27148, ~: 27148)
[PASS] testFuzz_revert_bridge_bridgePositions_dirtyRecipient(uint256) (runs: 256, μ: 31084, ~: 31084)
[PASS] test_CcipBridge_addAdmin() (gas: 21529)
[PASS] test_CcipBridge_adminCanPause() (gas: 55211)
[PASS] test_CcipBridge_adminCanUnpause() (gas: 46168)
[PASS] test_CcipBridge_initialize_setsOwnerAndAdmin() (gas: 22750)
[PASS] test_CcipBridge_quote_gasLimitOptions() (gas: 31215)
[PASS] test_CcipBridge_receive_rawBytes32Peer() (gas: 134311)
[PASS] test_CcipBridge_removeAdmin() (gas: 23217)
[PASS] test_CcipBridge_removePeer() (gas: 23898)
[PASS] test_CcipBridge_send_emitsRouterMessageId() (gas: 413431)
[PASS] test_CcipBridge_send_emptyOptions_defaultGasOutOfOrder() (gas: 423634)
[PASS] test_CcipBridge_send_gasLimitOptions_outOfOrder() (gas: 420921)
[PASS] test_CcipBridge_setPeer_afterRemoval() (gas: 27233)
[PASS] test_CcipBridge_setPeer_rawBytes32Peer() (gas: 26934)
[PASS] test_CcipBridge_upgrade_canChangeRouter() (gas: 5786512)
[PASS] test_CcipBridge_upgrade_ownerSucceedsAndPreservesStorage() (gas: 4226935)
[PASS] test_bridge_bridgeCollateral() (gas: 425216)
[PASS] test_bridge_bridgePositions() (gas: 519557)
[PASS] test_bridge_bridgePositions_noConditionPreparationNeeded() (gas: 475686)
[PASS] test_bridge_onERC1155Received() (gas: 13096)
```

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[PASS] test_bridge_pauseBoth() (gas: 44016)
[PASS] test_bridge_pauseChainReceive() (gas: 43143)
[PASS] test_bridge_pauseChainReceive_emitsEvent() (gas: 46315)
[PASS] test_bridge_pauseChainSend() (gas: 43229)
[PASS] test_bridge_pauseChainSend_emitsEvent() (gas: 46338)
[PASS] test_bridge_pauseReceive() (gas: 39626)
[PASS] test_bridge_pauseReceive_emitsEvent() (gas: 42868)
[PASS] test_bridge_pauseSend() (gas: 39502)
[PASS] test_bridge_pauseSend_emitsEvent() (gas: 42772)
[PASS] test_bridge_quoteBridgeCollateral() (gas: 32897)
[PASS] test_bridge_quoteBridgePositions() (gas: 38238)
[PASS] test_bridge_quoteBridgePositions_emptyArray() (gas: 17923)
[PASS] test_bridge_quoteBridgeResult() (gas: 109104)
[PASS] test_bridge_receiveCollateral() (gas: 119790)
[PASS] test_bridge_receiveFromOtherChainStillWorks_whenChainReceivePaused() (gas: 153702)
[PASS] test_bridge_receivePositions_binary() (gas: 119607)
[PASS] test_bridge_receivePositions_negRisk() (gas: 119774)
[PASS] test_bridge_receiveResult() (gas: 153113)
[PASS] test_bridge_receiveResumesAfterChainUnpause() (gas: 135457)
[PASS] test_bridge_receiveStillWorks_whenSendPaused() (gas: 148476)
[PASS] test_bridge_resumeAfterUnpause() (gas: 137163)
[PASS] test_bridge_sendResumesAfterChainUnpause() (gas: 434597)
[PASS] test_bridge_sendStillWorks_whenReceivePaused() (gas: 449515)
[PASS] test_bridge_sendToOtherChainStillWorks_whenChainSendPaused() (gas: 452710)
[PASS] test_bridge_setBatchModuleSupported() (gas: 124193)
[PASS] test_bridge_setModuleSupported() (gas: 61770)
[PASS] test_bridge_unpauseChainReceive() (gas: 33356)
[PASS] test_bridge_unpauseChainReceive_emitsEvent() (gas: 35962)
[PASS] test_bridge_unpauseChainSend() (gas: 33515)
[PASS] test_bridge_unpauseChainSend_emitsEvent() (gas: 36016)
[PASS] test_bridge_unpauseReceive() (gas: 29858)
[PASS] test_bridge_unpauseReceive_emitsEvent() (gas: 32432)
[PASS] test_bridge_unpauseSend() (gas: 29631)
[PASS] test_bridge_unpauseSend_emitsEvent() (gas: 32297)
[PASS] test_revert_CcipBridge_addAdmin_nonOwner() (gas: 20749)
[PASS] test_revert_CcipBridge_adminCannotSetModuleSupported() (gas: 29037)
[PASS] test_revert_CcipBridge_adminCannotSetPeer() (gas: 24446)
[PASS] test_revert_CcipBridge_bridgeCollateral_afterPeerRemoved() (gas: 96566)
[PASS] test_revert_CcipBridge_bridgeCollateral_peerNotSet() (gas: 91115)
[PASS] test_revert_CcipBridge_initialize_onImplementation() (gas: 4162264)
[PASS] test_revert_CcipBridge_initialize_twice() (gas: 23755)
[PASS] test_revert_CcipBridge_initialize_zeroAdmin() (gas: 4249252)
[PASS] test_revert_CcipBridge_initialize_zeroOwner() (gas: 4249241)
[PASS] test_revert_CcipBridge_pauseChainReceive_invalidChainSelector() (gas: 18289)
[PASS] test_revert_CcipBridge_pauseChainSend_invalidChainSelector() (gas: 18354)
[PASS] test_revert_CcipBridge_quoteBridgeCollateral_afterPeerRemoved() (gas: 26213)
[PASS] test_revert_CcipBridge_quoteBridgeCollateral_peerNotSet() (gas: 20806)
[PASS] test_revert_CcipBridge_quote_invalidOptions() (gas: 20881)
[PASS] test_revert_CcipBridge_receive_afterPeerRemoved() (gas: 57692)
[PASS] test_revert_CcipBridge_receive_dirtyUpperBitsSender() (gas: 54638)
[PASS] test_revert_CcipBridge_removeAdmin_nonOwner() (gas: 26370)
[PASS] test_revert_CcipBridge_removePeer_adminCannotRemove() (gas: 24070)
[PASS] test_revert_CcipBridge_removePeer_invalidChainSelector() (gas: 16158)
[PASS] test_revert_CcipBridge_removePeer_nonOwner() (gas: 18430)
[PASS] test_revert_CcipBridge_send_invalidOptions() (gas: 91972)
[PASS] test_revert_CcipBridge_setBatchModuleSupported_invalidChainSelector() (gas: 63868)
[PASS] test_revert_CcipBridge_setModuleSupported_invalidChainSelector() (gas: 19004)
[PASS] test_revert_CcipBridge_setPeer_invalidChainSelector() (gas: 16443)
[PASS] test_revert_CcipBridge_setPeer_zeroPeer() (gas: 16208)
[PASS] test_revert_CcipBridge_upgrade_unauthorized() (gas: 4172345)
[PASS] test_revert_bridge_bridgeCollateral_dirtyRecipient() (gas: 28606)
[PASS] test_revert_bridge_bridgeCollateral_whenChainSendPaused() (gas: 148467)
[PASS] test_revert_bridge_bridgeCollateral_whenSendPaused() (gas: 145194)
[PASS] test_revert_bridge_bridgeCollateral_zeroRecipient() (gas: 26398)
[PASS] test_revert_bridge_bridgePositions_combinatorialNotSupported() (gas: 6379969)
[PASS] test_revert_bridge_bridgePositions_differentModules() (gas: 152507)
[PASS] test_revert_bridge_bridgePositions_dirtyRecipient() (gas: 32544)
[PASS] test_revert_bridge_bridgePositions_emptyArray() (gas: 29464)
[PASS] test_revert_bridge_bridgePositions_lengthMismatch() (gas: 34975)
[PASS] test_revert_bridge_bridgePositions_routeNotSupported() (gas: 129818)
[PASS] test_revert_bridge_bridgePositions_whenSendPaused() (gas: 202771)
[PASS] test_revert_bridge_bridgePositions_zeroRecipient() (gas: 30270)
```

```
[PASS] test_revert_bridge_bridgeResult_moduleNotConfigured() (gas: 34485)
[PASS] test_revert_bridge_bridgeResult_resultNotReported() (gas: 50709)
[PASS] test_revert_bridge_bridgeResult_routeNotSupported() (gas: 142262)
[PASS] test_revert_bridge_bridgeResult_whenSendPaused() (gas: 149273)
[PASS] test_revert_bridge_pauseChainReceive_nonOwner() (gas: 22597)
[PASS] test_revert_bridge_pauseChainSend_nonOwner() (gas: 22683)
[PASS] test_revert_bridge_pauseReceive_nonOwner() (gas: 20172)
[PASS] test_revert_bridge_pauseSend_nonOwner() (gas: 20062)
[PASS] test_revert_bridge_quoteBridgePositions_lengthMismatch() (gas: 25963)
[PASS] test_revert_bridge_receiveEmptyMessage() (gas: 56251)
[PASS] test_revert_bridge_receiveFromUnauthorizedSender() (gas: 54522)
[PASS] test_revert_bridge_receivePositions_combinatorialNotSupported() (gas: 6370464)
[PASS] test_revert_bridge_receivePositions_differentModules() (gas: 120556)
[PASS] test_revert_bridge_receivePositions_moduleNotConfigured() (gas: 81594)
[PASS] test_revert_bridge_receivePositions_unknownModuleType() (gas: 62650)
[PASS] test_revert_bridge_receiveResult_moduleNotConfigured() (gas: 70544)
[PASS] test_revert_bridge_receiveTypeByteOnly() (gas: 56019)
[PASS] test_revert_bridge_receive_whenChainReceivePaused() (gas: 90449)
[PASS] test_revert_bridge_receive_whenReceivePaused() (gas: 85292)
[PASS] test_revert_bridge_setModuleSupported_invalidModule() (gas: 5694121)
[PASS] test_revert_bridge_setPeer_nonOwner() (gas: 18766)
[PASS] test_revert_bridge_unpauseChainReceive_nonOwner() (gas: 50380)
[PASS] test_revert_bridge_unpauseChainSend_nonOwner() (gas: 50515)
[PASS] test_revert_bridge_unpauseReceive_nonOwner() (gas: 46967)
[PASS] test_revert_bridge_unpauseSend_nonOwner() (gas: 46807)
Suite result: ok. 118 passed; 0 failed; 0 skipped; finished in 495.85ms (455.97ms CPU time)
```

```
Ran 1 test for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_complementaryNotCrossing
[PASS] test_revert_complementaryNotCrossing() (gas: 468068)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 43.58ms (3.90ms CPU time)
```

```
Ran 2 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_upgrade
[PASS] test_revert_upgrade_unauthorized() (gas: 5601346)
[PASS] test_upgradeToAndCall() (gas: 5605695)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 55.92ms (42.81ms CPU time)
```

```
Ran 2 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_validateOrderBranches_Migratable
[PASS] test_revert_validateOrder_alreadyFilled() (gas: 569912)
[PASS] test_revert_validateOrder_userPaused() (gas: 63974)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 28.91ms (8.61ms CPU time)
```

```
Ran 5 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_getPayout
[PASS] test_fullPayout() (gas: 163344)
[PASS] test_revert_conditionNotResolved() (gas: 49137)
[PASS] test_revert_invalidOutcomeIndex() (gas: 158318)
[PASS] test_revert_partialPayout() (gas: 126191)
[PASS] test_virtualNoPayout_afterOneYes() (gas: 154038)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 39.81ms (7.14ms CPU time)
```

```
Ran 3 tests for src/routers/test/RouterBinaryModule.t.sol:RouterBinaryModuleTest_events
[PASS] test_merge_emitsPositionsMerged() (gas: 431572)
[PASS] test_redeem_emitsPositionRedeemed() (gas: 462576)
[PASS] test_split_emitsPositionSplit() (gas: 287947)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 24.79ms (10.62ms CPU time)
```

```
Ran 2 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_hasResult
[PASS] test_hasResult_false() (gas: 45703)
[PASS] test_hasResult_true() (gas: 158331)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 20.48ms (4.79ms CPU time)
```

```
Ran 2 tests for src/routers/test/RouterBinaryModule.t.sol:RouterBinaryModuleTest_merge
[PASS] test_merge() (gas: 434788)
[PASS] test_merge_anyCondition() (gas: 215184)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 20.87ms (10.92ms CPU time)
```

```
Ran 3 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_userPauseBlockInterval
[PASS] test_revert_exceedsMaxPauseInterval() (gas: 18618)
[PASS] test_setUserPauseBlockInterval() (gas: 29092)
[PASS] test_setUserPauseBlockInterval_maxAllowed() (gas: 26163)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 30.35ms (9.40ms CPU time)
```



```
Ran 11 tests for
src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_feeValidation_Migratable
[PASS] test_maxFeeRateZero_disablesEnforcement() (gas: 18312507)
[PASS] test_revert_complementaryTakerFeeValidatedAgainstActualFill() (gas: 18275959)
[PASS] test_revert_feeExceedsMaxRate_buyOrder() (gas: 18276618)
[PASS] test_revert_feeExceedsMaxRate_sellOrder() (gas: 18303094)
[PASS] test_revert_feeExceedsProceeds_batchSell() (gas: 18412898)
[PASS] test_revert_feeExceedsProceeds_complementarySell() (gas: 18277959)
[PASS] test_revert_makerFeeExceedsMaxRate() (gas: 18198490)
[PASS] test_revert_marketBuyTakerFeeExceedsMaxRate() (gas: 18276575)
[PASS] test_revert_marketSellTakerFeeExceedsMaxRate() (gas: 18303116)
[PASS] test_takerSellSurplusFeeValidationUsesActualProceeds() (gas: 635324)
[PASS] test_validateFee_externalView() (gas: 11672)
Suite result: ok. 11 passed; 0 failed; 0 skipped; finished in 122.69ms (106.85ms CPU time)

Ran 1 test for src/exchange/test/MigratableExchangeMatrix.t.sol:PreapprovedTest_erc1271_Migratable
[PASS] test_signerInvalidated() (gas: 842837)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 30.96ms (5.63ms CPU time)

Ran 3 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_combinatorialModule
[PASS] test_implementationStoredInSlot() (gas: 7866)
[PASS] test_ownerStoredInSlot() (gas: 7939)
[PASS] test_roleStoredInSlot() (gas: 7938)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 61.75ms (20.92ms CPU time)

Ran 2 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_complementarySellNotCrossing
[PASS] test_revert_complementarySellNotCrossing_withFees() (gas: 484915)
[PASS] test_revert_complementarySellNotCrossing_zeroFee() (gas: 485030)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 46.80ms (18.08ms CPU time)

Ran 4 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_merge
[PASS] test_merge() (gas: 779826)
[PASS] test_mergeFees() (gas: 807721)
[PASS] test_revert_mergeCannotConsumeMoreThanTakerFill() (gas: 636055)
[PASS] test_revert_mixedBatchSellCannotConsumeMoreThanTakerFill() (gas: 808686)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 42.00ms (17.85ms CPU time)

Ran 2 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_horizontalMerge
[PASS] test_horizontalMerge() (gas: 532770)
[PASS] test_revert_invalidEventId() (gas: 18724)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 31.73ms (14.17ms CPU time)

Ran 2 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_userPauseEvents
[PASS] test_pauseUser_emitsEvent() (gas: 45974)
[PASS] test_unpauseUser_emitsEvent() (gas: 33279)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 29.72ms (8.84ms CPU time)

Ran 3 tests for src/exchange/test/ExchangeAdmin.t.sol:ExchangeAdminTest_validateOrder
[PASS] test_revert_validateOrder_invalidSignature() (gas: 32635)
[PASS] test_revert_validateOrder_userPaused() (gas: 63939)
[PASS] test_validateOrder_validOrder() (gas: 39296)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 28.74ms (13.29ms CPU time)

Ran 3 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_ctfRouter
[PASS] test_implementationStoredInSlot() (gas: 7866)
[PASS] test_ownerStoredInSlot() (gas: 7939)
[PASS] test_roleStoredInSlot() (gas: 45571)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 32.51ms (23.54ms CPU time)

Ran 4 tests for src/exchange/test/MigratableExchangeMatrix.t.sol:MatchOrdersTest_merge_Migratable
[PASS] test_merge() (gas: 782124)
[PASS] test_mergeFees() (gas: 810019)
[PASS] test_revert_mergeCannotConsumeMoreThanTakerFill() (gas: 638023)
[PASS] test_revert_mixedBatchSellCannotConsumeMoreThanTakerFill() (gas: 811631)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 48.85ms (34.80ms CPU time)

Ran 2 tests for src/modules/test/NegRiskModule.t.sol:NegRiskModuleTest_horizontalSplit
[PASS] test_horizontalSplit() (gas: 357740)
[PASS] test_revert_invalidEventId() (gas: 233199)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 21.53ms (7.70ms CPU time)

Ran 4 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_oracleAggregator
```

```
[PASS] test_globalPausedStoredInSlot() (gas: 67365)
[PASS] test_implementationStoredInSlot() (gas: 7888)
[PASS] test_ownerStoredInSlot() (gas: 7961)
[PASS] test_roleStoredInSlot() (gas: 45173)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 44.26ms (22.68ms CPU time)
```

```
Ran 4 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_mint
[PASS] test_mint() (gas: 572017)
[PASS] test_mintConsumesLimitFill_whenFundedAtExecutionPrice() (gas: 577516)
[PASS] test_mintFees() (gas: 599528)
[PASS] test_mintRefundedTakerEventReportsNetFill() (gas: 613756)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 47.21ms (30.93ms CPU time)
```

```
Ran 7 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_complementaryTakerSell
[PASS] test_complementaryTakerSell_withFees() (gas: 651188)
[PASS] test_complementaryTakerSell_zeroFee() (gas: 586778)
[PASS] test_complementaryTakerSell_zeroMakerFee() (gas: 629972)
[PASS] test_revert_complementaryTakerSell_feePathFillExceedsTakerFill() (gas: 558293)
[PASS] test_revert_complementaryTakerSell_makerFeeOnlyNotCrossing() (gas: 485024)
[PASS] test_revert_complementaryTakerSell_zeroFeeFillExceedsTakerFill() (gas: 556624)
[PASS] test_revert_complementaryTakerSell_zeroFeeNotCrossing() (gas: 485080)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 58.34ms (40.22ms CPU time)
```

```
Ran 3 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_exchange
[PASS] test_implementationStoredInSlot() (gas: 7866)
[PASS] test_ownerStoredInSlot() (gas: 7939)
[PASS] test_roleStoredInSlot() (gas: 7938)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 36.99ms (12.47ms CPU time)
```

```
Ran 5 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_soladySharedSlots
[PASS] test_erc1967ImplementationSlot() (gas: 457)
[PASS] test_handoverSlotSeed() (gas: 3117065)
[PASS] test_initializableSlot() (gas: 304)
[PASS] test_ownerSlot() (gas: 327)
[PASS] test_roleSlotSeed() (gas: 276)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 44.80ms (24.59ms CPU time)
```

```
Ran 1 test for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_partialFill
[PASS] test_partialFill() (gas: 573229)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 25.50ms (5.97ms CPU time)
```

```
Ran 6 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_positionManager
[PASS] test_erc1155ApprovalStoredInSlot() (gas: 43732)
[PASS] test_erc1155BalanceStoredInSlot() (gas: 54764)
[PASS] test_handoverStoredInSlot() (gas: 39906)
[PASS] test_implementationStoredInSlot() (gas: 7888)
[PASS] test_ownerStoredInSlot() (gas: 7961)
[PASS] test_roleStoredInSlot() (gas: 7983)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 43.86ms (23.84ms CPU time)
```

```
Ran 4 tests for src/exchange/test/ExchangeSnapshots.t.sol:ExchangeSnapshotsTest_complementary
[PASS] test_fiveMakers() (gas: 1658804)
[PASS] test_oneMaker() (gas: 564631)
[PASS] test_tenMakers() (gas: 3028582)
[PASS] test_twentyMakers() (gas: 5769742)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 66.52ms (53.04ms CPU time)
```

```
Ran 9 tests for src/positionManager/test/UnsafeTransfer.t.sol:UnsafeTransferTest_unsafeBatchTransferFrom
[PASS] test_approved() (gas: 118550)
[PASS] test_basic() (gas: 94786)
[PASS] test_emitsTransferBatch() (gas: 92171)
[PASS] test_emptyArrays() (gas: 23052)
[PASS] test_revert_insufficientBalance() (gas: 25659)
[PASS] test_revert_lengthMismatch() (gas: 25840)
[PASS] test_revert_notApproved() (gas: 25570)
[PASS] test_revert_toZeroAddress() (gas: 21267)
[PASS] test_toNonReceiver() (gas: 98759)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 8.08ms (5.19ms CPU time)
```

```
Ran 3 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_negRiskModule
[PASS] test_implementationStoredInSlot() (gas: 7866)
[PASS] test_ownerStoredInSlot() (gas: 7939)
```

```
[PASS] test_roleStoredInSlot() (gas: 7938)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 27.53ms (8.56ms CPU time)

Ran 9 tests for src/positionManager/test/UnsafeTransfer.t.sol:UnsafeTransferTest_unsafeTransferFrom
[PASS] test_approved() (gas: 85100)
[PASS] test_basic() (gas: 56292)
[PASS] test_emitsTransferSingle() (gas: 55391)
[PASS] test_revert_insufficientBalance() (gas: 24223)
[PASS] test_revert_notApproved() (gas: 24207)
[PASS] test_revert_toZeroAddress() (gas: 19926)
[PASS] test_selfTransfer() (gas: 27081)
[PASS] test_toNonReceiver() (gas: 96102)
[PASS] test_zeroAmount() (gas: 33570)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 18.58ms (14.84ms CPU time)

Ran 2 tests for src/libraries/test/Ids.t.sol:ConditionIdLibTest_isValidEventId
[PASS] test_isValidEventId_falseForSubcondition() (gas: 1818)
[PASS] test_isValidEventId_trueForEvent() (gas: 1830)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 1.54ms (182.63µs CPU time)

Ran 3 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_router
[PASS] test_implementationStoredInSlot() (gas: 7866)
[PASS] test_ownerStoredInSlot() (gas: 7939)
[PASS] test_roleStoredInSlot() (gas: 45549)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 31.50ms (4.68ms CPU time)

Ran 4 tests for src/exchange/test/ExchangeSnapshots.t.sol:ExchangeSnapshotsTest_mint
[PASS] test_fiveMakers() (gas: 1283944)
[PASS] test_oneMaker() (gas: 562213)
[PASS] test_tenMakers() (gas: 2188167)
[PASS] test_twentyMakers() (gas: 3998216)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 76.21ms (61.95ms CPU time)

Ran 1 test for src/libraries/test/Ids.t.sol:ConditionIdLibTest_positionId
[PASS] test_conditionToPosition(bytes32,uint256) (runs: 256, µ: 1159, ~: 1159)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 13.00ms (11.34ms CPU time)

Ran 4 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_proxyAndSafeSignatures
[PASS] test_polyGnosisSafeSignature() (gas: 652024)
[PASS] test_polyProxySignature() (gas: 622531)
[PASS] test_revert_polyGnosisSafeSignature_unrelatedMaker() (gas: 130145)
[PASS] test_revert_polyProxySignature_unrelatedMaker() (gas: 93363)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 57.43ms (19.61ms CPU time)

Ran 11 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_feeValidation
[PASS] test_maxFeeRateZero_disablesEnforcement() (gas: 6325250)
[PASS] test_revert_complementaryTakerFeeValidatedAgainstActualFill() (gas: 6289154)
[PASS] test_revert_feeExceedsMaxRate_buyOrder() (gas: 6289773)
[PASS] test_revert_feeExceedsMaxRate_sellOrder() (gas: 6316229)
[PASS] test_revert_feeExceedsProceeds_batchSell() (gas: 6426053)
[PASS] test_revert_feeExceedsProceeds_complementarySell() (gas: 6291094)
[PASS] test_revert_makerFeeExceedsMaxRate() (gas: 6211665)
[PASS] test_revert_marketBuyTakerFeeExceedsMaxRate() (gas: 6289770)
[PASS] test_revert_marketSellTakerFeeExceedsMaxRate() (gas: 6316271)
[PASS] test_takerSellSurplusFeeValidationUsesActualProceeds() (gas: 633026)
[PASS] test_validateFee_externalView() (gas: 11694)
Suite result: ok. 11 passed; 0 failed; 0 skipped; finished in 83.03ms (60.91ms CPU time)

Ran 4 tests for src/routers/test/RouterSnapshots.t.sol:RouterSnapshots_Test_bridge
[PASS] test_bridgeCollateral() (gas: 493674)
[PASS] test_bridgeCollateralTo() (gas: 495598)
[PASS] test_bridgePositions() (gas: 791015)
[PASS] test_bridgePositionsTo() (gas: 792904)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 23.80ms (9.80ms CPU time)

Ran 1 test for
src/exchange/test/MatchOrdersBalanceFlows.t.sol:MatchOrdersBalanceFlowsTest_takerPositionRefund
[PASS] test_batchSellTakerExcessPositionsRefunded() (gas: 601951)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 40.06ms (17.12ms CPU time)

Ran 3 tests for src/exchange/test/ExchangeSnapshots.t.sol:ExchangeSnapshotsTest_complementarySell
[PASS] test_oneMaker() (gas: 580212)
```



```
[PASS] test_tenMakers() (gas: 2184186)
[PASS] test_twentyMakers() (gas: 3969970)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 76.27ms (40.80ms CPU time)

Ran 4 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_ooReporterModule
[PASS] test_aggregatorStoredInSlotZero() (gas: 7865)
[PASS] test_implementationStoredInSlot() (gas: 7888)
[PASS] test_ownerStoredInSlot() (gas: 7961)
[PASS] test_roleStoredInSlot() (gas: 7938)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 76.90ms (35.27ms CPU time)

Ran 3 tests for src/routers/test/RouterSnapshots.t.sol:RouterSnapshots_Test_negRisk
[PASS] test_convert() (gas: 7290577)
[PASS] test_horizontalMerge_256() (gas: 10808022)
[PASS] test_horizontalSplit_256() (gas: 6417462)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 74.23ms (58.46ms CPU time)

Ran 2 tests for src/exchange/test/ExchangeSnapshots.t.sol:ExchangeSnapshotsTest_comboComplementaryMerge
[PASS] test_tenMakers() (gas: 2765614)
[PASS] test_twentyMakers() (gas: 5053253)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 84.02ms (47.57ms CPU time)

Ran 3 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_bridgeRouter
[PASS] test_implementationStoredInSlot() (gas: 7866)
[PASS] test_ownerStoredInSlot() (gas: 7939)
[PASS] test_roleStoredInSlot() (gas: 45527)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 131.33ms (50.30ms CPU time)

Ran 12 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_consumption
[PASS] test_matchOrders_partiallyFilledLegacyOrder_resumesFromLeftover() (gas: 917856)
[PASS] test_matchOrders_partiallyFilledLegacyTaker_resumesFromLeftover() (gas: 937321)
[PASS] test_nativeOrder_ignoresLegacyConsumptionState() (gas: 105845)
[PASS] test_partiallyFilledLegacyOrder_acceptedByValidateOrder() (gas: 168877)
[PASS] test_preapprovedFill_partiallyFilledLegacyOrder_resumesFromLeftover() (gas: 956967)
[PASS] test_revert_matchOrders_makerFilledOnLegacy() (gas: 859448)
[PASS] test_revert_matchOrders_partiallyFilledLegacyOrder_cappedAtLeftover() (gas: 798118)
[PASS] test_revert_matchOrders_partiallyFilledLegacyTaker_cappedAtLeftover() (gas: 786604)
[PASS] test_revert_preapprovedFill_consumedOnLegacy() (gas: 891295)
[PASS] test_revert_preapprovedFill_partiallyFilledLegacyOrder_cappedAtLeftover() (gas: 832100)
[PASS] test_revert_validateOrder_orderFilledOnLegacy() (gas: 206439)
[PASS] test_untouchedLegacyOrder_accepted() (gas: 143738)
Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 131.09ms (112.92ms CPU time)

Ran 12 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_marketOrders
[PASS] test_marketBuy_batchMixedWithRefund() (gas: 907612)
[PASS] test_marketBuy_complementary() (gas: 593861)
[PASS] test_marketBuy_fillsAtAnyPrice() (gas: 593830)
[PASS] test_marketBuy_mint() (gas: 594697)
[PASS] test_marketBuy_partialFill_thenComplete() (gas: 685638)
[PASS] test_marketBuy_withFee() (gas: 620419)
[PASS] test_marketOrderAsMaker_fillsAtZeroTaking() (gas: 578282)
[PASS] test_marketSell_batchMerge() (gas: 980972)
[PASS] test_marketSell_complementary() (gas: 609770)
[PASS] test_marketSell_withFee() (gas: 658921)
[PASS] test_revert_limitBuyTakerAmountOne_makingGtRemaining() (gas: 542320)
[PASS] test_revert_marketSellMakerFee_feeExceedsMaxRate() (gas: 479593)
Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 151.76ms (139.89ms CPU time)

Ran 4 tests for src/exchange/test/ExchangeSnapshots.t.sol:ExchangeSnapshotsTest_merge
[PASS] test_fiveMakers() (gas: 1941111)
[PASS] test_oneMaker() (gas: 773261)
[PASS] test_tenMakers() (gas: 3402981)
[PASS] test_twentyMakers() (gas: 6328389)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 135.63ms (93.75ms CPU time)

Ran 3 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_binaryModule
[PASS] test_implementationStoredInSlot() (gas: 7866)
[PASS] test_ownerStoredInSlot() (gas: 7939)
[PASS] test_roleStoredInSlot() (gas: 7938)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 83.33ms (18.23ms CPU time)

Ran 6 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_ccipBridge
```

```
[PASS] test_implementationStoredInSlot() (gas: 7888)
[PASS] test_moduleSupportedStoredInSlot() (gas: 62438)
[PASS] test_ownerStoredInSlot() (gas: 7961)
[PASS] test_pauseFlagsStoredInSlot() (gas: 45366)
[PASS] test_peerStoredAfterBridgeBaseGap() (gas: 45789)
[PASS] test_roleStoredInSlot() (gas: 7983)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 43.82ms (12.21ms CPU time)
```

```
Ran 14 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_reverts
[PASS] test_revert_invalidComplement_makerWrongCondition() (gas: 389774)
[PASS] test_revert_invalidSignature() (gas: 476251)
[PASS] test_revert_invalidSignature_eoaSignerMakerMismatch() (gas: 462308)
[PASS] test_revert_invalidTokenId_takerNonBinaryPosition() (gas: 455804)
[PASS] test_revert_makingGtRemaining() (gas: 463413)
[PASS] test_revert_mismatchedArrayLengths() (gas: 456340)
[PASS] test_revert_noMakerOrders() (gas: 211640)
[PASS] test_revert_notOperator() (gas: 38071)
[PASS] test_revert_orderAlreadyFilled() (gas: 575240)
[PASS] test_revert_paused() (gas: 475685)
[PASS] test_revert_takerReceiveAmountMismatch() (gas: 540329)
[PASS] test_revert_userIsPaused() (gas: 485594)
[PASS] test_revert_zeroMakerAmount() (gas: 75820)
[PASS] test_revert_zeroMakerAmount_taker() (gas: 351460)
Suite result: ok. 14 passed; 0 failed; 0 skipped; finished in 183.47ms (140.08ms CPU time)
```

```
Ran 2 tests for src/exchange/test/MatchOrders.t.sol:MatchOrdersTest_validateOrderBranches
[PASS] test_revert_validateOrder_alreadyFilled() (gas: 567534)
[PASS] test_revert_validateOrder_userPaused() (gas: 63939)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 50.38ms (5.08ms CPU time)
```

```
Ran 6 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_domain
[PASS] test_hashOrderLegacy_matchesIndependentComputation() (gas: 87210)
[PASS] test_legacyBinaryDomainSeparator_matchesIndependentComputation() (gas: 13812)
[PASS] test_legacyDomainSeparators_rebuildOnChainIdChange() (gas: 28569)
[PASS] test_legacyNegRiskDomainSeparator_matchesIndependentComputation() (gas: 16859)
[PASS] test_nativeAndLegacyDigests_differ() (gas: 65028)
[PASS] test_nativeDomain_unchanged() (gas: 12138)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 57.49ms (38.43ms CPU time)
```

```
Ran 1 test for src/routers/test/RouterNegRiskModule.t.sol:RouterNegRiskModule_Test_merge
[PASS] test_merge(uint256) (runs: 256,  $\mu$ : 417790,  $\sim$ : 417803)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 424.39ms (395.71ms CPU time)
```

```
Ran 2 tests for src/exchange/test/MatchOrdersBalanceFlows.t.sol:MatchOrdersBalanceFlowsTest_mint
[PASS] test_twoMakers() (gas: 757456)
[PASS] test_twoMakersWithFees() (gas: 787012)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 58.66ms (15.07ms CPU time)
```

```
Ran 7 tests for src/test/integration/StorageSlots.t.sol:StorageSlotsTest_collateralToken
[PASS] test_allowanceStoredInSlot() (gas: 44033)
[PASS] test_balanceStoredInSlot() (gas: 93352)
[PASS] test_implementationStoredInSlot() (gas: 7866)
[PASS] test_noncesStoredInSlot() (gas: 14456)
[PASS] test_ownerStoredInSlot() (gas: 7939)
[PASS] test_roleStoredInSlot() (gas: 45194)
[PASS] test_totalSupplyStoredInSlot() (gas: 93112)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 64.60ms (18.64ms CPU time)
```

```
Ran 4 tests for
src/routers/test/RouterCombinatorialCollateralReturn.t.sol:RouterCombinatorialCollateralReturnTest_validation
[PASS] test_allowsAnyCombinatorialModuleSelector() (gas: 98897)
[PASS] test_bubblesOperationRevert() (gas: 43724)
[PASS] test_revertsForMismatchedTransferInputs() (gas: 27665)
[PASS] test_revertsWhenCombinatorialModuleNotConfigured() (gas: 37796)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 26.67ms (7.40ms CPU time)
```

```
Ran 1 test for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_doubleFill
[PASS] test_noDoubleFill_acrossDomains() (gas: 1065919)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 34.60ms (11.06ms CPU time)
```

```
Ran 2 tests for src/exchange/test/MatchOrdersBalanceFlows.t.sol:MatchOrdersBalanceFlowsTest_merge
```

```
[PASS] test_twoMakers() (gas: 1079977)
[PASS] test_twoMakersWithFees() (gas: 1110196)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 41.51ms (20.62ms CPU time)

Ran 2 tests for
src/exchange/test/ExchangeSnapshots.t.sol:ExchangeSnapshotsTest_comboComplementaryMergeFees
[PASS] test_tenMakers() (gas: 3250894)
[PASS] test_twentyMakers() (gas: 6001831)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 126.20ms (34.95ms CPU time)

Ran 8 tests for src/exchange/test/MatchOrdersBalanceFlows.t.sol:MatchOrdersBalanceFlowsTest_complementary
[PASS] test_complementaryBuyMakerFeeOnly() (gas: 606825)
[PASS] test_complementaryBuyTakerFeeOnly() (gas: 606231)
[PASS] test_complementarySellMakerFeeOnly() (gas: 641103)
[PASS] test_complementarySellMixedMakerFees() (gas: 822177)
[PASS] test_takerBuyTwoMakers() (gas: 856080)
[PASS] test_takerBuyTwoMakersWithFees() (gas: 887204)
[PASS] test_takerSellTwoMakers() (gas: 776108)
[PASS] test_takerSellTwoMakersWithFees() (gas: 825595)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 66.36ms (38.56ms CPU time)

Ran 2 tests for src/exchange/test/ExchangeSnapshots.t.sol:ExchangeSnapshotsTest_comboComplementaryMint
[PASS] test_tenMakers() (gas: 2672198)
[PASS] test_twentyMakers() (gas: 4955342)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 114.89ms (46.84ms CPU time)

Ran 5 tests for
src/routers/test/RouterCombinatorialCollateralReturn.t.sol:RouterCombinatorialOperationsTest
[PASS] test_compress() (gas: 410840)
[PASS] test_extractInjectAndYesBasket() (gas: 856051)
[PASS] test_splitMergeAndConvertOnEvent() (gas: 1103367)
[PASS] test_splitMergeOnCondition() (gas: 570528)
[PASS] test_wrapAndUnwrap() (gas: 367293)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 38.15ms (20.22ms CPU time)

Ran 3 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_eoaReplay
[PASS] test_matchOrders_legacySigned_endToEnd() (gas: 1004573)
[PASS] test_validateOrder_acceptsLegacySignature() (gas: 143716)
[PASS] test_validateOrder_acceptsNativeSignature() (gas: 41895)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 78.38ms (16.78ms CPU time)

Ran 2 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_negative
[PASS] test_revert_legacySignature_wrongVerifyingContract() (gas: 41174)
[PASS] test_revert_unknownDomainSignature() (gas: 41285)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 42.29ms (9.23ms CPU time)

Ran 9 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_legacyUserPause
[PASS] test_isUserPaused_isNativeOnly() (gas: 89579)
[PASS] test_legacyOrder_v2PauseActivationDelayHonored() (gas: 241973)
[PASS] test_matchOrders_nativeOrder_makerPausedOnLegacy_stillFills() (gas: 854776)
[PASS] test_pendingLegacyPause_orderStillFillable() (gas: 1013276)
[PASS] test_revert_matchOrders_makerPausedOnLegacy() (gas: 236755)
[PASS] test_revert_matchOrders_preapprovedLegacyOrder_makerPausedAfterPreapproval() (gas: 276736)
[PASS] test_revert_matchOrders_takerPausedOnLegacy() (gas: 169089)
[PASS] test_revert_validateOrder_makerPausedOnLegacy() (gas: 117531)
[PASS] test_unpauseOnLegacy_reenablesLegacyOrderFill() (gas: 1094524)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 64.48ms (43.80ms CPU time)

Ran 3 tests for src/routers/test/RouterSnapshots.t.sol:RouterSnapshots_Test_binary
[PASS] test_merge() (gas: 406900)
[PASS] test_redeem() (gas: 440408)
[PASS] test_splitBinary() (gas: 288864)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 18.84ms (12.14ms CPU time)

Ran 1 test for src/routers/test/RouterNegRiskModule.t.sol:RouterNegRiskModule_Test_split
[PASS] test_split(uint256) (runs: 256, μ: 292889, ~: 292902)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 223.36ms (204.82ms CPU time)

Ran 8 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_negRiskVenue
[PASS]
testFuzz_hashOrderLegacy_negRisk_matchesRealVenue(uint256,address,address,uint256,uint256,uint8,uint8,uint256,bytes32,bytes32) (runs: 256, μ: 103056, ~: 103056)
```



```
[PASS] test_negRisk_carryOver_readsOwnVenueOnly() (gas: 369688)
[PASS] test_negRisk_matchOrders_endToEnd() (gas: 1260584)
[PASS] test_negRisk_partialFill_resumesFromVenueLeftover() (gas: 1311442)
[PASS] test_negRisk_pause_venueScoped() (gas: 314445)
[PASS] test_negRisk_realContract_validateOrder() (gas: 352195)
[PASS] test_negRisk_venuePreapproval_deferAndRevoke() (gas: 453294)
[PASS] test_revert_crossVenue_signatures() (gas: 353952)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 233.87ms (262.81ms CPU time)
```

```
Ran 7 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_preapprove
[PASS] test_deferredPreapproval_revokedOnV2_rejectedOnV3() (gas: 258994)
[PASS] test_deferredPreapproval_v2PreapprovedOrder_acceptedOnV3() (gas: 194677)
[PASS] test_preapprove_ignoresLegacyFillState() (gas: 122606)
[PASS] test_preapprove_nativeSignature() (gas: 73236)
[PASS] test_revert_emptySig_legacySignedButNotPreapprovedOnV2() (gas: 155793)
[PASS] test_revert_emptySignature_notPreapprovedAnywhere() (gas: 29904)
[PASS] test_revert_preapprove_legacySignature() (gas: 81829)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 312.98ms (41.95ms CPU time)
```

```
Ran 7 tests for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_migrationSequence
[PASS] test_deferredPreapproval_v2PreapprovedLegacyOrder_fillsViaEmptySig() (gas: 930151)
[PASS] test_downgrade_legacyPreapprovalInert_nativePreapprovalSurvives() (gas: 5842448)
[PASS] test_migrationPath_baseWhileLive_thenUpgradeForLegacy() (gas: 13711367)
[PASS] test_pausedLegacy_tradeNativePreapprovedThenLegacy() (gas: 1554084)
[PASS] test_revert_constructor_duplicateLegacyExchange() (gas: 208294)
[PASS] test_revert_constructor_zeroLegacyExchange() (gas: 211824)
[PASS] test_walletDerivation_matchesV2_divergesOnWrongConfig() (gas: 6614953)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 295.99ms (46.49ms CPU time)
```

```
Ran 4 tests for src/routers/test/RouterBinaryModule.t.sol:RouterBinaryModuleTest_redeem
[PASS] test_redeem(bool) (runs: 256, μ: 558314, ~: 558301)
[PASS] test_revert_invalidOutcomeIndex_aliased() (gas: 37382)
[PASS] test_revert_invalidOutcomeIndex_max(uint256) (runs: 256, μ: 33295, ~: 33272)
[PASS] test_revert_invalidOutcomeIndex_two() (gas: 29448)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 811.21ms (889.51ms CPU time)
```

```
Ran 1 test for src/exchange/test/MigratableExchange.t.sol:MigratableExchangeTest_fillConservation
[PASS] testFuzz_legacyMakerNeverOverfills(uint256,uint256,uint256) (runs: 256, μ: 859607, ~: 806825)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 395.42ms (378.40ms CPU time)
```

```
Ran 6 tests for src/exchange/test/MatchOrdersBalanceFlows.t.sol:MatchOrdersBalanceFlowsTest_mixedPath
[PASS] test_comboComplementaryMerge() (gas: 952389)
[PASS] test_comboComplementaryMergeWithFees() (gas: 982328)
[PASS] test_comboComplementaryMintNoFees() (gas: 859236)
[PASS] test_comboComplementaryMintWithFees() (gas: 6482596)
[PASS] test_threeMakersCompMergeWithFees() (gas: 1263510)
[PASS] test_threeMakersCompMintWithFees() (gas: 1074344)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 427.72ms (49.34ms CPU time)
```

```
Ran 2 tests for src/routers/test/RouterBinaryModule.t.sol:RouterBinaryModuleTest_split
[PASS] test_split() (gas: 311827)
[PASS] test_split_anyCondition() (gas: 291487)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 5.66ms (1.40ms CPU time)
```

```
Ran 3 tests for
src/routers/test/RouterCombinatorialCollateralReturn.t.sol:RouterCombinatorialCollateralReturnTest_execution
[PASS] test_routesIntermediateAndTerminalByproducts() (gas: 613245)
[PASS] test_splitAndMerge() (gas: 301115)
[PASS] test_transfersAllWalletPositionsBeforeCalls() (gas: 344578)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 6.32ms (2.35ms CPU time)
```

```
Ran 6 tests for src/exchange/test/MatchOrdersBalanceFlows.t.sol:MatchOrdersBalanceFlowsTest_surplusRefund
[PASS] test_batchBuyCollateralSurplusRefunded() (gas: 593257)
[PASS] test_batchBuyPositionSurplusRefunded() (gas: 884120)
[PASS] test_batchBuyPositionSurplusWithFees() (gas: 908421)
[PASS] test_batchMintOverfundedRefundedToTaker() (gas: 593259)
[PASS] test_batchSellPositiveSlippageRefundedToTaker() (gas: 955417)
[PASS] test_batchSellPositiveSlippageWithFees() (gas: 978929)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 593.32ms (24.56ms CPU time)
```

```
Ran 6 tests for src/exchange/test/ExchangeMath.t.sol:ExchangeMathTest
```

```
[PASS] test_calculateTakingAmount_fuzz(uint64,uint128,uint128) (runs: 256, μ: 11754, ~: 11754)
[PASS]
test_complementaryAggregateCrossingInvariant_fuzz(uint64,uint64,uint64,uint64,uint64,uint64,uint64,uint64
) (runs: 256, μ: 21356, ~: 21356)
[PASS] test_statusFillAmount_fuzz(uint64,uint64,uint128,uint128) (runs: 256, μ: 21678, ~: 21678)
[PASS] test_statusFillAmount_limitBuy() (gas: 14766)
[PASS] test_statusFillAmount_marketBuy() (gas: 14288)
[PASS] test_statusFillAmount_sell() (gas: 14312)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 941.74ms (1.32s CPU time)

Ran 3 tests for src/routers/test/RouterNegRiskModule.t.sol:RouterNegRiskModule_Test_convert
[PASS] test_convert(uint256) (runs: 256, μ: 8054272, ~: 8054272)
[PASS] test_revert_conditionIndexMaxUint16() (gas: 124052)
[PASS] test_revert_conditionIndexOutOfRange() (gas: 126578)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 583.69ms (575.15ms CPU time)

Ran 22 tests for src/bridge/test/CcipBridgeMigration.t.sol:CcipBridgeMigrationTest
[PASS] test_bridge_migration_binaryPositionId_matchesAcrossChains(bytes32,uint256) (runs: 256, μ: 10054,
~: 10083)
[PASS] test_bridge_migration_binaryPositions_hubToSpoke() (gas: 1102210)
[PASS] test_bridge_migration_binaryPositions_idempotentConditionPrep() (gas: 1632547)
[PASS] test_bridge_migration_binaryPositions_partialAmount() (gas: 1131922)
[PASS] test_bridge_migration_binaryPositions_roundTrip() (gas: 2094982)
[PASS] test_bridge_migration_binaryPositions_spokeToHub() (gas: 1568081)
[PASS] test_bridge_migration_binaryPositions_spokeToSpoke() (gas: 1782491)
[PASS] test_bridge_migration_binaryPositions_unpreparedMigrationCondition_noRevert() (gas: 493909)
[PASS] test_bridge_migration_binaryResult_hubToSpoke() (gas: 1549790)
[PASS] test_bridge_migration_binaryResult_spokeToSpoke() (gas: 2794781)
[PASS] test_bridge_migration_moduleId_matches() (gas: 44522)
[PASS] test_bridge_migration_negRiskDerivation_matchesAcrossChains(bytes32,uint8,uint256) (runs: 256, μ:
10850, ~: 11022)
[PASS] test_bridge_migration_negRiskNoResult_hubToSpoke_beforeEventFullyResolved() (gas: 1573146)
[PASS] test_bridge_migration_negRiskOtherResult_hubToSpoke_afterEventFullyResolved() (gas: 1732805)
[PASS] test_bridge_migration_negRiskPositions_hubToSpoke() (gas: 1434702)
[PASS] test_bridge_migration_negRiskPositions_idempotentConditionPrep() (gas: 1975293)
[PASS] test_bridge_migration_negRiskPositions_partialAmount() (gas: 1445645)
[PASS] test_bridge_migration_negRiskPositions_roundTrip() (gas: 2321482)
[PASS] test_bridge_migration_negRiskPositions_spokeToHub() (gas: 1968638)
[PASS] test_bridge_migration_negRiskPositions_spokeToSpoke() (gas: 2016160)
[PASS] test_bridge_migration_negRiskResult_hubToSpoke() (gas: 2205152)
[PASS] test_bridge_migration_negRiskResult_spokeToSpoke() (gas: 3178316)
Suite result: ok. 22 passed; 0 failed; 0 skipped; finished in 1.78s (771.23ms CPU time)

Ran 340 test suites in 2.44s (23.54s CPU time): 1544 tests passed, 0 failed, 3 skipped (1547 total tests)
```

Code Coverage

The test suite's coverage of the in scope contracts can be obtained by running the following command:

```
forge coverage --no-match-coverage "test|EOA|mocks|legacy|dev|collateral|Ccip|adapters|script"
```

File	% Lines	% Statements	% Branches	% Funcs
src/abstract/ERC1155TokenReceiver.sol	100.00% (7/7)	100.00% (6/6)	100.00% (0/0)	100.00% (3/3)
src/auth/InitializableRoles.sol	100.00% (18/18)	100.00% (9/9)	100.00% (0/0)	100.00% (9/9)
src/auth/Roles.sol	100.00% (32/32)	100.00% (16/16)	100.00% (0/0)	100.00% (16/16)
src/bridge/abstract/BridgeBase.sol	100.00% (152/152)	100.00% (164/164)	97.92% (47/48)	100.00% (28/28)

File	% Lines	% Statements	% Branches	% Funcs
src/exchange/EIP712WithLegacyDomain.sol	100.00% (38/38)	100.00% (37/37)	100.00% (4/4)	100.00% (7/7)
src/exchange/Exchange.sol	99.81% (513/514)	97.09% (601/619)	83.18% (89/107)	100.00% (67/67)
src/exchange/MigratableExchange.sol	99.21% (125/126)	99.40% (166/167)	100.00% (33/33)	100.00% (16/16)
src/libraries/BridgePayloads.sol	100.00% (6/6)	100.00% (6/6)	100.00% (0/0)	100.00% (3/3)
src/libraries/Ids.sol	96.15% (50/52)	96.49% (55/57)	100.00% (2/2)	95.83% (23/24)
src/modules/BinaryModule.sol	100.00% (33/33)	100.00% (50/50)	100.00% (7/7)	100.00% (6/6)
src/modules/CombinatorialModule.sol	99.30% (427/430)	99.04% (518/523)	82.78% (125/151)	100.00% (42/42)
src/modules/NegRiskModule.sol	100.00% (127/127)	99.46% (184/185)	94.59% (35/37)	100.00% (19/19)
src/modules/abstract/BaseModule.sol	100.00% (52/52)	100.00% (51/51)	81.82% (9/11)	100.00% (12/12)
src/modules/abstract/OracleModule.sol	92.86% (26/28)	94.44% (17/18)	100.00% (9/9)	90.00% (9/10)
src/modules/migration/BaseMigrationMixin.sol	100.00% (86/86)	100.00% (109/109)	96.77% (30/31)	100.00% (9/9)
src/modules/migration/BinaryMigrationMixin.sol	100.00% (28/28)	100.00% (21/21)	83.33% (5/6)	100.00% (11/11)
src/modules/migration/NegRiskMigrationMixin.sol	100.00% (53/53)	100.00% (63/63)	94.44% (17/18)	100.00% (13/13)
src/oracle/OracleAggregator.sol	96.28% (259/269)	96.23% (281/292)	93.75% (105/112)	90.70% (39/43)
src/oracle/abstract/OracleModuleBase.sol	100.00% (11/11)	100.00% (6/6)	100.00% (4/4)	100.00% (5/5)
src/oracle/libraries/OptimisticOraclePayoutLib.sol	100.00% (8/8)	100.00% (11/11)	100.00% (6/6)	100.00% (1/1)
src/oracle/mixins/Auth.sol	100.00% (12/12)	100.00% (6/6)	100.00% (0/0)	100.00% (6/6)
src/oracle/mixins/MarketDataRegistry.sol	100.00% (46/46)	100.00% (46/46)	100.00% (5/5)	100.00% (12/12)
src/oracle/mixins/Pausable.sol	100.00% (8/8)	100.00% (5/5)	100.00% (2/2)	100.00% (3/3)
src/oracle/modules/OOReporterModule.sol	100.00% (66/66)	100.00% (86/86)	100.00% (16/16)	100.00% (12/12)
src/oracle/modules/reporters/ChainlinkReporterModule.sol	98.65% (73/74)	100.00% (82/82)	100.00% (17/17)	93.75% (15/16)

File	% Lines	% Statements	% Branches	% Funcs
src/positionManager/PositionManager.sol	95.43% (167/175)	95.27% (161/169)	88.24% (30/34)	100.00% (17/17)
src/routers/BridgeRouter.sol	100.00% (18/18)	100.00% (11/11)	100.00% (0/0)	100.00% (7/7)
src/routers/CtfRouter.sol	100.00% (35/35)	100.00% (42/42)	100.00% (3/3)	100.00% (6/6)
src/routers/Router.sol	99.47% (186/187)	99.10% (221/223)	100.00% (9/9)	100.00% (30/30)
src/utils/AutoRedeemer.sol	100.00% (98/98)	98.32% (117/119)	100.00% (18/18)	100.00% (6/6)
Total	98.96% (2760/2789)	98.41% (3148/3199)	90.87% (627/690)	98.47% (452/459)

Changelog

- 2026-08-07 - Initial report
- 2026-08-20 - Final report
- 2026-08-25 - Updated report (inc PR-341)

About Quantstamp

Quantstamp is a global leader in blockchain security. Founded in 2017, Quantstamp's mission is to securely onboard the next billion users to Web3 through its best-in-class Web3 security products and services.

Quantstamp's team consists of cybersecurity experts hailing from globally recognized organizations including Microsoft, AWS, BMW, Meta, and the Ethereum Foundation. Quantstamp engineers hold PhDs or advanced computer science degrees, with decades of combined experience in formal verification, static analysis, blockchain audits, penetration testing, and original leading-edge research.

To date, Quantstamp has performed more than 1300 audits and secured over \$500 billion in digital asset risk from hackers. Quantstamp has worked with a diverse range of customers, including startups, category leaders and financial institutions. Brands that Quantstamp has worked with include Ethereum 2.0, Binance, Visa, PayPal, Solana, Canton, Polymarket, PancakeSwap, Virtuals Protocol, Wayfinder, Lido, TrustWallet, Alchemy, World Economic Forum.

Quantstamp's collaborations and partnerships showcase our commitment to world-class research, development and security. We're honored to work with some of the top names in the industry and proud to secure the future of web3.

Notable Collaborations & Customers:

- Blockchains: Ethereum 2.0, Solana, Polygon, TON, Cardano, Binance Smart Chain, Avalanche, Arbitrum, Flow
- DeFi: Lido, Ethena, Compound, Curve, Venus, PancakeSwap, Polymarket
- Infra/Staking: TrustWallet, Alchemy, Liquid Collective, Kiln, Galxe, API3, ssv.network, Luganodes
- Immersive: Virtuals Protocol, Wayfinder, OpenSea, Square Enix, Parallel, Camp, Decentraland
- Institutional: Visa, Circle, Revolut, Anthropic, OpenAI, Canton, Republic, Arkham, Sequoia

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